

Extended Data

Full appendix ledgers and benchmark manifests for
“Brain Researcher: AI-assisted research infrastructure workspace for neuroimaging analyses”

This document reproduces, in full, the data-card ledgers, the per-rule scientific-review registry, and the benchmark item/example/bundle manifests that were moved out of the lean reviewer-facing Supplementary Information to keep it short. The Supplementary Information retains a one-paragraph data card for each appendix (Appendix A–J) and the reviewer-facing Appendices G, I, J, and K; this Extended Data document carries the corresponding full ledgers. Appendix and section labels match the Supplementary Information. See `supplement_crosswalk.md` and `MANIFEST.json` in this package for the mapping and checksums.

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Appendix A. Episode and control-plane card

A1. Purpose

Appendix A records the fixed identity and control-plane configuration of each research episode. It answers: under what MCP implementation, model route, registry snapshot, BR-KG snapshot, policy profile, memory namespace, and episode-state history did this result occur?

A2. Episode identity and final state

Field	Example / value	Why it belongs in Appendix A
Episode ID	ep_hcp_predict.001	Stable identity linking the episode to all downstream cards.
Episode mode	interactive / benchmark / collaborator / bounded autonomous	Defines who exercises the commitment gate and how the episode is evaluated.
Scientific question	Predict behavioral score from HCP-YA resting-state FC features.	Defines the episode-level research target.
Episode source	collaborator case / benchmark task / autonomous campaign	Connects the episode to the evaluation arm.
Start and completion time	timestamp or run date	Supports reproducibility and run lookup.
Final state	completed / completed_with_qualifications / blocked / terminated	Summarizes the endpoint without replacing the run bundle.
Main output	scorecard, review card, claim card, report, figure	Points readers to the artifact families produced by the episode.

A3. Fixed model and prompt configuration

Field	Example / value	Notes
Model provider	fixed provider ID	Use a stable identifier for reported evaluations.
Model version	fixed model version	Avoid reporting claims from undocumented model drift.
Task route	planning / coding / review / summarization	Different tasks may be routed to different model classes.
Prompt-template version	planning_v3, review_v2	Records the prompt contract used for the episode.
Provider fallback allowed?	yes/no	For benchmark claims, fallback should usually be disabled or logged.
Evaluation locked?	yes/no	Indicates whether model/prompt/registry/KG snapshots were fixed.

```
model_route:
task_type: scientific_review
provider: fixed_provider_id
model_version: fixed_model_version
prompt_template: review_v2
evaluation_locked: true
```

A4. MCP implementation snapshot

Field	Value to record
Primary implementation	src/brain_researcher/services/mcp/server.py
Helper modules	execution_recipes.py; loop_primitives.py; research_summaries.py; sherlock_tools.py
Decorated MCP operations	87
Decorated MCP resources	3
Published schema entries	72
Claude Code hint metadata entries	83
Exposed plus workflow specs	177
Total registry specs	2,089
MCP-related test files	33
Default loop profile	external_coding_v1
Run root	recorded path or redacted path
Policy flags	network / dangerous / direct execution flags

A5. MCP capability-family summary

Capability family	Role in the episode
Tool/workflow discovery	Returns candidate tools and workflows rather than requiring the LLM to invent commands.
Dataset resolution	Resolves dataset identity, access, and asset readiness.
Planning/preflight	Identifies blockers, warnings, and the execution envelope.
Execution recipe	Converts a selected workflow into Python, Neurodesk, container, or Slurm instructions.
Run observability	Reads run status, logs, bundles, scorecards, and artifacts.
Artifact inspection	Reads produced files and metadata.
BR-KG access	Retrieves structured evidence and graph paths.
Grounding	Validates or downgrades evidence anchors.
Scientific review	Returns review verdicts and revision routes.
Memory	Reads or writes reviewed memory cards.
Slurm/HPC support	Renders or diagnoses cluster jobs.
Report generation	Produces manuscript/report artifacts.

A7. LLM/MCP decision boundary

Step	LLM / client role	MCP / backend role	Evidence returned
Interpret request	Proposes candidate modality, dataset, method, and constraints.	Receives structured query arguments.	Query/session context.
Discover tools	Asks for candidate tools or workflows.	Searches registry and returns ranked cards.	Tool/workflow cards.
Resolve selection	Chooses among returned options.	Resolves canonical ID and schema.	Metadata and parameter schema.
Preflight inputs	Asks whether the plan is feasible.	Checks resources, policy, and constraints.	Blockers and warnings.
Generate recipe	Requests a backend-specific plan.	Returns Python, Neurodesk, container, or Slurm recipe.	Recipe and expected outputs.
Execute work	Delegates to shell, container, scheduler, or admin gate.	Executes only if explicitly enabled and allowed.	Run ID, logs, artifacts.
Review outputs	Proposes interpretation.	Normalizes bundles, scorecards, and review outputs.	Review verdict and revision route.

Step	LLM / client role	MCP / backend role	Evidence returned
Explain or iterate	Summarizes evidence and selects next step.	Provides durable logs, memory, grounding, and run/session digests.	Final evidence basis or next-step plan.

A8. Policy and loop-profile summary

Field	Value to record
Loop profile	external_coding_v1 or configured alternative
Repository mutation inside MCP	disabled by default / enabled only under gated mode
Direct MCP execution	disabled unless admin gate is enabled
Network access	enabled/disabled
Dangerous tools	enabled/disabled
Allowed roots	recorded or redacted
Timeout policy	recorded timeout values
Auth mode	local stdio / HTTP bearer / JWT
Memory namespace	benchmark / collaborator / autonomous / development

A9. Checkpoint and state-transition excerpt

Step	State	Event
001	initialized	Episode record and run root created.
002	planning	Candidates, evidence bundle, and resource ledger populated.
003	committed or blocked	Preflight and commitment-gate verdict returned.
004	execution	Backend recipe delegated or run started.
005	review	Run bundle and candidate claims reviewed.
006	completed or terminated	Final state and writeback status recorded.

A10. Example episode-control card

episode_control_card: episode_id: ep_hcp_predict_001 mode: benchmark scientific_question: "Predict behavioral score from HCP-YA resting-state FC features." final_state: completed_with_qualifications model_route: planning_model: fixed_model_id review_model: fixed_model_id prompt_templates: planning: planning_v3 review: review_v2 snapshots: registry_snapshot: registry_YYYYMMDD brkg_snapshot: brkg_YYYYMMDD mcp_schema_version: mcp_schema_YYYYMMDD mcp_surface: operations: 87 exposed_plus_workflow_specs: 177 total_registry_specs: 2089 policy: loop_profile: external_coding_v1 network: true dangerous_tools: false direct_tool_execution: false memory_namespace: benchmark_isolated

A11. Architecture and Runtime Boundaries

Layer	Component	Responsibility	Notes
Transport	FastMCP	MCP protocol server over stdio, SSE, or streamable HTTP	mcp = FastMCP("brain-researcher")
HTTP wrapper	build_http_app	Starlette wrapper for auth, /healthz, /resolve, session bootstrap	Used for HTTP transports only
Tool surface	decorated functions in server.py	Public MCP tools and resources	87 tools, 3 resources
Discovery layer	UnifiedToolRegistry, tool_search, tool_get	Search BR tool specs and workflow specs	Quick self-test loaded 177 exposed+workflow specs and 2,089 all specs
Recipe layer	execution_recipes.py	Build stateless local execution recipes	Targets: python, neurodesk, container, slurm
Admin execution layer	tool_execute, pipeline_execute	Preview or execute gated tools/pipelines	Disabled by default for agents; allowlist and policy gates apply
Run store	data/runs/mcp_runs by default	Persist run records, traces, artifacts, bundles, research logs	Uses build_mcp_run_dir, iter_mcp_run_dir_candidates
Loop primitives	loop_primitives.py	Normalize run bundles, scorecards, comparisons	Used by external coding-agent loop
KG access	kg_* tools	Read-only Neo4j query helpers plus hypothesis/probe workflows	KG reads fail fast unless degraded mode is explicitly allowed
Dataset resolution	dataset_get_resources	Resolve local paths, BIDS paths, derivatives, URLs	Uses shared dataset mounts/resources
Grounding	grounding_resolve, grounding_gate_evidence_basis	Resolve and gate evidence anchors	Rejects malformed anchors, downgrades unresolved grounded anchors
Research logging	log_research_event, write_session_snapshot	Durable session start/note/snapshot record	Also attached to automatic tool-call telemetry

Layer	Component	Responsibility	Notes
Memory	MemoryStore, memory_* tools	Derived memory cards and relation events	Local store under run-root memory index/cards when present
Literature	Google tools, DeepXiv	File search, Gemini Deep Research, arXiv/PMC search/read	Network and API-key gated
Sherlock/HPC	sherlock_tools.py	Slurm script rendering, validation, job/log diagnostics	Advanced surface, not separate product
Docker package	infrastructure/docker/Dockerfile.mcp	Slim MCP runtime image	Exposes port 7000; leaves behavior-task heavy deps out
Neuroimaging registry	BR tool registry and workflow catalog	Stores the larger neuroimaging method/workflow surface that MCP can search, resolve, and package into recipes.	Inventory reports 177 exposed-plus-workflow specs and 2,089 total registry specs loaded by quick self-test.
Backend recipe targets	Python, Neurodesk, container, Slurm	Converts a selected registry tool/workflow into a backend-specific command, container invocation, or batch script.	Recipes are stateless handoffs unless the admin execution gate is explicitly enabled.
LLM client boundary	External coding agent, notebook, IDE, or web assistant	Interprets the goal, calls MCP tools, chooses among returned candidates, and explains results.	The LLM does not bypass MCP policy gates and is not the default repository mutator inside MCP.

A12. Compute Nodes

Node	Inputs	Outputs	Persistence
MCP transport	JSON-RPC over stdio, SSE, or streamable HTTP	tool list, tool result, resource result	none by itself
HTTP middleware	headers, auth token, host/origin, optional session id	authenticated request, session bootstrap, errors	in-memory session cache
tool list wrapper	FastMCP tool manager	MCP tool schemas plus optional Claude Code annotations/search hints	none
research telemetry wrapper	tool name, args, result, session binding	trace events, directives, session binding metadata	run root
discovery registry	tool registry configs, workflow catalog, query	ranked tool/workflow cards	in-memory caches and registry files
planning	user query, domain/modality, inputs	preflight facts, blockers, display plan, execution envelope	latest handoff cache
execution recipe	tool/workflow id, params, runtime target	command/script/container/Slurm recipe	response only unless caller persists
admin execution	gated plan/tool call, policy flags, allowlist, roots	run id, step records, artifacts, preflight issues	run root
run observability	run id	run status, logs, metrics, bundle, scorecard, compare	reads run root
KG access	KG ids/query/hypothesis	nodes, neighbors, paths, evidence, candidate cards	Neo4j read; optional run artifacts for workflows
grounding	reference anchors, evidence rows, resolvers	resolved support or downgraded evidence basis	response and optional run/session artifacts
research logging	session id, event/snapshot payload	research event log, session snapshot, digest	run root
memory	card payloads, query filters	memory cards, relation events, search hits	run-root memory store

A13. High-Level Directed Edges

Edge	Trigger	Main function/tool	Main output
client -> discovery	user asks what to use	tool_search, workflow_search, tool_get	candidate tools/workflows
discovery -> recipe	selected tool/workflow	get_execution_recipe	local execution instructions
query -> preflight -> plan	user asks for a plan	plan_preflight, plan_create	display plan and execution envelope
plan -> admin execution	explicit manual/admin approval	pipeline_plan_validate, pipeline_execute	persisted run
tool id -> admin execution	explicitly allowed single-tool call	tool_execute	preview or persisted run
run -> bundle	user/agent needs evidence	run_bundle_get	normalized run observation
bundle -> scorecard	quality inspection	run_scorecard	artifact completeness and quality signals
baseline + candidate -> compare	keep/discard decision	run_compare	directional metric comparison
query/entities -> KG	research grounding/hypothesis work	kg_* tools	KG evidence and structural signals
KG/literature -> grounding	final-answer evidence check	grounding_resolve, grounding_gate_evidence_basis	valid, downgraded, or rejected anchors
work session -> durable memory	start/note/close	log_research_event, write_session_snapshot	session digest and resumable snapshot

Appendix B. Evidence bundle and BR-KG atlas card

Purpose

Appendix B should record the evidence substrate available to Brain Researcher during planning and review. It links the episode query to resolved entities, ONVOC normalization, graph paths, evidence tiers, source/provenance coverage, connector failures, and release-readiness caveats.

B1. Headline values

Metric	Value	Evidence / caveat
Snapshot date	2026-05-03	Live query against production Neo4j pod
Query target	production Neo4j via brain-researcher-BR-KG-0	Production connection target
Unique nodes	694,135	MATCH (n) RETURN count(n)
Edges	2,423,334	MATCH ()-[r]->() RETURN count(r)
Active node labels	75	Labels present on at least one node
Token-store node labels	87	Includes zero-node labels
Active relationship types	85	Types present on at least one edge
Token-store relationship types	86	Includes zero-edge relationship types
Canonical schema triples	151	Source-label / relationship / target-label patterns
Property keys	626	CALL db.propertyKeys()
Indexes	87	SHOW INDEXES
Constraints	27	SHOW CONSTRAINTS
Degree min / median / mean / p95 / p99 / max	0 / 1 / 6.9740 / 25 / 71 / 36,737	Undirected degree summary
StatsMap nodes / distinct runs	35,240 / 18	MATCH (s:StatsMap)
Text embeddings	23,865, dim=4096, model=BrainGPT-7B-v0.2, source=niclip	Embedding nodes
Task embeddings	44 text, 44 behavior, 44 both / 34,926 Task nodes	Task node properties

B2. BR-KG snapshot identity

Field	Example / value
BR-KG snapshot ID	brkg_YYYYMMDD
Graph database	Neo4j snapshot or production snapshot
Snapshot date	date
Node count	694,135
Relationship count	2,423,334
Active node labels	75
Active relationship types	85
Canonical schema triples	151
Orphan nodes	52,029 / 7.50%
Index / constraint state	87 indexes / 27 constraints
Release status	internal / draft / release-audit pending

B3. BR-KG visual atlas

The rendered atlas appears in Supplementary Figs. S4–S7. The table below maps the atlas panels to the appendix cards where each evidence surface is audited.

Figure	Placement	Why it belongs here
Node-type composition	Appendix B	Shows what dominates the graph surface.
Relationship-type composition	Appendix B	Shows which edge families carry most relationship volume.
Dominant schema meta-graph	Appendix B	Explains schema topology more clearly than a triples table.
Edge density matrix	Appendix B	Shows dense and sparse label-pair connections.
Literature peak	Appendix B	Shows publication to coordinate to term evidence entry.
Saved statistical map	Appendix B	Shows map to contrast/resource/model/region provenance.
Task to construct	Appendix B	Shows task as a bridge to construct and contrast definitions.
NeuroVault map asset	Appendix B	Shows why source-specific provenance remains visible.
Multi-hop query path	Appendix B	Shows why BR-KG is more than keyword search.
Dataset coverage	Appendix C	Belongs to dataset/resource readiness rather than the BR-KG evidence card.
Tool/workflow contract	Appendix D	Belongs to the tool registry and specification ledger.

B4. Node-label and relationship-type composition

Top node label	Count	Share
Coordinate	447,499	64.47%
Publication	49,744	7.17%
Collection	48,009	6.92%
StatsMap	35,240	5.08%
Task	34,926	5.03%
Embedding	23,865	3.44%
StatisticalMap	21,283	3.07%
DataResource	9,282	1.34%

Top relationship type	Count	Share
BELONGS_TO	1,077,573	44.47%
HAS_COORDINATE	447,499	18.47%
HAS_TERM	358,050	14.78%
IN_REGION	121,261	5.00%
ABOUT	75,922	3.13%
IN_ONVOC	63,160	2.61%

B5. Source and provenance coverage

Coverage item	Present	Total	Coverage
Nodes with id	693,919	694,135	99.97%
Nodes with source	687,246	694,135	99.01%
Edges with source-like value	2,381,941	2,423,334	98.29%
Edges with confidence	318,609	2,423,334	13.15%
Edges with weight	480,061	2,423,334	19.81%

B6. Source-like value table

Source-like value	Node count	Edge count with same source-like value
neurosynth	464,946	358,050
neurostore	97,238	0
openneuro_glmfitlins	45,395	266,193
neurovault	37,467	1,076,839

Source-like value	Node count	Edge count with same source-like value
niclip	23,865	0
missing	6,889	41,393
cognitive_atlas	4,214	14,591

B7. Evidence bundle schema

Field	Meaning
input_query	Original or normalized query for the episode.
resolved_entities	Task, contrast, construct, region, dataset, and tool entities.
ONVOC mappings	Free text mapped to typed vocabulary records.
graph_paths	Representative BR-KG paths used during planning or review.
evidence_tiers	Accepted graph, batch-extracted, candidate, real-time retrieval, or failed connector.
literature_results	Retrieved papers or retrieval summaries.
connector_failures	Timeouts, missing API results, or failed connectors.
dataset_links	Dataset/resource candidates passed to Appendix C.
tool_links	Tool/workflow candidates passed to Appendix D.
prior_findings	Relevant memory or BR-KG findings.
method-condition records	Known failure modes or method constraints passed to Appendix E.
coverage_notes	Sparse domain, missing confidence, source gap, or retrieval caveat.
evidence_basis_verdict	sufficient / partial / weak / blocked.

B8. ONVOC normalization example

onvoc_normalization: raw_terms: - "N-back" - "WM.BACK" - "working-memory task" - "verbal working memory" normalized_task_family: ONVOC:N_BACK linked_constructs: - working_memory non_collapsed_fields: - contrast_definition - modality - stimulus_type - coordinate_space
--

B9. Representative graph paths

Publication -> HAS_COORDINATE -> Coordinate -> IN_REGION -> BrainRegion StatsMap -> GENERATED_FROM -> ModelSpec -> HAS_CONTRAST -> TaskContrast Task -> IN_ONVOC -> OnvocClass -> MAPS_TO -> CognitiveConcept Dataset -> HAS_TASK -> Task -> MEASURES/SUGGESTS_MEASURES -> Construct Tool -> HAS_CONTRACT -> MethodCondition -> KNOWN_TO_FAIL_WHEN -> Constraint
--

B10. Evidence-tier legend

Evidence tier	Meaning	Can it support a claim?
Accepted graph evidence	Source-backed and accepted BR-KG record.	Yes, with source citation and review.
Batch-extracted evidence	Structured extraction with provenance; may be confidence-marked.	Partial; usually needs caveat/review.

Evidence tier	Meaning	Can it support a claim?
Candidate evidence	Generated or unaccepted KGGGen/GABRIEL/ retrieval record.	No; planning/review only.
Real-time retrieval	Connector output from PubMed, literature, or web-like tools.	Only after source citation and review.
Failed connector	Missing, timeout, or error record.	No; triggers coverage caveat.

B11. Embedding-lane metadata

Lane	Usage surface	Model	Dimension	Snapshot / index state
tool-description lane	tool/workflow retrieval	model ID	dim	snapshot; ready/ partial
brain-text lane	task/construct/region simi- larity	model ID	dim	snapshot; ready/ partial
claim-memory lane	prior claim retrieval	model ID	dim	snapshot; ready/ partial
graph-storage/source lane	graph/source-specific em- beddings	BrainGPT/niclip etc.	4096 if applica- ble	snapshot; ready/ partial

B12. Release-readiness and quality blockers

Issue	Current value	Manuscript implication
Unattributed source-like records	6,889 nodes / 41,393 edges	Document, attribute, exclude, or mark before release.
Connected components	Not measured	Do not report connected-component claims.
Edge confidence coverage	13.15%	Do not imply all edges carry confidence scores.
Edge weight coverage	19.81%	Weights exist only for selected relationship families.
Gabriel/KGGGen live source marking	0 marked nodes / 0 marked edges	Treat as candidate or comparison lanes unless accepted records are source-marked.
License/citation matrix	Not release-verified	Needed before public release and final manuscript claims.

B13. Example evidence-bundle excerpt

```

evidence_bundle:
episode.id: ep_hcp_predict.001
input_query: "Predict behavioral score from HCP-YA resting-state FC features."
resolved_entities:
dataset: HCP-YA
modality: resting_state_fmri
feature_type: functional_connectivity
target_type: behavioral_score
evidence_tiers:
accepted_graph_evidence:
- brkg_path: Dataset -> Task -> Construct
- brkg_path: Tool -> MethodCondition -> Constraint
batch_extracted_evidence: []
candidate_evidence: []
real_time_retrieval:
- literature_search_summary
connector_failures:
- connector: PubMed
status: timeout
consequence: "coverage caveat added"
downstream_links:
dataset_card: Appendix C
tool_ledger: Appendix D
constraint_card: Appendix E

```

B14. Full BR-KG active node-label counts

Node label	Count	Share of unique nodes
Coordinate	447,499	64.47%
Publication	49,744	7.17%
Collection	48,009	6.92%
StatsMap	35,240	5.08%
Task	34,926	5.03%
Embedding	23,865	3.44%
StatisticalMap	21,283	3.07%
DataResource	9,282	1.34%
OpenNeuro	7,619	1.10%
ToolVersion	4,142	0.60%
Term	3,228	0.47%
Concept	2,336	0.34%
Contrast	2,206	0.32%
BrainRegion	2,140	0.31%
Dataset	2,136	0.31%
Tool	2,084	0.30%
Phenotype	1,218	0.18%
Citation	1,201	0.17%
Subject	1,139	0.16%
TaskCondition	807	0.12%
OntologyConcept	752	0.11%
OnvocClass	752	0.11%
GLMContrast	621	0.09%
TaskIndicator	584	0.08%
SubjectGroup	402	0.06%
Condition	310	0.04%
GLMDesignPrior	216	0.03%
Author	172	0.02%
TaskFamily	138	0.02%
Institution	119	0.02%
TaskSpec	110	0.02%

Node label	Count	Share of unique nodes
ReviewSchemaField	106	0.02%
ResourceType	102	0.01%
ToolFamily	96	0.01%
Repository	88	0.01%
BrainAnnotation	86	0.01%
ReviewImplementationRule	80	0.01%
ModelSpec	78	0.01%
Experiment	76	0.01%
Psych101Experiment	76	0.01%
TaskAnalysis	72	0.01%
ReviewRule	71	0.01%
ReviewCalibrationCase	60	0.01%
StatMap	60	0.01%
Consortium	38	0.01%
Study	33	0.00%
Modality	28	0.00%
Parcellation	28	0.00%
Battery	27	0.00%
Parcel	26	0.00%
Finding	16	0.00%
Species	14	0.00%
ReviewRuleGroup	13	0.00%
IngestionRun	12	0.00%
ConceptClass	10	0.00%
Process	10	0.00%
TemplateSpace	10	0.00%
Region	8	0.00%
ReviewReasonTag	8	0.00%
GLMVariant	7	0.00%
ReviewPolicyDecision	6	0.00%
ReviewLifecycleStatus	5	0.00%
ReviewSensitivityTemplate	5	0.00%
ReviewValidityLayer	5	0.00%
ExecutionFailure	4	0.00%
Run	4	0.00%
Atlas	2	0.00%
ReviewSeverity	2	0.00%
GLMRun	1	0.00%
Psych101Dataset	1	0.00%
ResultSummary	1	0.00%
ReviewImplementationRuleCatalog	1	0.00%
ReviewPositiveModifier	1	0.00%
ReviewRuleRegistry	1	0.00%
ToolEvidence	1	0.00%

B15. Full BR-KG active relationship-type counts

Relationship type	Count	Share of edges
BELONGS_TO	1,077,573	44.47%
HAS_COORDINATE	447,499	18.47%
HAS_TERM	358,050	14.78%
IN_REGION	121,261	5.00%
ABOUT	75,922	3.13%
IN_ONVOC	63,160	2.61%

Relationship type	Count	Share of edges
IN_DOMAIN	52,117	2.15%
IN_SPACE	35,199	1.45%
COMPUTED_WITH	30,880	1.27%
GENERATED_FROM	30,880	1.27%
DERIVED_FROM	30,163	1.24%
MAPS_TO	17,667	0.73%
HAS_RESOURCE	14,728	0.61%
HAS_TEXT_EMBEDDING	12,958	0.53%
MEASURES	10,875	0.45%
IMPLEMENTS_FAMILY	4,294	0.18%
HAS_VERSION	4,142	0.17%
CITES	3,397	0.14%
SUPPORTS_MODALITY	2,655	0.11%
SUGGESTS_MEASURES	2,514	0.10%
HAS_CONTRAST	2,464	0.10%
HAS_REGION	2,123	0.09%
PARTICIPATES_IN	1,982	0.08%
HAS_MODALITY	1,839	0.08%
HOSTED_AT	1,694	0.07%
HAS_TASK	1,679	0.07%
PART_OF	1,425	0.06%
HASCITATION	1,293	0.05%
ASSERTS	1,237	0.05%
HAS_PHENOTYPE	1,218	0.05%
MEASUREDBY	1,218	0.05%
INVOLVES_SPECIES	990	0.04%
CLASSIFIED_UNDER	863	0.04%
HASCONDITION	807	0.03%
USES_CONDITION	750	0.03%
HASINDICATOR	584	0.02%
CLASSIFIEDUNDER	523	0.02%
KINDOF	429	0.02%
INCLUDES	402	0.02%
PRODUCES_RESOURCE	362	0.01%
CONSUMES_RESOURCE	338	0.01%
HAS_CONDITION	310	0.01%
AFFILIATED_WITH	292	0.01%
PARTOF	263	0.01%
AUTHORED_BY	232	0.01%
HAS_GLM_PRIOR	224	0.01%
RELATED_TO	190	0.01%
BELONGS_TO_FAMILY	176	0.01%
REQUIRES_FIELD	163	0.01%
USES_TASK	152	0.01%
INBATTERY	134	0.01%
HAS_VALIDITY_LAYER	102	0.00%
HAS_REASON_TAG	81	0.00%
CONTAINS_IMPLEMENTATION_RULE	80	0.00%
HAS_EXPERIMENT	76	0.00%
DESCRIBES_TASK	72	0.00%
CONTAINS_RULE	71	0.00%
HAS_LIFECYCLE_STATUS	71	0.00%
HAS_SEVERITY	71	0.00%
IN_RULE_GROUP	71	0.00%
CONTAINS_CALIBRATION_CASE	60	0.00%

Relationship type	Count	Share of edges
CALIBRATES_RULE	55	0.00%
CONTRAST_OF	53	0.00%
CITED_BY	49	0.00%
MAPPED_TO_IMPLEMENTATION	28	0.00%
HAS_PARCEL	26	0.00%
IN_PARCELLATION	17	0.00%
VALIDATED_ON	13	0.00%
HAD_FAILURE	7	0.00%
HAS_VARIANT	7	0.00%
ACTIVATES	6	0.00%
HAS_POLICY_DECISION	6	0.00%
TRIGGERS_SENSITIVITY	5	0.00%
FAILED_ON	3	0.00%
DOCUMENTED_IN	2	0.00%
HAS_GLM_RUN	2	0.00%
HAS_PARCELLATION	2	0.00%
CALIBRATES_MODIFIER	1	0.00%
CONTAINS_MODIFIER	1	0.00%
HAS_EVIDENCE	1	0.00%
HAS_SUMMARY	1	0.00%
LOCATED_IN	1	0.00%
MEASURED_BY	1	0.00%
MENTIONS_CONCEPT	1	0.00%
SIMILAR_TO	1	0.00%

B16. Full BR-KG source-like node and edge counts from JSON

Source-like property value	Node count	Node share	Edge count	Edge share
neurovault	37,467	5.40%	1,076,839	44.44%
neurosynth	464,946	66.98%	358,050	14.78%
neurosynth_v7	0	0.00%	447,498	18.47%
openneuro_glmfitlins	45,395	6.54%	266,193	10.98%
config_text_backfill	0	0.00%	104,603	4.32%
neurostore	97,238	14.01%	0	0.00%
neurostore_metadata	0	0.00%	52,117	2.15%
<missing>	6,889	0.99%	41,393	1.71%
onvoc_linker	0	0.00%	34,445	1.42%
niclip	23,865	3.44%	0	0.00%
cognitive_atlas	4,214	0.61%	14,591	0.60%
neurostore_taxonomy	0	0.00%	9,143	0.38%
openneuro_glmfitlins_inferred	0	0.00%	6,436	0.27%
scholarly_metadata	333	0.05%	3,919	0.16%
neurobagel:OpenNeuro	1,580	0.23%	1,259	0.05%
Allen Brain Atlas	1,329	0.19%	1,328	0.05%
scholarly_metadata_stub	2,557	0.37%	0	0.00%
capabilities.merged.yaml	2,073	0.30%	0	0.00%
cognitive_atlas_niclip	1,772	0.26%	0	0.00%
nilearn	816	0.12%	810	0.03%
onvoc	752	0.11%	765	0.03%
Allen CCFv3	0	0.00%	1,326	0.05%
multiverse_fitlins_runonly	60	0.01%	1,020	0.04%

Source-like property value	Node count	Node share	Edge count	Edge share
pubmed.api	1,047	0.15%	0	0.00%
neurobagel:International Neuroimaging Data-sharing Initiative	442	0.06%	361	0.01%
cognitive_atlas.cao	645	0.09%	0	0.00%
taxonomy_rule	0	0.00%	612	0.03%
Psych-101	127	0.02%	375	0.02%
task_families	314	0.05%	0	0.00%
PubMed	161	0.02%	5	0.00%
task_family_enrichment	14	0.00%	126	0.01%
neuromaps	86	0.01%	0	0.00%
openneuro_glmfitlins_manual	0	0.00%	50	0.00%
disease_path_backfill	0	0.00%	34	0.00%
psych-101_taxonomy	0	0.00%	13	0.00%
openneuro_glmfitlins_taxonomy	8	0.00%	0	0.00%
GraphQL API	0	0.00%	5	0.00%
CogAtlas	0	0.00%	5	0.00%
seed	0	0.00%	4	0.00%
taxonomy_surface_rules	4	0.00%	0	0.00%
Manual	0	0.00%	2	0.00%
psych101_curated_registry	0	0.00%	2	0.00%
bulk_loader	0	0.00%	1	0.00%
OpenNeuro	0	0.00%	1	0.00%
Test Suite	0	0.00%	1	0.00%
Test	0	0.00%	1	0.00%
Integration Test	0	0.00%	1	0.00%
Yeo2011	1	0.00%	0	0.00%

B17. Full BR-KG source-by-label rows

Node label	Source-like value	Count
Atlas	Yeo2011	1
Atlas	Allen Brain Atlas	1
Author	scholarly_metadata	172
Battery	cognitive_atlas	27
BrainAnnotation	neuromaps	86
BrainRegion	Allen Brain Atlas	1,327
BrainRegion	nilearn	796
BrainRegion	<missing>	17
Citation	cognitive_atlas	1,201
Collection	neurostore	31,885
Collection	neurovault	16,124
Concept	cognitive_atlas_niclip	919
Concept	onvoc	752
Concept	cognitive_atlas.cao	645
Concept	<missing>	20
ConceptClass	cognitive_atlas	10
Condition	openneuro_glmfitlins	310
Consortium	<missing>	38
Contrast	cognitive_atlas	1,585
Contrast	openneuro_glmfitlins	621
Coordinate	neurosynth	447,498

Node label	Source-like value	Count
Coordinate	<missing>	1
DataResource	openneuro_glmfitlins	7,581
DataResource	<missing>	1,695
DataResource	nilearn	6
Dataset	<missing>	1,695
Dataset	neurobagel:OpenNeuro	321
Dataset	neurobagel:International Neuroimaging Data-sharing Initiative	81
Dataset	openneuro_glmfitlins	38
Dataset	Psych-101	1
Embedding	niclip	23,865
ExecutionFailure	<missing>	4
Experiment	Psych-101	76
Finding	<missing>	16
GLMContrast	openneuro_glmfitlins	621
GLMDesignPrior	openneuro_glmfitlins	216
GLMRun	<missing>	1
GLMVariant	<missing>	7
IngestionRun	<missing>	12
Institution	scholarly_metadata	119
Modality	<missing>	28
ModelSpec	openneuro_glmfitlins	78
OntologyConcept	onvoc	752
OnvocClass	onvoc	752
OpenNeuro	openneuro_glmfitlins	7,619
Parcel	<missing>	26
Parcellation	<missing>	20
Parcellation	nilearn	8
Phenotype	neurobagel:OpenNeuro	938
Phenotype	neurobagel:International Neuroimaging Data-sharing Initiative	280
Process	cognitive_atlas	10
Psych101Dataset	Psych-101	1
Psych101Experiment	Psych-101	76
Publication	neurostore	31,662
Publication	neurosynth	14,220
Publication	scholarly_metadata_stub	2,557
Publication	pubmed_api	1,047
Publication	PubMed	161
Publication	openneuro_glmfitlins	49
Publication	scholarly_metadata	42
Publication	<missing>	6
Region	<missing>	8
Repository	<missing>	88
ResourceType	<missing>	102
ResultSummary	<missing>	1
ReviewCalibrationCase	<missing>	60
ReviewImplementationRule	<missing>	80
ReviewImplementationRuleCatalog	<missing>	1
ReviewLifecycleStatus	<missing>	5
ReviewPolicyDecision	<missing>	6
ReviewPositiveModifier	<missing>	1
ReviewReasonTag	<missing>	8
ReviewRule	<missing>	71
ReviewRuleGroup	<missing>	13
ReviewRuleRegistry	<missing>	1
ReviewSchemaField	<missing>	106

Node label	Source-like value	Count
ReviewSensitivityTemplate	<missing>	5
ReviewSeverity	<missing>	2
ReviewValidityLayer	<missing>	5
Run	<missing>	4
Species	<missing>	14
StatMap	neurovault	60
StatisticalMap	neurovault	21,283
StatsMap	openneuro_glmfitlins	35,180
StatsMap	multiverse_fitlins_runonly	60
Study	<missing>	33
Subject	openneuro_glmfitlins	1,139
SubjectGroup	neurobagel:OpenNeuro	321
SubjectGroup	neurobagel:International Neuroimaging Data-sharing Initiative	81
Task	neurostore	33,691
Task	cognitive_atlas_niclip	853
Task	task_families	314
Task	Psych-101	44
Task	<missing>	12
Task	openneuro_glmfitlins_taxonomy	8
Task	taxonomy_surface_rules	4
TaskAnalysis	openneuro_glmfitlins	72
TaskCondition	cognitive_atlas	807
TaskFamily	<missing>	118
TaskFamily	task_family_enrichment	14
TaskFamily	Psych-101	6
TaskIndicator	cognitive_atlas	584
TaskSpec	openneuro_glmfitlins	110
TemplateSpace	nilearn	6
TemplateSpace	<missing>	2
TemplateSpace	openneuro_glmfitlins	1
TemplateSpace	Allen Brain Atlas	1
Term	neurosynth	3,228
Tool	capabilities.merged.yaml	2,073
Tool	<missing>	11
ToolEvidence	<missing>	1
ToolFamily	<missing>	96
ToolVersion	<missing>	4,142

B18. Full BR-KG ingestion source

Configured sources with live values

Configured source	Mode	Live source value(s)	Live count summary	Release status
cognitive_atlas	full	cognitive_atlas, cognitive_atlas_niclip, cognitive_atlas_cao	6,631 nodes / 14,591 edges	present; aliases need release normalization
nilearn_atlases	full	nilearn	816 nodes / 810 edges	present under source alias
neurobagel	full	neurobagel:OpenNeuro, neurobagel:International Neuroimaging Data-sharing Initiative	2,022 nodes / 1,620 edges	present under source-qualified aliases

Configured source	Mode	Live source value(s)	Live count summary	Release status
onvoc	full	onvoc, onvoc_linker	752 nodes / 35,210 edges	present; node source and linker source should be separated in release notes
openneuro_glmfitlins	full	openneuro_glmfitlins, openneuro_glmfitlins_inferred, openneuro_glmfitlins_manual, openneuro_glmfitlins_taxonomy	45,403 nodes / 272,679 edges	present; split measured, inferred, manual, and taxonomy lanes
pubmed	spine	PubMed, pubmed_api	1,208 nodes / 5 edges	present under two source strings
neurosynth	spine	neurosynth, neurosynth_v7	464,946 nodes / 805,548 edges	present; source alias should record Neurosynth snapshot/version
neurovault	spine	neurovault	37,467 nodes / 1,076,839 edges	present
openneuro	spine	OpenNeuro, neurobagel:OpenNeuro, openneuro_glmfitlins*	46,975+ nodes / 273,939+ edges across related values	present through several lanes; release table should avoid double counting multi-source surfaces
niclip	spine	niclip, cognitive_atlas_niclip	25,637 nodes / 0 source-valued edges	present; distinguish embedding product from Cognitive Atlas enrichment
neuromaps	spine	neuromaps	86 nodes / 0 edges	present
neurostore	spine	neurostore, neurostore_metadata, neurostore_taxonomy	97,238 nodes / 61,260 edges	present across metadata and taxonomy lanes
scholarly_metadata	on_demand	scholarly_metadata, scholarly_metadata_stub	2,890 nodes / 3,919 edges	present despite being configured on-demand; release status needs clarification

Unattributed source values

Source-like value	Node count	Edge count	Risk	Required audit
<missing>	6,889	41,393	largest provenance gap; release users cannot trace these objects to an upstream source or build lane	group by node label, relationship type, and creation/loading fields; backfill source / edge_source where possible

Internal, generated, manual, and test lanes

Source-like value	Node count	Edge count	Proposed release class	Release treatment
config_text_backfill	0	104,603	internal generated enrichment	document generation rule and upstream fields; do not list as external source
capabilities.merged.yaml	2,073	0	internal tool/capability catalog	document as BR capability registry, not neuroimaging upstream data
taxonomy_rule	0	612	internal generated taxonomy rule	document rule source and version
taxonomy_surface_rules	4	0	internal generated taxonomy surface	document rule source and version

Source-like value	Node count	Edge count	Proposed release class	Release treatment
task_families	314	0	internal/generated task-family catalog	document source file and curation status
task_family_enrichment	14	126	internal generated enrichment	document enrichment script and confidence semantics
disease_path_backfill	0	34	internal generated backfill	document backfill rule and affected relationship types
seed	0	4	seed/manual bootstrap	document seed artifact or remove from public dump if test-only
bulk_loader	0	1	loader artifact	classify as operational provenance or remove if accidental
GraphQL API	0	5	runtime/API-created records	audit whether these are production user actions, tests, or seed data
Manual	0	2	manual curation	document curator workflow and source evidence
openneuro_glmfitlins_manual	0	50	manual OpenNeuro/GLMFitLins lane	document manual decision criteria
psych101_curated_registry	0	2	curated registry	document registry artifact and curator status
Integration Test	0	1	test artifact	exclude from release or mark internal-test-only
Test	0	1	test artifact	exclude from release or mark internal-test-only
Test Suite	0	1	test artifact	exclude from release or mark internal-test-only

Release field coverage gaps

Release field	Current state in this document	Required release action
Snapshot date	not stored consistently in live source fields	record source-specific snapshot date or API pull date for every external source
Upstream URL	not measured from live graph	add canonical upstream URL per source in Section 2.1 and data card
License	not measured from live graph	fill source license and redistribution/commercial-use constraints from upstream license pages
Required citation	not measured from live graph	add citation strings or citation keys and propagate them into release notes
Loader path	partially inferable from repo loader names, not release-ready	map each source to exact loader/module/script path and command
Data artifact path	partially present for Gabriel/KGGen and benchmark artifacts, not complete	map each source to raw/staged/build artifact paths or mark on-demand-only
Source class	mixed in live graph	classify each source-like value as external upstream, ontology, internal generated, manual, runtime, or test

Gabriel and KGGen status

Lane	Repo artifacts found	Live source count	Current interpretation	Required follow-up
Gabriel manifests	32 data/BR-KG/raw/gabriel/runs/*/manifest.json files	0 source-marked nodes/edges	Gabriel exists as raw/run artifacts, but is not visible as a marked live KG source in this scan	decide whether Gabriel was not ingested, was candidate-only, or was ingested without a source=gabriel marker

Lane	Repo artifacts found	Live source count	Current interpretation	Required follow-up
Gabriel review queues	12 data/BR-KG/raw/gabriel/runs/*/review_queue*.jsonl files	0 source-marked nodes/edges	review/candidate artifacts exist outside live source counts	document accepted vs candidate-only vs rejected lanes before release
KGGen summaries	75 data/BR-KG/raw/kggen/**/*summary.json files	0 source-marked nodes/edges	KGGen exists as comparison/generation artifacts, not as a marked live KG source	keep KGGen as research/comparison lane unless accepted records are explicitly ingested and marked
KGGen JSONL outputs	81 data/BR-KG/raw/kggen/**/*jsonl files	0 source-marked nodes/edges	generated candidate files exist	document model/prompt/validation before any public KG claim

B19. Source/lane summary and release disposition

Audit source or lane	Live source value(s)	Class	Config/source state	Nodes	Edges	Release disposition
Neurosynth	neurosynth; neurosynth_v7	external upstream meta-analysis corpus	configured spine; live	464,946	805,548	releasable only after source-specific snapshot, license, and citation are pinned
NeuroVault	neurovault	external upstream image/stat-map repository	configured spine; live	37,467	1,076,839	releasable only after collection/image license matrix and pull date are pinned
Neurostore	neurostore; neurostore_metadata; neurostore_taxonomy	external upstream study/analysis meta-data plus taxonomy lane	configured spine; live	97,238	61,260	releasable after Neurostore snapshot/version, license, and citation are pinned
Cognitive Atlas	cognitive_atlas; cognitive_atlas_cao; CogAtlas	external ontology/task/concept catalog	configured full; live	4,859	14,596	releasable after snapshot and citation pin; normalize CogAtlas alias
NiCLIP / Cognitive Atlas enrichment	niclip; cognitive_atlas_niclip	derived embedding/enrichment lane	configured spine for niclip; live	25,637	0	publish as derived lane only after model, asset, and upstream-source citations are pinned
ONVOC	onvoc; onvoc_linker	local/open ontology and linker lane	configured full; live	752	35,210	releasable after ONVOC version, ontology license, and linker-rule provenance are pinned
OpenNeuro GLMFitLins / BIDS-derived maps	openneuro_glmfitlins; openneuro_glmfitlins_inferred; openneuro_glmfitlins_manual; openneuro_glmfitlins_taxonomy; OpenNeuro	external datasets plus derived GLM/stat-map lane	configured full for openneuro_glmfitlins, spine for openneuro; live	45,403	272,680	releasable only with per-dataset license matrix, statsmodel copy date, manifest hash, and derivative citation policy

Audit source or lane	Live source value(s)	Class	Config/source state	Nodes	Edges	Release disposition
BIDS literal source	none for literal bids	schema/metadata standard, represented indirectly	configured spine; absent as live source value	0	0	document as standard/format dependency, not a separate persisted upstream source unless source markers are added
Multiverse FitLins run-only	multiverse_fitlins_runonly	internal derived analysis-run lane	live, not configured as upstream source	60	1,020	classify as derived internal lane; include only with run manifests and upstream licenses
Neurobagel	neurobagel; OpenNeuro; neurobagel: International Neuroimaging Data-sharing Initiative	external federated dataset metadata	configured full; live	2,022	1,620	releasable after node-level snapshot dates and source-node attribution/license matrix are pinned
PubMed	PubMed; pubmed_api	external bibliographic metadata	configured spine as pubmed; live	1,208	5	releasable after API pull date, NCBI attribution, and citation policy are pinned
PubMed Central / PMC	none	external full-text repository, currently inventory-only	listed as source target; not found in live source scan	0	0	mark absent unless PMC full-text artifacts are intentionally ingested and licensed
Scholarly metadata / Crossref / OpenAlex	scholarly_metadata; scholarly_metadata_stub	external bibliographic metadata and stub lane	configured on-demand; live persisted values exist	2,890	3,919	keep as bibliographic metadata lane; release after per-provider attribution and cache snapshot policy are pinned
Allen Human Brain Atlas / Allen CCF	Allen Brain Atlas; Allen CCFv3	external atlas/expression resource plus atlas alias	configured spine as allen_hba; live under aliases	1,329	2,654	normalize alias to allen_hba; audit whether records are real upstream, CCF atlas, or synthetic/sample before release
Nilearn atlases / Yeo2011	nilearn; Yeo2011	external atlas definitions bundled/fetched through Nilearn	configured full as nilearn_atlases; live under aliases	817	810	releasable after per-atlas license/citation/version matrix is pinned
Neuromaps	neuromaps	external annotation/atlas map library	configured spine; live	86	0	releasable after annotation-level license/citation table is pinned
NeuroQuery	none for literal neuroquery	external/on-demand evidence source	configured on-demand; absent as persisted live source	0	0	document as on-demand adapter unless cached outputs are released

Audit source or lane	Live source value(s)	Class	Config/source state	Nodes	Edges	Release disposition
NiMARE	none for literal nimare	software-derived meta-analysis evidence lane	configured on-demand; absent as persisted live source	0	0	document as software/derived evidence dependency; release only generated priors with inputs and method version
NeuroScout	none for literal neuroscout	external/on-demand feature source	configured on-demand; absent as persisted live source	0	0	document as on-demand adapter unless feature summaries are released
NIDM results	none for literal nidm_results	external/standardized results metadata	configured on-demand; absent as persisted live source	0	0	mark adapter/source configured but not persisted; release only with NIDM artifact manifest
BrainMap	none for literal brainmap	external/licensed coordinate database/tooling lane	configured spine; absent as persisted live source	0	0	do not release as KG source unless licensed data and citation obligations are confirmed
Wikidata	none for literal wikidata	external linked-data ontology/entity source	configured spine; absent as persisted live source	0	0	mark configured but absent unless source markers or artifacts are added
Virtual Brain	none for literal virtual_brain	external software/simulation lane	configured spine; absent as persisted live source	0	0	document as configured/adapter lane, not released KG content unless simulation reports are included
GWAS Catalog / OpenMed / PGC	none in 2026-05-03 source-like scan under OpenMed/PGC/GWAS source names	external genetics metadata and top-loci lane	documented ingest, but not visible in current source scan	0 in current source scan	0 in current source scan	reconcile current live graph with 2026-04-08 validation counts before release claims
Psych-101 / psychology task registry	Psych-101; psych-101_taxonomy; psych101_curated_registry	external/curated behavioral task registry and taxonomy lane	live, not in ingestion_modes.yaml source list	127	390	release as curated registry only after source snapshot, curator status, and license are pinned
Tool/capability catalog	capabilities.merged.yaml	internal BR capability registry	live internal source lane	2,073	0	include only if public KG intentionally includes BR runtime/tool nodes; otherwise split from BR-KG public dump
Config text backfill	config_text_backfill	internal generated enrichment	live internal source lane	0	104,603	keep as internal generated lane; release only with rule definition and affected edge-type audit

Audit source or lane	Live source value(s)	Class	Config/source state	Nodes	Edges	Release disposition
Taxonomy rules	taxonomy_rule; taxonomy_surface_rules	internal generated taxonomy/rule lane	live internal source lane	4	612	release with rule/version manifest or exclude from upstream-source table
Task family enrichment	task_families; task_family_enrichment	internal/generated task-family catalog	live internal source lane	328	126	release with task-family registry version and curator/source evidence
Disease path backfill	disease_path_backfill	internal generated backfill	live internal source lane	0	34	audit before release; document rule or remove accidental backfill edges
Seed/bootstrap lane	seed	seed/manual bootstrap	live internal source lane	0	4	audit and either attribute seed rows or exclude from public dump
Bulk loader operational lane	bulk_loader	operational loader marker	live internal source lane	0	1	audit whether this should be provenance metadata or excluded
GraphQL API runtime lane	GraphQL API	runtime/API-created records	live runtime lane	0	5	audit as possible production mutation/test residue before release
Manual lane	Manual	manual curation lane	live manual lane	0	2	release only after curator workflow and evidence provenance are documented
Test lanes	Integration Test; Test; Test Suite	test artifacts	live test lane	0	3	exclude from public dump or mark internal-test-only
Unattributed source	<missing>	unresolved provenance gap	live source-like missing value	6,889	41,393	release blocker: backfill, exclude, or explicitly document each bucket
Gabriel paper-mining lane	no live gabriel source marker	LLM/heuristic candidate extraction lane	repo artifacts exist; absent as live source marker	0	0	candidate-only unless accepted records are explicitly ingested with source markers and validation evidence
KGGen comparison/generation lane	no live kggen source marker	LLM/generated comparison/candidate lane	repo artifacts exist; absent as live source marker	0	0	keep as comparison/candidate lane unless promoted records are validated and source-marked

Scientific review / governance registry	no live source string identified in current scan	internal governance/rule registry	repo source exists; not detected in source-like scan	0	0	document separately from BR-KG upstream source list unless records are included in public dump
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B20. Evidence, artifact, loader, and license details

Audit source or lane	Snapshot/version evidence	URL/artifact evidence	Loader/builder evidence	License/citation status
Neurosynth	live graph snapshot counted on 2026-05-03; local files encode version-7; source-specific pull date not stored in live graph	upstream homepage https://neurosynth.org ; local artifacts include data/neurosynth_nimare/neurosynth_data-neurosynth_version-7.* and data/neurosynth_maps	src/brain_researcher/core/ingestion/loaders/neurosynth_unified.py; src/brain_researcher/services/BR-KG/etl/loaders/neurosynth_loader.py; scripts/tools/ingest/download_neurosynth_dataset.py; scripts/tools/ingest/neurosynth_spine.py	license and required citation not release-verified in repo; citation must include Neurosynth/NiMARE source used for v7 assets
NeuroVault	live graph snapshot counted on 2026-05-03; source-specific API pull date not stored in live graph	upstream homepage https://neurovault.org ; local scripts reference collection downloads; no complete raw artifact path measured in current pass	src/brain_researcher/core/ingestion/loaders/neurovault_unified.py; src/brain_researcher/services/BR-KG/etl/loaders/neurovault_loader.py; scripts/tools/etl/neurovault_fetch_inventory.py; scripts/tools/etl/neurovault_ingest_filtered.py	license and required citation not release-verified in repo; per-image licenses may vary
Neurostore	live graph snapshot counted on 2026-05-03; source-specific snapshot not stored in live graph	upstream homepage https://neurostore.org ; no complete raw artifact path measured in current pass	src/brain_researcher/core/ingestion/loaders/neurostore_unified.py; src/brain_researcher/services/BR-KG/evidence/connectors/neurostore.py; scripts/ingest_neurostore_tasks.py	license and citation not release-verified in repo
Cognitive Atlas	live graph snapshot counted on 2026-05-03; source-specific pull date not stored in live graph	upstream homepage https://www.cognitiveatlas.org ; local artifacts include data/cognitive_atlas when present	src/brain_researcher/core/ingestion/loaders/cognitive_atlas_unified.py; src/brain_researcher/services/BR-KG/etl/loaders/cognitive_atlas_loader.py; src/brain_researcher/services/BR-KG/etl/loaders/cogatlas_loader.py; scripts/tools/etl/ingest_cogatlas.py; scripts/tools/once/parse_cogat_owl.py	license and citation not release-verified in repo
NiCLIP / Cognitive Atlas enrichment	live graph snapshot counted on 2026-05-03; embedding asset version not stored in live graph	local docs docs/NiCLIP_CONFIGURATION.md; docs/services/BR-KG/niclip_integration_features.md; local vector caches under data/BR-KG/vector_cache/niclip and data/atlas/niclip	src/brain_researcher/services/BR-KG/etl/loaders/niclip_loader.py; src/brain_researcher/services/BR-KG/etl/integrate_niclip_mappings.py; scripts/BR-KG/materialize_niclip_similarity.py	release license/citation depends on NiCLIP model/assets and underlying Cognitive Atlas/Neurosynth sources; not release-verified in repo

Audit source or lane	Snapshot/version evidence	URL/artifact evidence	Loader/builder evidence	License/citation status
ONVOC	live graph snapshot counted on 2026-05-03; ontology build/version not stored in live graph	repo artifacts include docs/vocab/ONVOC.md, configs/onvoc.tree.yaml, docs/proposals/onvoc_*	src/brain_researcher/core/ingestion/loaders/onvoc_unified.py; scripts/tools/once/parse_onvoc.owl.py; scripts/tools/ontologies/build_onvoc_tree.py; scripts/tools/etl/backfill_onvoc.links.py; scripts/link_tasks_onvoc.py; scripts/link_terms_onvoc.py	local ontology/ license and citation propagation not release-verified
OpenNeuro GLM-FitLins / BIDS-derived maps	live graph snapshot counted on 2026-05-03; data/openneuro_glmfitlins/README.md says statsmodel specs copied on 2025-10-27; map mount paths external to repo	upstream homepage https://openneuro.org ; local artifacts include data/openneuro_glmfitlins/statsmodel_specs, data/openneuro_glmfitlins/summaries/yeo17_summary.csv, and manifest paths described in docs/runbooks/openneuro_glmfitlins_ingest.md	src/brain_researcher/core/ingestion/loaders/openneuro_glm_loader.py; src/brain_researcher/services/BR-KG/etl/glmfitlins_ingest/load_to_BR-KG.py; scripts/run_openneuro_glmfitlins_ingest_neo4j.sh; scripts/tools/once/build_openneuro_glm_manifest.py; scripts/BR-KG/load_openneuro_fitlins.py	OpenNeuro dataset licenses are dataset-specific; GLMFitLins derivative license/ citation not release-verified in repo
BIDS literal source	configured source absent in 2026-05-03 live scan	upstream homepage https://bids.neuroimaging.io ; BIDS-derived metadata appears through OpenNeuro/GLMFitLins lanes	src/brain_researcher/core/ingestion/loaders/bids_unified.py; src/brain_researcher/services/BR-KG/extractors/bids_events.py; src/brain_researcher/services/tools/bids_tools.py	BIDS standard citation/ license and dataset-specific licenses not release-verified in repo
Multiverse FitLins run-only	live graph snapshot counted on 2026-05-03; run-specific snapshot not stored in source field	repo docs docs/multiverse_glmfitlins.md; artifact paths not fully measured in current pass	src/brain_researcher/services/BR-KG/etl/glm_runs/ingest_glm_run.py; scripts/ingest_openneuro_glmfitlins_yeo17.py	inherits OpenNeuro/GLMFitLins and local analysis provenance; not independently release-verified
Neurobagel	live graph snapshot counted on 2026-05-03; data/BR-KG/raw/neurobagel_public/summary.json records node API URLs and dataset counts	upstream homepage https://neurobagel.org ; local artifacts include data/BR-KG/raw/neurobagel_public/*/datasets.json and meta-data files	src/brain_researcher/services/BR-KG/etl/loaders/neurobagel_loader.py; src/brain_researcher/services/BR-KG/etl/loaders/neurobagel_public_loader.py; scripts/ingest_neurobagel_datasets.py	Neurobagel and source-node licenses/ citations not release-verified in repo
PubMed	live graph snapshot counted on 2026-05-03; source-specific API pull date not stored in live graph	upstream homepage https://pubmed.ncbi.nlm.nih.gov ; local marker data/knowledge/db/pubmed_last_run.txt exists	src/brain_researcher/core/ingestion/loaders/pubmed_unified.py; src/brain_researcher/services/BR-KG/etl/loaders/pubmed_loader.py; src/brain_researcher/services/BR-KG/evidence/connectors/pubmed.py	NCBI/PubMed citation and API policy not release-verified in repo
PubMed Central / PMC	no persisted live source value in 2026-05-03 scan	upstream homepage https://pmc.ncbi.nlm.nih.gov ; no repo artifact measured in current pass	no dedicated PMC loader identified in current pass	license varies by article; not release-verified
Scholarly metadata / Crossref / OpenAlex	live graph snapshot counted on 2026-05-03; source-specific API pull dates not stored in live graph	local artifacts include many data/BR-KG/raw/scholarly_metadata/openalex_*.json and crossref_*.json files	src/brain_researcher/services/BR-KG/etl/loaders/scholarly_metadata_loader.py	Crossref/OpenAlex terms, citation, and redistribution policy not release-verified in repo

Audit source or lane	Snapshot/version evidence	URL/artifact evidence	Loader/builder evidence	License/citation status
Allen Human Brain Atlas / Allen CCF	live graph snapshot counted on 2026-05-03; local data/BR-KG/raw/allen_hba/manifest.json contains synthetic sample marker rather than confirmed full HBA snapshot	upstream homepage https://human.brain-map.org/ ; local artifacts include data/BR-KG/raw/allen_hba, data/allen_hba, and data/BR-KG/raw/evidence/allen_hba_sample.json	src/brain_researcher/core/ingestion/loaders/allen_hba_loader.py; src/brain_researcher/core/ingestion/loaders/allen_brain_unified.py; src/brain_researcher/services/BR-KG/etl/adapters/allen_hba_adapter.py	Allen license/citation not release-verified; synthetic/sample marker must be clarified before claims
Nilearn atlases / Yeo2011	live graph snapshot counted on 2026-05-03; local README says folder is Nilearn download cache; per-atlas versions not stored in live source field	upstream docs https://nilearn.github.io/ ; local artifacts include data/BR-KG/raw/nilearn_atlases, data/atlas/nilearn, and data/BR-KG/raw/Yeo_JNeurophysiol11_FreeSurfer.zip	src/brain_researcher/core/ingestion/loaders/nilearn_atlas_unified.py; scripts/BR-KG/load_neuromaps_parcellations.py for related parcellation loading	Nilearn software license and per-atlas licenses/citations not release-verified in repo
Neuromaps	live graph snapshot counted on 2026-05-03; source-specific fetch date not stored in live graph	upstream docs https://netneurolab.github.io/neuromaps/ ; local artifacts include data/BR-KG/raw/neuromaps and data/atlas/neuromaps	src/brain_researcher/core/ingestion/loaders/neuromaps_unified.py; scripts/BR-KG/fetch_all_neuromaps.py; scripts/BR-KG/load_neuromaps_parcellations.py	neuromaps and annotation-specific licenses/citations not release-verified in repo
NeuroQuery	no persisted live source value in 2026-05-03 scan	upstream homepage https://neuroquery.org/ ; sample path configured as data/BR-KG/raw/evidence/neuroquery_sample.json	src/brain_researcher/services/BR-KG/etl/adapters/neuroquery_adapter.py; adapter configured in configs/BR-KG/ingestion_modes.yaml	license/citation not release-verified in repo
NiMARE	no persisted live source value in 2026-05-03 scan; local Neurosynth assets use NiMARE fetch wrappers	upstream docs https://nimare.readthedocs.io/ ; sample path configured as data/BR-KG/raw/evidence/nimare_sample.json; local artifacts include data/neurosynth_nimare	src/brain_researcher/services/BR-KG/etl/adapters/nimare_adapter.py; scripts/data_processing/train_nimare_lda.py; scripts/BR-KG/install_nimare_templates.py	NiMARE software citation/license and upstream dataset licenses not release-verified
NeuroScout	no persisted live source value in 2026-05-03 scan	upstream homepage https://neuroscout.org/ ; sample path configured as data/BR-KG/raw/evidence/neuroscout_features.json	src/brain_researcher/services/BR-KG/etl/adapters/neuroscout_adapter.py; adapter configured in configs/BR-KG/ingestion_modes.yaml	license/citation not release-verified in repo
NIDM results	no persisted live source value in 2026-05-03 scan	upstream project URL not release-verified; local artifacts include data/BR-KG/raw/nidm	src/brain_researcher/services/BR-KG/etl/loaders/nidm_results_loader.py	license/citation not release-verified in repo
BrainMap	no persisted live source value in 2026-05-03 scan	upstream homepage https://brainmap.org/ ; local directories/scripts include data/brainmap and scripts/brainmap/*	src/brain_researcher/core/ingestion/loaders/brainmap_unified.py; src/brain_researcher/core/ingestion/parsers/brainmap_parser.py; src/brain_researcher/services/tools/meta_brainmap_tool.py	BrainMap license/redistribution terms not release-verified and likely require explicit review

Audit source or lane	Snapshot/version evidence	URL/artifact evidence	Loader/builder evidence	License/citation status
Wikidata	no persisted live source value in 2026-05-03 scan	upstream home-page https://www.wikidata.org ; no raw artifact path measured in current pass	src/brain_researcher/core/ingestion/loaders/wikidata_unified.py; src/brain_researcher/services/BR-KG/etl/loaders/wikidata_loader.py; scripts/BR-KG/load_wikidata_brain_regions.py	Wikidata license/citation not release-verified in repo
Virtual Brain	no persisted live source value in 2026-05-03 scan	upstream home-page https://www.thevirtualbrain.org ; local service paths under src/brain_researcher/services/virtual_brain	src/brain_researcher/core/ingestion/loaders/virtual_brain_loader.py; src/brain_researcher/services/BR-KG/etl/adapters/virtual_brain_adapter.py; scripts/virtual_brain/create_simulation_links.py	software/data license and citation not release-verified in repo
GWAS Catalog / OpenMed / PGC	docs/BR-KG/openmed_pgc.gwas_ingest.md reports validation counts as of 2026-04-08, but those source strings are absent from the current source scan	upstream sources are Hugging Face OpenMed/pgc-* repos and GWAS Catalog REST API; artifact paths not measured in current pass	src/brain_researcher/services/BR-KG/etl/loaders/openmed_pgc_hf_loader.py; src/brain_researcher/services/BR-KG/etl/loaders/gwas_catalog_top_loci_loader.py; scripts/BR-KG/run_openmed_pgc_live_ingest.py	OpenMed/HF and GWAS Catalog license/citation not release-verified in repo
Psych-101 / psychology task registry	live graph snapshot counted on 2026-05-03; source-specific snapshot not stored in live source field	local docs/runbooks include docs/runbooks/workflow_psych101_*; artifacts not fully measured in current pass	src/brain_researcher/services/BR-KG/etl/loaders/psych101_loader.py; src/brain_researcher/services/BR-KG/etl/loaders/psych101_hf_loader.py; scripts/BR-KG/audit_psych101_task_fmri_bridge.py	upstream and curated-registry license/citation not release-verified in repo
Tool/capability catalog	live graph snapshot counted on 2026-05-03; config version not embedded in source string	repo config configs/catalog/capabilities.generated.yaml; source config configs/BR-KG/sources/tools_catalog.yaml	src/brain_researcher/services/BR-KG/loader/tools_catalog_loader.py	internal repo artifact, not external data license; citation not applicable except BR software/data card
Config text backfill	live graph snapshot counted on 2026-05-03; rule version not embedded in source string	artifact path not measured in current pass	loader/script path not mapped in current pass	inherits upstream/source-node licenses; generated rule citation not applicable
Taxonomy rules	live graph snapshot counted on 2026-05-03; rule version not embedded in source string	local configs include configs/taxonomy/* and configs/BR-KG/scientific_review_rule_registry.yaml	src/brain_researcher/services/BR-KG/etl/linkers/taxonomy_linker.py; src/brain_researcher/services/BR-KG/etl/loaders/scientific_review_rule_registry_loader.py	internal/generated; cite local method/config version, not external upstream
Task family enrichment	live graph snapshot counted on 2026-05-03; family registry version not embedded in source string	local configs include configs/taxonomy/families and related crosswalks	src/brain_researcher/services/BR-KG/task_family_matcher.py; taxonomy/linker scripts under scripts/analysis and scripts/tools/ontologies	internal/generated; cite local curation/method version
Disease path backfill	live graph snapshot counted on 2026-05-03; backfill version not embedded in source string	artifact path not measured in current pass	script path not mapped in current pass	inherits upstream disease/source licenses; no separate external citation

Audit source or lane	Snapshot/version evidence	URL/artifact evidence	Loader/builder evidence	License/citation status
Seed/bootstrap lane	live graph snapshot counted on 2026-05-03; seed artifact not mapped	artifact path not measured in current pass	builder path not mapped in current pass	internal/manual; no upstream license without seed evidence
Bulk loader operational lane	live graph snapshot counted on 2026-05-03	artifact path not measured in current pass	src/brain_researcher/services/BR-KG/bulk_loader.py	internal operational marker
GraphQL API runtime lane	live graph snapshot counted on 2026-05-03	artifact path not measured in current pass	src/brain_researcher/services/BR-KG/api/graphql_enhanced.py; src/brain_researcher/services/BR-KG/api/graphql_optimized.py	runtime/manual; upstream evidence unknown
Manual lane	live graph snapshot counted on 2026-05-03	manual artifact path not measured in current pass	curation API exists under src/brain_researcher/services/BR-KG/api/curation.py and src/brain_researcher/services/BR-KG/curation/curation_api.py	manual curation must cite underlying evidence, not just curator action
Test lanes	live graph snapshot counted on 2026-05-03	artifact path not measured in current pass	test source not mapped in current pass	not release data
Unattributed source	live graph snapshot counted on 2026-05-03	not applicable until attributed	audit by label/type and loader-created fields	not releasable as-is
Gabriel paper-mining lane	32 run manifests and 12 review queues found under data/BR-KG/raw/gabriel/runs	local docs docs/BR-KG/gabriel.full_pipeline.md; raw artifacts under data/BR-KG/raw/gabriel	src/brain_researcher/services/BR-KG/etl/gabriel_generator.py; src/brain_researcher/services/BR-KG/etl/loaders/gabriel_loader.py; scripts/deep_research_to_gabriel_manifest.py	upstream paper licenses, model/prompt versions, and validation/citation status not release-verified
KGGen comparison/generation lane	75 *.summary.json files and 81 *.jsonl files found under data/BR-KG/raw/kggen	raw artifacts under data/BR-KG/raw/kggen; linked from docs/BR-KG/gabriel.full_pipeline.md	src/brain_researcher/services/BR-KG/etl/evaluation/gabriel_kggen_eval.py; scripts/kggen_generate_from_manifest.py; scripts/BR-KG/build_kggen_task_mapping_salvage_pack.py	model/prompt/license/citation and validation status not release-verified
Scientific review / governance registry	current source-like scan did not find a registry source value	local config configs/BR-KG/scientific_review_rule_registry.yaml	src/brain_researcher/services/BR-KG/etl/loaders/scientific_review_rule_registry_loader.py	internal policy/config, not external upstream data

B21. Full canonical schema-triple counts

Rank	Source label set	Relationship	Target label set	Edge count	Share of edges
1	['StatisticalMap']	BELONGS_TO	['Collection']	1,075,385	44.38%
2	['Publication']	HAS_COORDINATE	['Coordinate']	447,499	18.47%
3	['Publication']	HAS_TERM	['Term']	358,050	14.78%
4	['StatsMap']	IN_REGION	['BrainRegion']	121,261	5.00%
5	['Publication']	ABOUT	['Concept', 'OntologyConcept', 'OnvocalClass']	67,785	2.80%
6	['Task']	IN_DOMAIN	['Process', 'ConceptClass']	52,117	2.15%
7	['StatsMap']	IN_ONVOC	['Concept', 'OntologyConcept', 'OnvocalClass']	43,958	1.81%
8	['StatsMap']	IN_SPACE	['TemplateSpace']	35,180	1.45%
9	['StatsMap']	COMPUTED_WITH	['ModelSpec']	30,880	1.27%

Rank	Source label set	Relationship	Target label set	Edge count	Share of edges
10	['StatsMap']	GENERATED_FROM	['TaskAnalysis']	30,880	1.27%
11	['StatsMap']	DERIVED_FROM	['Contrast', 'GLMContrast']	30,120	1.24%
12	['StatsMap']	HAS_RESOURCE	['DataResource', 'OpenNeuro']	14,146	0.58%
13	['Publication']	HAS_TEXT_EMBEDDING	['Embedding']	12,958	0.53%
14	['Task']	MAPS_TO	['Task']	11,062	0.46%
15	['Task']	IN_ONVOC	['Concept', 'OntologyConcept', 'OnvocClass']	10,894	0.45%
16	['StatisticalMap']	IN_ONVOC	['Concept', 'OntologyConcept', 'OnvocClass']	7,331	0.30%
17	['StatsMap']	MEASURES	['Concept']	4,502	0.19%
18	['Tool']	HAS_VERSION	['ToolVersion']	4,142	0.17%
19	['DataResource', 'OpenNeuro']	ABOUT	['Concept', 'OntologyConcept', 'OnvocClass']	3,689	0.15%
20	['Task']	MEASURES	['Task']	3,560	0.15%
21	['Publication']	CITES	['Publication']	3,395	0.14%
22	['Concept']	MAPS_TO	['Concept']	3,145	0.13%
23	['Dataset', 'DataResource']	ABOUT	['Concept', 'OntologyConcept', 'OnvocClass']	2,807	0.12%
24	['Task']	MEASURES	['Concept']	2,773	0.11%
25	['Tool']	SUPPORTS_MODALITY	['Modality']	2,655	0.11%
26	['Tool']	IMPLEMENTS_FAMILY	['TaskFamily']	2,167	0.09%
27	['Tool']	IMPLEMENTS_FAMILY	['ToolFamily']	2,127	0.09%
28	['Subject']	PARTICIPATES_IN	['TaskAnalysis']	1,982	0.08%
29	['StatsMap']	SUGGESTS_MEASURES	['Concept']	1,902	0.08%
30	['Dataset', 'DataResource']	HAS_MODALITY	['Modality']	1,839	0.08%
31	['Dataset', 'DataResource']	HOSTED_AT	['Repository']	1,694	0.07%
32	['Task']	HAS_CONTRAST	['Contrast']	1,585	0.07%
33	['Dataset', 'DataResource']	HAS_TASK	['Task']	1,569	0.06%
34	['StatisticalMap']	BELONGS_TO	['StatisticalMap']	1,454	0.06%
35	['Tool']	ABOUT	['Concept', 'OntologyConcept', 'OnvocClass']	1,337	0.06%
36	['Atlas']	HAS_REGION	['BrainRegion']	1,327	0.05%
37	['BrainRegion']	PART_OF	['BrainRegion']	1,326	0.05%
38	['Concept']	MAPS_TO	['Task']	1,312	0.05%
39	['Task']	MAPS_TO	['Concept']	1,289	0.05%
40	['Task']	ASSERTS	['Concept']	1,237	0.05%
41	['SubjectGroup']	HAS_PHENOTYPE	['Phenotype']	1,218	0.05%
42	['Concept']	MEASUREDBY	['Task']	1,218	0.05%
43	['Task']	HASCITATION	['Citation']	1,091	0.05%
44	['Dataset', 'DataResource']	INVOLVES_SPECIES	['Species']	990	0.04%
45	['Task']	HASCONDITION	['TaskCondition']	807	0.03%
46	['Parcellation']	HAS_REGION	['BrainRegion']	796	0.03%
47	['Concept', 'OntologyConcept', 'OnvocClass']	CLASSIFIED_UNDER	['Concept', 'OntologyConcept', 'OnvocClass']	765	0.03%
48	['Contrast', 'GLMContrast']	USES_CONDITION	['Condition']	750	0.03%
49	['Contrast', 'GLMContrast']	MAPS_TO	['Contrast', 'GLMContrast']	621	0.03%
50	['Task']	SUGGESTS_MEASURES	['Task']	612	0.03%

Rank	Source label set	Relationship	Target label set	Edge count	Share of edges
51	['Task']	HASINDICATOR	['TaskIndicator']	584	0.02%
52	['Concept']	CLASSIFIEDUNDER	['Process', 'ConceptClass']	523	0.02%
53	['Contrast', 'GLM-Contrast']	BELONGS_TO	['Publication']	517	0.02%
54	['Contrast']	IN_ONVOC	['Concept', 'OntologyConcept', 'OnvocClass']	468	0.02%
55	['TaskAnalysis']	HAS_RESOURCE	['DataResource', 'OpenNeuro']	432	0.02%
56	['Concept']	KINDOF	['Concept']	429	0.02%
57	['Dataset']	INCLUDES	['SubjectGroup']	402	0.02%
58	['ModelSpec']	HAS_CONTRAST	['Contrast', 'GLMContrast']	393	0.02%
59	['Dataset', 'DataResource']	IN_ONVOC	['Concept', 'OntologyConcept', 'OnvocClass']	367	0.02%
60	['ToolVersion']	PRODUCES_RESOURCE	['ResourceType']	362	0.01%
61	['ToolVersion']	CONSUMES_RESOURCE	['ResourceType']	338	0.01%
62	['TaskAnalysis']	HAS_CONDITION	['Condition']	310	0.01%
63	['Author']	AFFILIATED_WITH	['Institution']	292	0.01%
64	['Concept']	PARTOF	['Concept']	263	0.01%
65	['Dataset']	ABOUT	['Concept', 'OntologyConcept', 'OnvocClass']	254	0.01%
66	['Dataset', 'OpenNeuro']	HAS_CONTRAST	['Contrast', 'GLMContrast']	243	0.01%
67	['TaskSpec']	HAS_CONTRAST	['Contrast', 'GLMContrast']	243	0.01%
68	['Publication']	AUTHORED_BY	['Author']	232	0.01%
69	['Contrast', 'GLM-Contrast']	BELONGS_TO	['Study']	217	0.01%
70	['Concept']	HASCITATION	['Citation']	202	0.01%
71	['Task']	RELATED_TO	['Concept']	185	0.01%
72	['Task']	BELONGS_TO_FAMILY	['TaskFamily']	176	0.01%
73	['ReviewRule']	REQUIRES_FIELD	['ReviewSchemaField']	162	0.01%
74	['Experiment', 'Psych101 Experiment']	USES_TASK	['Task']	151	0.01%
75	['TaskSpec']	HAS_GLM_PRIOR	['GLMDesignPrior']	148	0.01%
76	['ModelSpec']	HAS_RESOURCE	['DataResource', 'OpenNeuro']	144	0.01%
77	['Task']	INBATTERY	['Battery']	134	0.01%
78	['ReviewRule']	HAS_VALIDITY_LAYER	['ReviewValidityLayer']	102	0.00%
79	['Dataset', 'DataResource']	PART_OF	['Consortium']	99	0.00%
80	['Experiment', 'Psych101 Experiment']	CLASSIFIED_UNDER	['TaskFamily']	98	0.00%
81	['ReviewRule']	HAS_REASON_TAG	['ReviewReasonTag']	81	0.00%
82	['ReviewImplementationRuleCatalog']	CONTAINS_IMPLEMENTATION_RULE	['ReviewImplementationRule']	80	0.00%
83	['Dataset', 'Psych101Dataset']	HAS_EXPERIMENT	['Experiment', 'Psych101 Experiment']	76	0.00%
84	['ModelSpec']	DESCRIBES_TASK	['TaskAnalysis']	72	0.00%
85	['Dataset', 'OpenNeuro']	HAS_GLM_PRIOR	['GLMDesignPrior']	72	0.00%
86	['Dataset', 'DataResource']	HAS_TASK	['TaskSpec']	72	0.00%
87	['ReviewRuleRegistry']	CONTAINS_RULE	['ReviewRule']	71	0.00%

Rank	Source label set	Relationship	Target label set	Edge count	Share of edges
88	['ReviewRule']	HAS_LIFECYCLE_STATUS	['ReviewLifecycleStatus']	71	0.00%
89	['ReviewRule']	HAS_SEVERITY	['ReviewSeverity']	71	0.00%
90	['ReviewRule']	IN_RULE_GROUP	['ReviewRuleGroup']	71	0.00%
91	['Collection']	MAPS_TO	['Dataset', 'DataResource']	61	0.00%
92	['ReviewRuleRegistry']	CONTAINS_CALIBRATION_CASE	['ReviewCalibrationCase']	60	0.00%
93	['Concept']	MAPS_TO	['Concept', 'OntologyConcept', 'OnvocalClass']	60	0.00%
94	['ReviewCalibrationCase']	CALIBRATES_RULE	['ReviewRule']	55	0.00%
95	['Contrast', 'GLMContrast']	CONTRAST_OF	['Concept']	52	0.00%
96	['Dataset', 'DataResource']	CITED_BY	['Publication']	49	0.00%
97	['Dataset', 'OpenNeuro']	ABOUT	['Concept', 'OntologyConcept', 'OnvocalClass']	40	0.00%
98	['Contrast', 'GLMContrast']	IN_ONVOC	['Concept', 'OntologyConcept', 'OnvocalClass']	40	0.00%
99	['StatMap']	IN_ONVOC	['Concept', 'OntologyConcept', 'OnvocalClass']	40	0.00%
100	['Dataset', 'OpenNeuro']	HAS_TASK	['TaskSpec']	38	0.00%
101	['Dataset', 'OpenNeuro']	MAPS_TO	['Dataset', 'OpenNeuro']	38	0.00%
102	['StatMap']	DERIVED_FROM	['Contrast']	37	0.00%
103	['TaskAnalysis']	MEASURES	['Concept']	32	0.00%
104	['ReviewRule']	MAPPED_TO_IMPLEMENTATION	['ReviewImplementationRule']	28	0.00%
105	['Parcellation']	HAS_PARCEL	['Parcel']	26	0.00%
106	['TaskAnalysis']	IN_ONVOC	['Concept', 'OntologyConcept', 'OnvocalClass']	23	0.00%
107	['Concept']	IN_ONVOC	['Concept', 'OntologyConcept', 'OnvocalClass']	22	0.00%
108	['TaskAnalysis']	MAPS_TO	['Task']	19	0.00%
109	['StatisticalMap']	MAPS_TO	['Dataset', 'DataResource']	19	0.00%
110	['Contrast']	MAPS_TO	['Contrast', 'GLMContrast']	19	0.00%
111	['Parcellation']	IN_SPACE	['TemplateSpace']	18	0.00%
112	['TaskSpec']	IN_ONVOC	['Concept', 'OntologyConcept', 'OnvocalClass']	17	0.00%
113	['BrainRegion']	IN_PARCELLATION	['Parcellation']	17	0.00%
114	['Collection']	MAPS_TO	['Dataset']	14	0.00%
115	['Tool']	VALIDATED_ON	['Dataset', 'DataResource']	11	0.00%
116	['Study']	ABOUT	['Concept', 'OntologyConcept', 'OnvocalClass']	10	0.00%
117	['Concept']	MEASURES	['Task']	8	0.00%
118	['GLMRun']	HAS_VARIANT	['GLMVariant']	7	0.00%
119	['StatMap']	DERIVED_FROM	['Contrast', 'GLMContrast']	6	0.00%
120	['ReviewRuleRegistry']	HAS_POLICY_DECISION	['ReviewPolicyDecision']	6	0.00%
121	['Parcellation']	HAS_RESOURCE	['DataResource']	6	0.00%
122	['Task']	ACTIVATES	['Region']	5	0.00%
123	['StatisticalMap']	MAPS_TO	['Dataset']	5	0.00%
124	['ReviewRule']	TRIGGERS_SENSITIVITY	['ReviewSensitivityTemplate']	5	0.00%
125	['Run']	HAD_FAILURE	['ExecutionFailure']	4	0.00%
126	['Task']	HAS_GLM_PRIOR	['GLMDesignPrior']	4	0.00%
127	['Tool']	FAILED_ON	['Dataset', 'DataResource']	3	0.00%

Rank	Source label set	Relationship	Target label set	Edge count	Share of edges
128	['Tool']	HAD_FAILURE	['ExecutionFailure']	3	0.00%
129	['TaskAnalysis']	MAPS_TO	['Concept']	3	0.00%
130	['Concept']	RELATED_TO	['Task']	3	0.00%
131	['Tool']	DOCUMENTED_IN	['Publication']	2	0.00%
132	['Atlas']	HAS_PARCELLATION	['Parcellation']	2	0.00%
133	['Concept']	RELATED_TO	['Concept']	2	0.00%
134	['Tool']	VALIDATED_ON	['DataResource']	2	0.00%
135	['Concept']	ACTIVATES	['Task']	1	0.00%
136	['ReviewCalibrationCase']	CALIBRATES_MODIFIER	['ReviewPositiveModifier']	1	0.00%
137	['Atlas']	CITES	['Publication']	1	0.00%
138	['Concept']	CITES	['Task']	1	0.00%
139	['ReviewRuleRegistry']	CONTAINS_MODIFIER	['ReviewPositiveModifier']	1	0.00%
140	['Contrast', 'GLM-Contrast']	CONTRAST_OF	['Concept', 'OntologyConcept', 'OnvocClass']	1	0.00%
141	['Tool']	HAS_EVIDENCE	['ToolEvidence']	1	0.00%
142	['Dataset', 'Open-Neuro']	HAS_GLM_RUN	['GLMRun']	1	0.00%
143	['TaskSpec']	HAS_GLM_RUN	['GLMRun']	1	0.00%
144	['GLMRun']	HAS_SUMMARY	['ResultSummary']	1	0.00%
145	['Atlas']	IN_SPACE	['TemplateSpace']	1	0.00%
146	['Coordinate']	LOCATED_IN	['Region']	1	0.00%
147	['Concept']	MEASURED_BY	['Task']	1	0.00%
148	['Publication']	MENTIONS_CONCEPT	['Concept']	1	0.00%
149	['ReviewPositive-Modifier']	REQUIRES_FIELD	['ReviewSchemaField']	1	0.00%
150	['Concept']	SIMILAR_TO	['Task']	1	0.00%
151	['Dataset']	USES_TASK	['Task']	1	0.00%

B22. Node property coverage by label and property

Node label	Nodes	Property	Present	Missing	Present %
Coordinate	447,499	id	447,499	0	100.0000%
Coordinate	447,499	labels	447,499	0	100.0000%
Coordinate	447,499	space	447,499	0	100.0000%
Coordinate	447,499	x	447,499	0	100.0000%
Coordinate	447,499	y	447,499	0	100.0000%
Coordinate	447,499	z	447,499	0	100.0000%
Coordinate	447,499	round_mm	447,498	1	99.9998%
Coordinate	447,499	source	447,498	1	99.9998%
Publication	49,744	source	49,738	6	99.9879%
Publication	49,744	id	49,580	164	99.6703%
Publication	49,744	labels	49,580	164	99.6703%
Publication	49,744	doi	48,499	1,245	97.4972%
Publication	49,744	title	47,042	2,702	94.5682%
Publication	49,744	pmid	32,873	16,871	66.0844%
Publication	49,744	journal	15,324	34,420	30.8057%
Publication	49,744	year	14,223	35,521	28.5924%
Publication	49,744	authors	14,222	35,522	28.5904%
Publication	49,744	neurosynth_id	14,222	35,522	28.5904%
Publication	49,744	space	14,222	35,522	28.5904%
Publication	49,744	abstract	493	49,251	0.9911%

Node label	Nodes	Property	Present	Missing	Present %
Publication	49,744	concepts	493	49,251	0.9911%
Publication	49,744	mesh_terms	493	49,251	0.9911%
Publication	49,744	issn	57	49,687	0.1146%
Publication	49,744	keywords	57	49,687	0.1146%
Publication	49,744	openalex_id	57	49,687	0.1146%
Publication	49,744	publication_date	57	49,687	0.1146%
Publication	49,744	publication_year	57	49,687	0.1146%
Publication	49,744	dataset_id	49	49,695	0.0985%
Publication	49,744	reference	49	49,695	0.0985%
Publication	49,744	is_dataset_doi	6	49,738	0.0121%
Publication	49,744	url	1	49,743	0.0020%
Collection	48,009	id	48,009	0	100.0000%
Collection	48,009	labels	48,009	0	100.0000%
Collection	48,009	name	48,009	0	100.0000%
Collection	48,009	source	48,009	0	100.0000%
Collection	48,009	publication_id	31,885	16,124	66.4146%
Collection	48,009	study_id	31,885	16,124	66.4146%
Collection	48,009	modalities	31,884	16,125	66.4125%
Collection	48,009	study_objective	31,729	16,280	66.0897%
Collection	48,009	title	31,729	16,280	66.0897%
Collection	48,009	owner_name	16,124	31,885	33.5854%
Collection	48,009	url	16,124	31,885	33.5854%
Collection	48,009	add_date	15,991	32,018	33.3083%
Collection	48,009	communities	15,991	32,018	33.3083%
Collection	48,009	contributors	15,991	32,018	33.3083%
Collection	48,009	download_url	15,991	32,018	33.3083%
Collection	48,009	modify_date	15,991	32,018	33.3083%
Collection	48,009	number_of_images	15,991	32,018	33.3083%
Collection	48,009	owner	15,991	32,018	33.3083%
Collection	48,009	private	15,991	32,018	33.3083%
Collection	48,009	full_dataset_url	10,845	37,164	22.5895%
Collection	48,009	description	6,784	41,225	14.1307%
Collection	48,009	paper_url	2,710	45,299	5.6448%
Collection	48,009	journal_name	2,705	45,304	5.6344%
Collection	48,009	authors	2,696	45,313	5.6156%
Collection	48,009	software_package	2,650	45,359	5.5198%
Collection	48,009	scanner_make	2,645	45,364	5.5094%
Collection	48,009	scanner_model	2,641	45,368	5.5011%
Collection	48,009	intrasubject_ model_type	2,637	45,372	5.4927%
Collection	48,009	software_version	2,637	45,372	5.4927%
Collection	48,009	intersubject_ registration_ software	2,635	45,374	5.4886%
Collection	48,009	pulse_sequence	2,635	45,374	5.4886%
Collection	48,009	hemodynamic_ response_function	2,633	45,376	5.4844%
Collection	48,009	smoothing_type	2,633	45,376	5.4844%
Collection	48,009	group_description	2,632	45,377	5.4823%
Collection	48,009	intrasubject_ modeling_software	2,631	45,378	5.4802%
Collection	48,009	acquisition_ orientation	2,630	45,379	5.4781%
Collection	48,009	group_model_type	2,630	45,379	5.4781%
Collection	48,009	group_modeling_ software	2,630	45,379	5.4781%
Collection	48,009	inclusion_ exclusion_criteria	2,630	45,379	5.4781%

Node label	Nodes	Property	Present	Missing	Present %
Collection	48,009	high_pass_filter_ method	2,629	45,380	5.4761%
Collection	48,009	order_of_ preprocessing_ operations	2,625	45,384	5.4677%
Collection	48,009	autocorrelation_ model	2,624	45,385	5.4656%
Collection	48,009	motion_correction_ software	2,624	45,385	5.4656%
Collection	48,009	intrasubject_ estimation_type	2,623	45,386	5.4636%
Collection	48,009	object_image_type	2,623	45,386	5.4636%
Collection	48,009	parallel_imaging	2,622	45,387	5.4615%
Collection	48,009	group_estimation_ type	2,621	45,388	5.4594%
Collection	48,009	quality_control	2,620	45,389	5.4573%
Collection	48,009	functional_ coregistration_ method	2,619	45,390	5.4552%
Collection	48,009	motion_correction_ reference	2,619	45,390	5.4552%
Collection	48,009	b0_unwarping_ software	2,618	45,391	5.4531%
Collection	48,009	group_repeated_ measures_method	2,616	45,393	5.4490%
Collection	48,009	nonlinear_ transform_type	2,615	45,394	5.4469%
Collection	48,009	slice_timing_ correction_software	2,615	45,394	5.4469%
Collection	48,009	group_model_ multilevel	2,614	45,395	5.4448%
Collection	48,009	interpolation_ method	2,614	45,395	5.4448%
Collection	48,009	motion_correction_ interpolation	2,614	45,395	5.4448%
Collection	48,009	motion_correction_ metric	2,614	45,395	5.4448%
Collection	48,009	optimization_ method	2,614	45,395	5.4448%
Collection	48,009	orthogonalization_ description	2,614	45,395	5.4448%
Collection	48,009	transform_ similarity_metric	2,612	45,397	5.4406%
Collection	48,009	target_template_ image	2,611	45,398	5.4386%
Collection	48,009	length_of_trials	2,498	45,511	5.2032%
Collection	48,009	nutbrain_food_ viewing_conditions	1,576	46,433	3.2827%
Collection	48,009	nutbrain_food_ choice_type	1,574	46,435	3.2786%
Collection	48,009	nutbrain_odor_ conditions	1,574	46,435	3.2786%
Collection	48,009	nutbrain_taste_ conditions	1,572	46,437	3.2744%
Collection	48,009	doi_add_date	931	47,078	1.9392%
Collection	48,009	DOI	924	47,085	1.9246%
Collection	48,009	group_comparison	463	47,546	0.9644%
Collection	48,009	type_of_design	441	47,568	0.9186%
Collection	48,009	handedness	396	47,613	0.8248%
Collection	48,009	subject_age_mean	348	47,661	0.7249%
Collection	48,009	field_strength	330	47,679	0.6874%
Collection	48,009	used_intersubject_ registration	325	47,684	0.6770%
Collection	48,009	proportion_male_ subjects	322	47,687	0.6707%
Collection	48,009	publication_status	303	47,706	0.6311%

Node label	Nodes	Property	Present	Missing	Present %
Collection	48,009	subject_age_min	303	47,706	0.6311%
Collection	48,009	coordinate_space	301	47,708	0.6270%
Collection	48,009	subject_age_max	300	47,709	0.6249%
Collection	48,009	used_smoothing	299	47,710	0.6228%
Collection	48,009	used_motion_correction	287	47,722	0.5978%
Collection	48,009	functional_coregistered_to_structural	280	47,729	0.5832%
Collection	48,009	used_slice_timing_correction	266	47,743	0.5541%
Collection	48,009	repetition_time	263	47,746	0.5478%
Collection	48,009	number_of_imaging_runs	261	47,748	0.5436%
Collection	48,009	used_motion_regressors	255	47,754	0.5312%
Collection	48,009	intersubject_transformation_type	253	47,756	0.5270%
Collection	48,009	order_of_acquisition	239	47,770	0.4978%
Collection	48,009	echo_time	232	47,777	0.4832%
Collection	48,009	number_of_rejected_subjects	230	47,779	0.4791%
Collection	48,009	group_inference_type	226	47,783	0.4707%
Collection	48,009	used_reaction_time_regressor	222	47,787	0.4624%
Collection	48,009	flip_angle	220	47,789	0.4582%
Collection	48,009	smoothing_fwhm	217	47,792	0.4520%
Collection	48,009	used_temporal_derivatives	216	47,793	0.4499%
Collection	48,009	used_b0_unwarping	215	47,794	0.4478%
Collection	48,009	group_repeated_measures	214	47,795	0.4457%
Collection	48,009	slice_thickness	213	47,796	0.4437%
Collection	48,009	number_of_experimental_units	204	47,805	0.4249%
Collection	48,009	used_orthogonalization	192	47,817	0.3999%
Collection	48,009	optimization	190	47,819	0.3958%
Collection	48,009	used_dispersion_derivatives	190	47,819	0.3958%
Collection	48,009	field_of_view	184	47,825	0.3833%
Collection	48,009	used_high_pass_filter	167	47,842	0.3479%
Collection	48,009	matrix_size	157	47,852	0.3270%
Collection	48,009	length_of_runs	149	47,860	0.3104%
Collection	48,009	resampled_voxel_size	149	47,860	0.3104%
Collection	48,009	skip_distance	137	47,872	0.2854%
Collection	48,009	used_motion_susceptibility_correction	125	47,884	0.2604%
Collection	48,009	target_resolution	111	47,898	0.2312%
Collection	48,009	length_of_blocks	51	47,958	0.1062%
Collection	48,009	nutbrain_hunger_state	26	47,983	0.0542%
Collection	48,009	preprint_DOI	15	47,994	0.0312%
StatsMap	35,240	id	35,240	0	100.0000%
StatsMap	35,240	source	35,240	0	100.0000%
StatsMap	35,240	analysis_level	35,180	60	99.8297%
StatsMap	35,240	contrast	35,180	60	99.8297%
StatsMap	35,240	dataset_id	35,180	60	99.8297%

Node label	Nodes	Property	Present	Missing	Present %
StatsMap	35,240	format	35,180	60	99.8297%
StatsMap	35,240	ingested_at	35,180	60	99.8297%
StatsMap	35,240	is_symlink	35,180	60	99.8297%
StatsMap	35,240	labels	35,180	60	99.8297%
StatsMap	35,240	metadata.json	35,180	60	99.8297%
StatsMap	35,240	node_name	35,180	60	99.8297%
StatsMap	35,240	path	35,180	60	99.8297%
StatsMap	35,240	provided_by	35,180	60	99.8297%
StatsMap	35,240	relative_path	35,180	60	99.8297%
StatsMap	35,240	space	35,180	60	99.8297%
StatsMap	35,240	stat_type	35,180	60	99.8297%
StatsMap	35,240	task	35,180	60	99.8297%
StatsMap	35,240	file_size_bytes	35,178	62	99.8241%
StatsMap	35,240	subject	34,156	1,084	96.9240%
StatsMap	35,240	primary_onvoc_confidence	32,020	3,220	90.8627%
StatsMap	35,240	primary_onvoc_id	32,020	3,220	90.8627%
StatsMap	35,240	symlink_target	29,045	6,195	82.4205%
StatsMap	35,240	run	25,558	9,682	72.5255%
StatsMap	35,240	session	2,640	32,600	7.4915%
StatsMap	35,240	dataset_folder	2,048	33,192	5.8116%
StatsMap	35,240	template_space	1,910	33,330	5.4200%
Task	34,926	id	34,926	0	100.0000%
Task	34,926	labels	34,926	0	100.0000%
Task	34,926	name	34,926	0	100.0000%
Task	34,926	source	34,914	12	99.9656%
Task	34,926	aliases	34,870	56	99.8397%
Task	34,926	alias	1,167	33,759	3.3414%
Task	34,926	batteries	853	34,073	2.4423%
Task	34,926	citations	853	34,073	2.4423%
Task	34,926	concept_relations	853	34,073	2.4423%
Task	34,926	conditions	853	34,073	2.4423%
Task	34,926	contrasts	853	34,073	2.4423%
Task	34,926	creation_time	853	34,073	2.4423%
Task	34,926	definition	853	34,073	2.4423%
Task	34,926	disorders	853	34,073	2.4423%
Task	34,926	external_datasets	853	34,073	2.4423%
Task	34,926	implementations	853	34,073	2.4423%
Task	34,926	indicators	853	34,073	2.4423%
Task	34,926	last_updated	853	34,073	2.4423%
Task	34,926	metadata	853	34,073	2.4423%
Task	34,926	family_id	470	34,456	1.3457%
Task	34,926	family_label	470	34,456	1.3457%
Task	34,926	subfamily_id	470	34,456	1.3457%
Task	34,926	subfamily_label	470	34,456	1.3457%
Task	34,926	created_at	314	34,612	0.8990%
Task	34,926	created_from	314	34,612	0.8990%
Task	34,926	label	314	34,612	0.8990%
Task	34,926	updated_at	314	34,612	0.8990%
Task	34,926	primary_onvoc_confidence	147	34,779	0.4209%
Task	34,926	primary_onvoc_id	147	34,779	0.4209%
Task	34,926	paradigm_name	126	34,800	0.3608%
Task	34,926	task_family_match_method	126	34,800	0.3608%
Task	34,926	task_family_match_score	126	34,800	0.3608%

Node label	Nodes	Property	Present	Missing	Present %
Task	34,926	task_family_match_source	126	34,800	0.3608%
Task	34,926	description	48	34,878	0.1374%
Task	34,926	description_source	44	34,882	0.1260%
Task	34,926	embedding_centaur_behavior_v1	44	34,882	0.1260%
Task	34,926	embedding_centaur_behavior_v1_backend	44	34,882	0.1260%
Task	34,926	embedding_centaur_behavior_v1_dim	44	34,882	0.1260%
Task	34,926	embedding_centaur_behavior_v1_model	44	34,882	0.1260%
Task	34,926	embedding_centaur_behavior_v1_pooling	44	34,882	0.1260%
Task	34,926	embedding_centaur_behavior_v1_source	44	34,882	0.1260%
Task	34,926	embedding_centaur_behavior_v1_updated_at	44	34,882	0.1260%
Task	34,926	embedding.text.v1	44	34,882	0.1260%
Task	34,926	embedding.text.v1_backend	44	34,882	0.1260%
Task	34,926	embedding.text.v1_dim	44	34,882	0.1260%
Task	34,926	embedding.text.v1_model	44	34,882	0.1260%
Task	34,926	embedding.text.v1_pooling	44	34,882	0.1260%
Task	34,926	embedding.text.v1_source	44	34,882	0.1260%
Task	34,926	embedding.text.v1_template.version	44	34,882	0.1260%
Task	34,926	embedding.text.v1_updated_at	44	34,882	0.1260%
Task	34,926	schema_version	44	34,882	0.1260%
Task	34,926	canonical_name	39	34,887	0.1117%
Task	34,926	ontology_match_method	39	34,887	0.1117%
Task	34,926	ontology_match_score	39	34,887	0.1117%
Task	34,926	canonical_task_id	16	34,910	0.0458%
Task	34,926	canonical_task_name	16	34,910	0.0458%
Task	34,926	links_json	8	34,918	0.0229%
Task	34,926	canonical_task_cogat_id	7	34,919	0.0200%
Task	34,926	canonical_definition	6	34,920	0.0172%
Task	34,926	canonical_id	4	34,922	0.0115%
Embedding	23,865	dimension	23,865	0	100.0000%
Embedding	23,865	id	23,865	0	100.0000%
Embedding	23,865	kind	23,865	0	100.0000%
Embedding	23,865	labels	23,865	0	100.0000%
Embedding	23,865	model	23,865	0	100.0000%
Embedding	23,865	normalization	23,865	0	100.0000%
Embedding	23,865	owner_id	23,865	0	100.0000%
Embedding	23,865	source	23,865	0	100.0000%
Embedding	23,865	storage_index	23,865	0	100.0000%
Embedding	23,865	storage_path	23,865	0	100.0000%

Node label	Nodes	Property	Present	Missing	Present %
Embedding	23,865	text_section	23,865	0	100.0000%
Embedding	23,865	vector_norm	23,865	0	100.0000%
StatisticalMap	21,283	add_date	21,283	0	100.0000%
StatisticalMap	21,283	brain_coverage	21,283	0	100.0000%
StatisticalMap	21,283	collection	21,283	0	100.0000%
StatisticalMap	21,283	collection_id	21,283	0	100.0000%
StatisticalMap	21,283	data_origin	21,283	0	100.0000%
StatisticalMap	21,283	file	21,283	0	100.0000%
StatisticalMap	21,283	file_size	21,283	0	100.0000%
StatisticalMap	21,283	id	21,283	0	100.0000%
StatisticalMap	21,283	image_type	21,283	0	100.0000%
StatisticalMap	21,283	is_valid	21,283	0	100.0000%
StatisticalMap	21,283	map_type	21,283	0	100.0000%
StatisticalMap	21,283	modify_date	21,283	0	100.0000%
StatisticalMap	21,283	name	21,283	0	100.0000%
StatisticalMap	21,283	not_mni	21,283	0	100.0000%
StatisticalMap	21,283	perc_bad_voxels	21,283	0	100.0000%
StatisticalMap	21,283	source	21,283	0	100.0000%
StatisticalMap	21,283	target_template_image	21,283	0	100.0000%
StatisticalMap	21,283	url	21,283	0	100.0000%
StatisticalMap	21,283	qa_score	21,282	1	99.9953%
StatisticalMap	21,283	qa_status	21,282	1	99.9953%
StatisticalMap	21,283	labels	21,281	2	99.9906%
StatisticalMap	21,283	modality	21,139	144	99.3234%
StatisticalMap	21,283	perc_voxels_outside	21,115	168	99.2106%
StatisticalMap	21,283	is_thresholded	20,203	1,080	94.9255%
StatisticalMap	21,283	thumbnail	17,237	4,046	80.9895%
StatisticalMap	21,283	analysis_level	16,782	4,501	78.8517%
StatisticalMap	21,283	qa_reasons	15,783	5,500	74.1578%
StatisticalMap	21,283	surface_left_file	15,543	5,740	73.0301%
StatisticalMap	21,283	surface_right_file	15,531	5,752	72.9737%
StatisticalMap	21,283	cognitive_paradigm_cogatlas	9,868	11,415	46.3656%
StatisticalMap	21,283	cognitive_paradigm_cogatlas_id	9,868	11,415	46.3656%
StatisticalMap	21,283	number_of_subjects	5,574	15,709	26.1899%
StatisticalMap	21,283	description	5,364	15,919	25.2032%
StatisticalMap	21,283	reduced_representation	4,753	16,530	22.3324%
StatisticalMap	21,283	cognitive_contrast_cogatlas	3,529	17,754	16.5813%
StatisticalMap	21,283	cognitive_contrast_cogatlas_id	3,529	17,754	16.5813%
StatisticalMap	21,283	qa_is_primary	2,108	19,175	9.9046%
StatisticalMap	21,283	contrast_definition	1,127	20,156	5.2953%
StatisticalMap	21,283	figure	836	20,447	3.9280%
StatisticalMap	21,283	subject_species	69	21,214	0.3242%
StatisticalMap	21,283	contrast_definition_cogatlas	63	21,220	0.2960%
StatisticalMap	21,283	smoothness_fwhm	5	21,278	0.0235%
StatisticalMap	21,283	statistic_parameters	1	21,282	0.0047%
DataResource	9,282	id	9,280	2	99.9785%
DataResource	9,282	labels	9,280	2	99.9785%
DataResource	9,282	name	9,274	8	99.9138%
DataResource	9,282	format	7,587	1,695	81.7388%

Node label	Nodes	Property	Present	Missing	Present %
DataResource	9,282	path	7,587	1,695	81.7388%
DataResource	9,282	source	7,587	1,695	81.7388%
DataResource	9,282	file_size_bytes	7,581	1,701	81.6742%
DataResource	9,282	is_symlink	7,581	1,701	81.6742%
DataResource	9,282	relative_path	7,581	1,701	81.6742%
DataResource	9,282	access_type	1,693	7,589	18.2396%
DataResource	9,282	acquisitions	1,693	7,589	18.2396%
DataResource	9,282	alias	1,693	7,589	18.2396%
DataResource	9,282	aliases	1,693	7,589	18.2396%
DataResource	9,282	category	1,693	7,589	18.2396%
DataResource	9,282	category_source	1,693	7,589	18.2396%
DataResource	9,282	created_from	1,693	7,589	18.2396%
DataResource	9,282	dataset_id	1,693	7,589	18.2396%
DataResource	9,282	description	1,693	7,589	18.2396%
DataResource	9,282	disease_flags	1,693	7,589	18.2396%
DataResource	9,282	has_derivatives	1,693	7,589	18.2396%
DataResource	9,282	is_openneuro	1,693	7,589	18.2396%
DataResource	9,282	license	1,693	7,589	18.2396%
DataResource	9,282	modalities	1,693	7,589	18.2396%
DataResource	9,282	preview_media	1,693	7,589	18.2396%
DataResource	9,282	primary_url	1,693	7,589	18.2396%
DataResource	9,282	search_blob	1,693	7,589	18.2396%
DataResource	9,282	short_name	1,693	7,589	18.2396%
DataResource	9,282	source_repo	1,693	7,589	18.2396%
DataResource	9,282	source_repo_bucket	1,693	7,589	18.2396%
DataResource	9,282	source_repo_id	1,693	7,589	18.2396%
DataResource	9,282	species	1,693	7,589	18.2396%
DataResource	9,282	tags	1,693	7,589	18.2396%
DataResource	9,282	tasks	1,693	7,589	18.2396%
DataResource	9,282	modalities_notes	1,595	7,687	17.1838%
DataResource	9,282	source_version	1,594	7,688	17.1730%
DataResource	9,282	subjects_count	713	8,569	7.6815%
DataResource	9,282	age_range	541	8,741	5.8285%
DataResource	9,282	sessions_count	356	8,926	3.8354%
DataResource	9,282	primary_onvoc_confidence	355	8,927	3.8246%
DataResource	9,282	primary_onvoc_id	355	8,927	3.8246%
DataResource	9,282	center	100	9,182	1.0774%
DataResource	9,282	consortium	99	9,183	1.0666%
DataResource	9,282	principal_investigator	99	9,183	1.0666%
DataResource	9,282	size_human	99	9,183	1.0666%
DataResource	9,282	atlas_slug	6	9,276	0.0646%
DataResource	9,282	resource_id	3	9,279	0.0323%
OpenNeuro	7,619	id	7,619	0	100.0000%
OpenNeuro	7,619	labels	7,619	0	100.0000%
OpenNeuro	7,619	name	7,619	0	100.0000%
OpenNeuro	7,619	source	7,619	0	100.0000%
OpenNeuro	7,619	file_size_bytes	7,581	38	99.5012%
OpenNeuro	7,619	format	7,581	38	99.5012%
OpenNeuro	7,619	is_symlink	7,581	38	99.5012%
OpenNeuro	7,619	path	7,581	38	99.5012%
OpenNeuro	7,619	relative_path	7,581	38	99.5012%
OpenNeuro	7,619	dataset_id	38	7,581	0.4988%
OpenNeuro	7,619	doi	38	7,581	0.4988%
OpenNeuro	7,619	spec_hash	38	7,581	0.4988%
ToolVersion	4,142	id	4,142	0	100.0000%

Node label	Nodes	Property	Present	Missing	Present %
ToolVersion	4,142	tool_id	4,142	0	100.0000%
ToolVersion	4,142	version_id	4,142	0	100.0000%
ToolVersion	4,142	labels	2,581	1,561	62.3129%
ToolVersion	4,142	op	2,084	2,058	50.3139%
ToolVersion	4,142	software	2,084	2,058	50.3139%
ToolVersion	4,142	container_image	1,974	2,168	47.6581%
ToolVersion	4,142	version	1,905	2,237	45.9923%
ToolVersion	4,142	python_module	182	3,960	4.3940%
ToolVersion	4,142	python_function	98	4,044	2.3660%
Term	3,228	id	3,228	0	100.0000%
Term	3,228	labels	3,228	0	100.0000%
Term	3,228	name	3,228	0	100.0000%
Term	3,228	source	3,228	0	100.0000%
Concept	2,336	id	2,336	0	100.0000%
Concept	2,336	name	2,336	0	100.0000%
Concept	2,336	labels	2,331	5	99.7860%
Concept	2,336	aliases	2,321	15	99.3579%
Concept	2,336	definition	2,320	16	99.3151%
Concept	2,336	source	2,316	20	99.1438%
Concept	2,336	label	2,264	72	96.9178%
Concept	2,336	concept_classes	1,564	772	66.9521%
Concept	2,336	synonyms	1,505	831	64.4264%
Concept	2,336	alias	919	1,417	39.3408%
Concept	2,336	citations	919	1,417	39.3408%
Concept	2,336	concept_class	919	1,417	39.3408%
Concept	2,336	contrast_links	919	1,417	39.3408%
Concept	2,336	creation_time	919	1,417	39.3408%
Concept	2,336	last_updated	919	1,417	39.3408%
Concept	2,336	metadata	919	1,417	39.3408%
Concept	2,336	related_concepts	919	1,417	39.3408%
Concept	2,336	is_top_concept	752	1,584	32.1918%
Concept	2,336	scheme	752	1,584	32.1918%
Concept	2,336	top_schemes	752	1,584	32.1918%
Concept	2,336	uri	752	1,584	32.1918%
Concept	2,336	primary_onvoc_confidence	22	2,314	0.9418%
Concept	2,336	primary_onvoc_id	22	2,314	0.9418%
Contrast	2,206	id	2,206	0	100.0000%
Contrast	2,206	labels	2,206	0	100.0000%
Contrast	2,206	name	2,206	0	100.0000%
Contrast	2,206	source	2,206	0	100.0000%
Contrast	2,206	conditions	1,585	621	71.8495%
Contrast	2,206	creation_time	1,585	621	71.8495%
Contrast	2,206	last_updated	1,585	621	71.8495%
Contrast	2,206	task_id	1,585	621	71.8495%
Contrast	2,206	event_stamp	1,476	730	66.9084%
Contrast	2,206	id_user	1,476	730	66.9084%
Contrast	2,206	dataset_id	621	1,585	28.1505%
Contrast	2,206	condition_list	393	1,813	17.8150%
Contrast	2,206	ingested_at	393	1,813	17.8150%
Contrast	2,206	task	393	1,813	17.8150%
Contrast	2,206	test	393	1,813	17.8150%
Contrast	2,206	weights	393	1,813	17.8150%
Contrast	2,206	task_label	228	1,978	10.3354%
Contrast	2,206	primary_onvoc_confidence	3	2,203	0.1360%
Contrast	2,206	primary_onvoc_id	3	2,203	0.1360%

Node label	Nodes	Property	Present	Missing	Present %
BrainRegion	2,140	atlas	2,140	0	100.0000%
BrainRegion	2,140	id	2,140	0	100.0000%
BrainRegion	2,140	name	2,140	0	100.0000%
BrainRegion	2,140	space	2,140	0	100.0000%
BrainRegion	2,140	atlas_slug	2,123	17	99.2056%
BrainRegion	2,140	labels	2,123	17	99.2056%
BrainRegion	2,140	source	2,123	17	99.2056%
BrainRegion	2,140	label_index	1,344	796	62.8037%
BrainRegion	2,140	abbreviation	1,327	813	62.0093%
BrainRegion	2,140	acronym	1,327	813	62.0093%
BrainRegion	2,140	depth	1,327	813	62.0093%
BrainRegion	2,140	graph_order	1,327	813	62.0093%
BrainRegion	2,140	species	1,327	813	62.0093%
BrainRegion	2,140	structure_id	1,327	813	62.0093%
BrainRegion	2,140	structure_id_path	1,327	813	62.0093%
BrainRegion	2,140	parent_structure_id	1,326	814	61.9626%
BrainRegion	2,140	index	796	1,344	37.1963%
BrainRegion	2,140	label_original	796	1,344	37.1963%
BrainRegion	2,140	modality	796	1,344	37.1963%
BrainRegion	2,140	parcellation_id	796	1,344	37.1963%
BrainRegion	2,140	parcellation_slug	796	1,344	37.1963%
BrainRegion	2,140	region_id	796	1,344	37.1963%
BrainRegion	2,140	map_file	727	1,413	33.9720%
BrainRegion	2,140	scale	538	1,602	25.1402%
BrainRegion	2,140	space_resolution_mm	524	1,616	24.4860%
BrainRegion	2,140	space_variant	457	1,683	21.3551%
BrainRegion	2,140	hemisphere	408	1,732	19.0654%
BrainRegion	2,140	n_rois	300	1,840	14.0187%
BrainRegion	2,140	network	300	1,840	14.0187%
BrainRegion	2,140	resolution_mm	300	1,840	14.0187%
BrainRegion	2,140	yeo_network_set	300	1,840	14.0187%
BrainRegion	2,140	yeo_networks	300	1,840	14.0187%
BrainRegion	2,140	atlas_variant	191	1,949	8.9252%
BrainRegion	2,140	roi_number	56	2,084	2.6168%
Dataset	2,136	id	2,136	0	100.0000%
Dataset	2,136	labels	2,136	0	100.0000%
Dataset	2,136	name	2,136	0	100.0000%
Dataset	2,136	aliases	2,095	41	98.0805%
Dataset	2,136	dataset_id	1,732	404	81.0861%
Dataset	2,136	description	1,694	442	79.3071%
Dataset	2,136	license	1,694	442	79.3071%
Dataset	2,136	tags	1,694	442	79.3071%
Dataset	2,136	access_type	1,693	443	79.2603%
Dataset	2,136	acquisitions	1,693	443	79.2603%
Dataset	2,136	alias	1,693	443	79.2603%
Dataset	2,136	category	1,693	443	79.2603%
Dataset	2,136	category_source	1,693	443	79.2603%
Dataset	2,136	created_from	1,693	443	79.2603%
Dataset	2,136	disease_flags	1,693	443	79.2603%
Dataset	2,136	has_derivatives	1,693	443	79.2603%
Dataset	2,136	is_openneuro	1,693	443	79.2603%
Dataset	2,136	modalities	1,693	443	79.2603%
Dataset	2,136	preview_media	1,693	443	79.2603%
Dataset	2,136	primary_url	1,693	443	79.2603%
Dataset	2,136	search_blob	1,693	443	79.2603%

Node label	Nodes	Property	Present	Missing	Present %
Dataset	2,136	short_name	1,693	443	79.2603%
Dataset	2,136	source_repo	1,693	443	79.2603%
Dataset	2,136	source_repo_bucket	1,693	443	79.2603%
Dataset	2,136	source_repo_id	1,693	443	79.2603%
Dataset	2,136	species	1,693	443	79.2603%
Dataset	2,136	tasks	1,693	443	79.2603%
Dataset	2,136	modalities_notes	1,595	541	74.6723%
Dataset	2,136	source_version	1,594	542	74.6255%
Dataset	2,136	subjects_count	713	1,423	33.3801%
Dataset	2,136	age_range	541	1,595	25.3277%
Dataset	2,136	source	441	1,695	20.6461%
Dataset	2,136	dataset_uuid	402	1,734	18.8202%
Dataset	2,136	portal_uri	402	1,734	18.8202%
Dataset	2,136	total_subjects_reported	402	1,734	18.8202%
Dataset	2,136	sessions_count	356	1,780	16.6667%
Dataset	2,136	primary_onvoc_confidence	355	1,781	16.6199%
Dataset	2,136	primary_onvoc_id	355	1,781	16.6199%
Dataset	2,136	openneuro_id	321	1,815	15.0281%
Dataset	2,136	center	100	2,036	4.6816%
Dataset	2,136	consortium	99	2,037	4.6348%
Dataset	2,136	principal_investigator	99	2,037	4.6348%
Dataset	2,136	size_human	99	2,037	4.6348%
Dataset	2,136	doi	38	2,098	1.7790%
Dataset	2,136	spec_hash	38	2,098	1.7790%
Dataset	2,136	accession	2	2,134	0.0936%
Dataset	2,136	subject_count	2	2,134	0.0936%
Dataset	2,136	n_experiments	1	2,135	0.0468%
Dataset	2,136	n_participants	1	2,135	0.0468%
Dataset	2,136	provenance	1	2,135	0.0468%
Dataset	2,136	resource_id	1	2,135	0.0468%
Dataset	2,136	schema_version	1	2,135	0.0468%
Dataset	2,136	url	1	2,135	0.0468%
Tool	2,084	category	2,084	0	100.0000%
Tool	2,084	default_group_id	2,084	0	100.0000%
Tool	2,084	description	2,084	0	100.0000%
Tool	2,084	domain	2,084	0	100.0000%
Tool	2,084	exposed	2,084	0	100.0000%
Tool	2,084	exposure_group	2,084	0	100.0000%
Tool	2,084	id	2,084	0	100.0000%
Tool	2,084	is_default	2,084	0	100.0000%
Tool	2,084	name	2,084	0	100.0000%
Tool	2,084	op	2,084	0	100.0000%
Tool	2,084	op_key	2,084	0	100.0000%
Tool	2,084	primary_intent	2,084	0	100.0000%
Tool	2,084	runtime_kind	2,084	0	100.0000%
Tool	2,084	software	2,084	0	100.0000%
Tool	2,084	tool_id	2,084	0	100.0000%
Tool	2,084	intents	2,078	6	99.7121%
Tool	2,084	display_name	2,073	11	99.4722%
Tool	2,084	runtime	2,073	11	99.4722%
Tool	2,084	source	2,073	11	99.4722%
Tool	2,084	labels	2,070	14	99.3282%
Tool	2,084	version	1,905	179	91.4107%
Tool	2,084	kind	1,154	930	55.3743%

Node label	Nodes	Property	Present	Missing	Present %
Phenotype	1,218	category	1,218	0	100.0000%
Phenotype	1,218	dataset_uuid	1,218	0	100.0000%
Phenotype	1,218	id	1,218	0	100.0000%
Phenotype	1,218	labels	1,218	0	100.0000%
Phenotype	1,218	measurement_type	1,218	0	100.0000%
Phenotype	1,218	name	1,218	0	100.0000%
Phenotype	1,218	source	1,218	0	100.0000%
Phenotype	1,218	total_observations	1,218	0	100.0000%
Phenotype	1,218	value_counts	885	333	72.6601%
Phenotype	1,218	numeric_summary	333	885	27.3399%
Citation	1,201	citation_authors	1,201	0	100.0000%
Citation	1,201	citation_comment	1,201	0	100.0000%
Citation	1,201	citation_desc	1,201	0	100.0000%
Citation	1,201	citation_pmid	1,201	0	100.0000%
Citation	1,201	citation_pubdate	1,201	0	100.0000%
Citation	1,201	citation_pubname	1,201	0	100.0000%
Citation	1,201	citation_source	1,201	0	100.0000%
Citation	1,201	citation_type	1,201	0	100.0000%
Citation	1,201	citation_url	1,201	0	100.0000%
Citation	1,201	id	1,201	0	100.0000%
Citation	1,201	labels	1,201	0	100.0000%
Citation	1,201	relationship	1,201	0	100.0000%
Citation	1,201	source	1,201	0	100.0000%
Citation	1,201	event_stamp	1,116	85	92.9226%
Citation	1,201	id_user	1,116	85	92.9226%
Citation	1,201	creation_time	85	1,116	7.0774%
Citation	1,201	doi	85	1,116	7.0774%
Citation	1,201	last_updated	85	1,116	7.0774%
Citation	1,201	name	85	1,116	7.0774%
Subject	1,139	dataset_id	1,139	0	100.0000%
Subject	1,139	id	1,139	0	100.0000%
Subject	1,139	labels	1,139	0	100.0000%
Subject	1,139	source	1,139	0	100.0000%
Subject	1,139	subject_code	1,139	0	100.0000%
TaskCondition	807	creation_time	807	0	100.0000%
TaskCondition	807	id	807	0	100.0000%
TaskCondition	807	labels	807	0	100.0000%
TaskCondition	807	last_updated	807	0	100.0000%
TaskCondition	807	name	807	0	100.0000%
TaskCondition	807	source	807	0	100.0000%
TaskCondition	807	task_id	807	0	100.0000%
TaskCondition	807	relationship	791	16	98.0173%
TaskCondition	807	condition_description	662	145	82.0322%
TaskCondition	807	condition_text	662	145	82.0322%
TaskCondition	807	event_stamp	662	145	82.0322%
TaskCondition	807	id_user	662	145	82.0322%
OntologyConcept	752	aliases	752	0	100.0000%
OntologyConcept	752	definition	752	0	100.0000%
OntologyConcept	752	id	752	0	100.0000%
OntologyConcept	752	is_top_concept	752	0	100.0000%
OntologyConcept	752	label	752	0	100.0000%
OntologyConcept	752	labels	752	0	100.0000%
OntologyConcept	752	name	752	0	100.0000%
OntologyConcept	752	scheme	752	0	100.0000%
OntologyConcept	752	source	752	0	100.0000%

Node label	Nodes	Property	Present	Missing	Present %
OntologyConcept	752	synonyms	752	0	100.0000%
OntologyConcept	752	top_schemes	752	0	100.0000%
OntologyConcept	752	uri	752	0	100.0000%
OnvocClass	752	aliases	752	0	100.0000%
OnvocClass	752	definition	752	0	100.0000%
OnvocClass	752	id	752	0	100.0000%
OnvocClass	752	is_top_concept	752	0	100.0000%
OnvocClass	752	label	752	0	100.0000%
OnvocClass	752	labels	752	0	100.0000%
OnvocClass	752	name	752	0	100.0000%
OnvocClass	752	scheme	752	0	100.0000%
OnvocClass	752	source	752	0	100.0000%
OnvocClass	752	synonyms	752	0	100.0000%
OnvocClass	752	top_schemes	752	0	100.0000%
OnvocClass	752	uri	752	0	100.0000%
GLMContrast	621	dataset_id	621	0	100.0000%
GLMContrast	621	id	621	0	100.0000%
GLMContrast	621	labels	621	0	100.0000%
GLMContrast	621	name	621	0	100.0000%
GLMContrast	621	source	621	0	100.0000%
GLMContrast	621	condition_list	393	228	63.2850%
GLMContrast	621	ingested_at	393	228	63.2850%
GLMContrast	621	task	393	228	63.2850%
GLMContrast	621	test	393	228	63.2850%
GLMContrast	621	weights	393	228	63.2850%
GLMContrast	621	task_label	228	393	36.7150%
GLMContrast	621	primary_onvoc_confidence	3	618	0.4831%
GLMContrast	621	primary_onvoc_id	3	618	0.4831%
TaskIndicator	584	id	584	0	100.0000%
TaskIndicator	584	labels	584	0	100.0000%
TaskIndicator	584	relationship	584	0	100.0000%
TaskIndicator	584	source	584	0	100.0000%
TaskIndicator	584	task_id	584	0	100.0000%
TaskIndicator	584	type	584	0	100.0000%
TaskIndicator	584	creation_time	2	582	0.3425%
TaskIndicator	584	last_updated	2	582	0.3425%
TaskIndicator	584	name	2	582	0.3425%
SubjectGroup	402	dataset_uuid	402	0	100.0000%
SubjectGroup	402	id	402	0	100.0000%
SubjectGroup	402	imaging_sessions	402	0	100.0000%
SubjectGroup	402	labels	402	0	100.0000%
SubjectGroup	402	name	402	0	100.0000%
SubjectGroup	402	phenotypic_sessions	402	0	100.0000%
SubjectGroup	402	records_protected	402	0	100.0000%
SubjectGroup	402	source	402	0	100.0000%
SubjectGroup	402	unique_subjects	402	0	100.0000%
Condition	310	dataset_id	310	0	100.0000%
Condition	310	id	310	0	100.0000%
Condition	310	ingested_at	310	0	100.0000%
Condition	310	labels	310	0	100.0000%
Condition	310	name	310	0	100.0000%
Condition	310	order_index	310	0	100.0000%
Condition	310	source	310	0	100.0000%
Condition	310	task	310	0	100.0000%
GLMDesignPrior	216	axes	216	0	100.0000%

Node label	Nodes	Property	Present	Missing	Present %
GLMDesignPrior	216	confounds	216	0	100.0000%
GLMDesignPrior	216	coverage	216	0	100.0000%
GLMDesignPrior	216	high_pass	216	0	100.0000%
GLMDesignPrior	216	hrf_basis	216	0	100.0000%
GLMDesignPrior	216	id	216	0	100.0000%
GLMDesignPrior	216	labels	216	0	100.0000%
GLMDesignPrior	216	n_specs	216	0	100.0000%
GLMDesignPrior	216	source	216	0	100.0000%
GLMDesignPrior	216	support	216	0	100.0000%
GLMDesignPrior	216	task	216	0	100.0000%
GLMDesignPrior	216	dataset_id	110	106	50.9259%
Author	172	id	172	0	100.0000%
Author	172	labels	172	0	100.0000%
Author	172	name	172	0	100.0000%
Author	172	roles	172	0	100.0000%
Author	172	source	172	0	100.0000%
Author	172	orcid	141	31	81.9767%
TaskFamily	138	id	138	0	100.0000%
TaskFamily	138	name	138	0	100.0000%
TaskFamily	138	labels	43	95	31.1594%
TaskFamily	138	source	20	118	14.4928%
TaskFamily	138	family_id	15	123	10.8696%
TaskFamily	138	family_label	15	123	10.8696%
TaskFamily	138	family_description	14	124	10.1449%
TaskFamily	138	schema_version	14	124	10.1449%
TaskFamily	138	ontology_source	9	129	6.5217%
TaskFamily	138	subfamily_id	9	129	6.5217%
TaskFamily	138	subfamily_label	9	129	6.5217%
Institution	119	country	119	0	100.0000%
Institution	119	id	119	0	100.0000%
Institution	119	labels	119	0	100.0000%
Institution	119	name	119	0	100.0000%
Institution	119	ror	119	0	100.0000%
Institution	119	source	119	0	100.0000%
TaskSpec	110	dataset	110	0	100.0000%
TaskSpec	110	id	110	0	100.0000%
TaskSpec	110	labels	110	0	100.0000%
TaskSpec	110	name	110	0	100.0000%
TaskSpec	110	source	110	0	100.0000%
TaskSpec	110	aliases	72	38	65.4545%
TaskSpec	110	bids_model_version	72	38	65.4545%
TaskSpec	110	column_names	72	38	65.4545%
TaskSpec	110	dummy_volumes	72	38	65.4545%
TaskSpec	110	group_by	72	38	65.4545%
TaskSpec	110	ingested_at	72	38	65.4545%
TaskSpec	110	model_name	72	38	65.4545%
TaskSpec	110	n_subjects	72	38	65.4545%
TaskSpec	110	task_metadata_ json	72	38	65.4545%
TaskSpec	110	bold_volumes	70	40	63.6364%
TaskSpec	110	cite_links	64	46	58.1818%
Re- viewSchemaField	106	field_path	106	0	100.0000%
Re- viewSchemaField	106	id	106	0	100.0000%
Re- viewSchemaField	106	labels	106	0	100.0000%
ResourceType	102	id	102	0	100.0000%

Node label	Nodes	Property	Present	Missing	Present %
ResourceType	102	name	102	0	100.0000%
ResourceType	102	labels	83	19	81.3725%
ToolFamily	96	id	96	0	100.0000%
ToolFamily	96	labels	96	0	100.0000%
ToolFamily	96	name	96	0	100.0000%
Repository	88	id	88	0	100.0000%
Repository	88	labels	88	0	100.0000%
Repository	88	name	88	0	100.0000%
BrainAnnotation	86	dataset	86	0	100.0000%
BrainAnnotation	86	density	86	0	100.0000%
BrainAnnotation	86	description	86	0	100.0000%
BrainAnnotation	86	files	86	0	100.0000%
BrainAnnotation	86	formats	86	0	100.0000%
BrainAnnotation	86	id	86	0	100.0000%
BrainAnnotation	86	labels	86	0	100.0000%
BrainAnnotation	86	primary_references	86	0	100.0000%
BrainAnnotation	86	restricted	86	0	100.0000%
BrainAnnotation	86	source	86	0	100.0000%
BrainAnnotation	86	space	86	0	100.0000%
BrainAnnotation	86	summary	86	0	100.0000%
BrainAnnotation	86	tags	86	0	100.0000%
BrainAnnotation	86	urls	86	0	100.0000%
BrainAnnotation	86	sample_size	78	8	90.6977%
BrainAnnotation	86	age_mean	54	32	62.7907%
BrainAnnotation	86	age_sd	54	32	62.7907%
BrainAnnotation	86	resolution	47	39	54.6512%
BrainAnnotation	86	hemispheres	39	47	45.3488%
BrainAnnotation	86	secondary_references	10	76	11.6279%
ReviewImplementationRule	80	action	80	0	100.0000%
ReviewImplementationRule	80	applies_to	80	0	100.0000%
ReviewImplementationRule	80	description	80	0	100.0000%
ReviewImplementationRule	80	id	80	0	100.0000%
ReviewImplementationRule	80	labels	80	0	100.0000%
ReviewImplementationRule	80	message	80	0	100.0000%
ReviewImplementationRule	80	reason_tags	80	0	100.0000%
ReviewImplementationRule	80	review_mode	80	0	100.0000%
ReviewImplementationRule	80	rule_id	80	0	100.0000%
ReviewImplementationRule	80	severity	80	0	100.0000%
ReviewImplementationRule	80	source_path	80	0	100.0000%
ReviewImplementationRule	80	stage	80	0	100.0000%
ReviewImplementationRule	80	suggested_fix	80	0	100.0000%
ReviewImplementationRule	80	tags	80	0	100.0000%
ReviewImplementationRule	80	check_fn	61	19	76.2500%
ReviewImplementationRule	80	comparator	26	54	32.5000%
ReviewImplementationRule	80	metric	26	54	32.5000%

Node label	Nodes	Property	Present	Missing	Present %
ReviewImplementationRule	80	threshold	26	54	32.5000%
ReviewImplementationRule	80	tool_filter	9	71	11.2500%
ReviewImplementationRule	80	kg_lookup	1	79	1.2500%
ModelSpec	78	dataset_id	78	0	100.0000%
ModelSpec	78	hash	78	0	100.0000%
ModelSpec	78	id	78	0	100.0000%
ModelSpec	78	labels	78	0	100.0000%
ModelSpec	78	path	78	0	100.0000%
ModelSpec	78	source	78	0	100.0000%
ModelSpec	78	task	78	0	100.0000%
ModelSpec	78	bids_model_version	72	6	92.3077%
ModelSpec	78	convolve_input	72	6	92.3077%
ModelSpec	78	fitlins_params_json	72	6	92.3077%
ModelSpec	78	group_by	72	6	92.3077%
ModelSpec	78	hrf_model	72	6	92.3077%
ModelSpec	78	ingested_at	72	6	92.3077%
ModelSpec	78	model_name	72	6	92.3077%
ModelSpec	78	model_type	72	6	92.3077%
ModelSpec	78	confounds_terms	71	7	91.0256%
ModelSpec	78	hrf_derivative	15	63	19.2308%
ModelSpec	78	hrf_dispersion	7	71	8.9744%
Experiment	76	dataset_id	76	0	100.0000%
Experiment	76	description	76	0	100.0000%
Experiment	76	experiment_id	76	0	100.0000%
Experiment	76	experiment_name	76	0	100.0000%
Experiment	76	experiment_path	76	0	100.0000%
Experiment	76	experiment_slug	76	0	100.0000%
Experiment	76	id	76	0	100.0000%
Experiment	76	labels	76	0	100.0000%
Experiment	76	n_participants	76	0	100.0000%
Experiment	76	n_trials	76	0	100.0000%
Experiment	76	name	76	0	100.0000%
Experiment	76	provenance	76	0	100.0000%
Experiment	76	raw	76	0	100.0000%
Experiment	76	schema_version	76	0	100.0000%
Experiment	76	source	76	0	100.0000%
Experiment	76	task_label	76	0	100.0000%
Experiment	76	task_labels	76	0	100.0000%
Experiment	76	task_ontology	76	0	100.0000%
Experiment	76	task_ontology_evidence	76	0	100.0000%
Experiment	76	task_families	62	14	81.5789%
Experiment	76	task_family_id	35	41	46.0526%
Experiment	76	task_family_label	35	41	46.0526%
Experiment	76	task_ontology_match_field	35	41	46.0526%
Experiment	76	task_ontology_match_method	35	41	46.0526%
Experiment	76	task_ontology_match_score	35	41	46.0526%
Experiment	76	task_ontology_match_text	35	41	46.0526%
Experiment	76	task_paradigm_name	35	41	46.0526%
Experiment	76	task_paradigms	35	41	46.0526%
Experiment	76	task_subfamilies	35	41	46.0526%
Experiment	76	task_subfamily_id	35	41	46.0526%

Node label	Nodes	Property	Present	Missing	Present %
Experiment	76	task_subfamily_label	35	41	46.0526%
Experiment	76	canonical_task_cogat_id	5	71	6.5789%
Experiment	76	canonical_task_id	5	71	6.5789%
Experiment	76	canonical_task_label	5	71	6.5789%
Psych101 Experiment	76	dataset_id	76	0	100.0000%
Psych101 Experiment	76	description	76	0	100.0000%
Psych101 Experiment	76	experiment_id	76	0	100.0000%
Psych101 Experiment	76	experiment_name	76	0	100.0000%
Psych101 Experiment	76	experiment_path	76	0	100.0000%
Psych101 Experiment	76	experiment_slug	76	0	100.0000%
Psych101 Experiment	76	id	76	0	100.0000%
Psych101 Experiment	76	labels	76	0	100.0000%
Psych101 Experiment	76	n_participants	76	0	100.0000%
Psych101 Experiment	76	n_trials	76	0	100.0000%
Psych101 Experiment	76	name	76	0	100.0000%
Psych101 Experiment	76	provenance	76	0	100.0000%
Psych101 Experiment	76	raw	76	0	100.0000%
Psych101 Experiment	76	schema_version	76	0	100.0000%
Psych101 Experiment	76	source	76	0	100.0000%
Psych101 Experiment	76	task_label	76	0	100.0000%
Psych101 Experiment	76	task_labels	76	0	100.0000%
Psych101 Experiment	76	task_ontology	76	0	100.0000%
Psych101 Experiment	76	task_ontology_evidence	76	0	100.0000%
Psych101 Experiment	76	task_families	62	14	81.5789%
Psych101 Experiment	76	task_family_id	35	41	46.0526%
Psych101 Experiment	76	task_family_label	35	41	46.0526%
Psych101 Experiment	76	task_ontology_match_field	35	41	46.0526%
Psych101 Experiment	76	task_ontology_match_method	35	41	46.0526%
Psych101 Experiment	76	task_ontology_match_score	35	41	46.0526%
Psych101 Experiment	76	task_ontology_match_text	35	41	46.0526%
Psych101 Experiment	76	task_paradigm_name	35	41	46.0526%
Psych101 Experiment	76	task_paradigms	35	41	46.0526%
Psych101 Experiment	76	task_subfamilies	35	41	46.0526%
Psych101 Experiment	76	task_subfamily_id	35	41	46.0526%

Node label	Nodes	Property	Present	Missing	Present %
Psych101 Experiment	76	task_subfamily_label	35	41	46.0526%
Psych101 Experiment	76	canonical_task_cogat_id	5	71	6.5789%
Psych101 Experiment	76	canonical_task_id	5	71	6.5789%
Psych101 Experiment	76	canonical_task_label	5	71	6.5789%
TaskAnalysis	72	bids_model_version	72	0	100.0000%
TaskAnalysis	72	convolve_input	72	0	100.0000%
TaskAnalysis	72	dataset_id	72	0	100.0000%
TaskAnalysis	72	fitlins_params_json	72	0	100.0000%
TaskAnalysis	72	group_by	72	0	100.0000%
TaskAnalysis	72	hrf_model	72	0	100.0000%
TaskAnalysis	72	id	72	0	100.0000%
TaskAnalysis	72	ingested_at	72	0	100.0000%
TaskAnalysis	72	labels	72	0	100.0000%
TaskAnalysis	72	model_name	72	0	100.0000%
TaskAnalysis	72	model_type	72	0	100.0000%
TaskAnalysis	72	n_subjects	72	0	100.0000%
TaskAnalysis	72	source	72	0	100.0000%
TaskAnalysis	72	subjects	72	0	100.0000%
TaskAnalysis	72	task	72	0	100.0000%
TaskAnalysis	72	task_metadata_json	72	0	100.0000%
TaskAnalysis	72	confounds_terms	71	1	98.6111%
TaskAnalysis	72	hrf_derivative	15	57	20.8333%
TaskAnalysis	72	primary_onvoc_confidence	10	62	13.8889%
TaskAnalysis	72	primary_onvoc_id	10	62	13.8889%
TaskAnalysis	72	hrf_dispersion	7	65	9.7222%
ReviewRule	71	description	71	0	100.0000%
ReviewRule	71	detection	71	0	100.0000%
ReviewRule	71	exemptions	71	0	100.0000%
ReviewRule	71	id	71	0	100.0000%
ReviewRule	71	implementation_rule_ids	71	0	100.0000%
ReviewRule	71	labels	71	0	100.0000%
ReviewRule	71	lifecycle_status	71	0	100.0000%
ReviewRule	71	rule_id	71	0	100.0000%
ReviewRule	71	severity	71	0	100.0000%
ReviewRule	71	novelty	3	68	4.2254%
ReviewCalibrationCase	60	case_id	60	0	100.0000%
ReviewCalibrationCase	60	id	60	0	100.0000%
ReviewCalibrationCase	60	labels	60	0	100.0000%
ReviewCalibrationCase	60	scenario	60	0	100.0000%
ReviewCalibrationCase	60	severity	60	0	100.0000%
ReviewCalibrationCase	60	novelty	4	56	6.6667%
StatMap	60	cognitive_paradigm_cogatlas	60	0	100.0000%
StatMap	60	collection_id	60	0	100.0000%
StatMap	60	collection_name	60	0	100.0000%
StatMap	60	description	60	0	100.0000%
StatMap	60	doi	60	0	100.0000%
StatMap	60	file_url	60	0	100.0000%

Node label	Nodes	Property	Present	Missing	Present %
StatMap	60	id	60	0	100.0000%
StatMap	60	labels	60	0	100.0000%
StatMap	60	map_type	60	0	100.0000%
StatMap	60	name	60	0	100.0000%
StatMap	60	source	60	0	100.0000%
StatMap	60	analysis_level	36	24	60.0000%
StatMap	60	cognitive_contrast_ cogatlas	9	51	15.0000%
Consortium	38	id	38	0	100.0000%
Consortium	38	labels	38	0	100.0000%
Consortium	38	name	38	0	100.0000%
Study	33	doi	33	0	100.0000%
Study	33	id	33	0	100.0000%
Study	33	labels	33	0	100.0000%
Study	33	title	33	0	100.0000%
Modality	28	id	28	0	100.0000%
Modality	28	name	28	0	100.0000%
Modality	28	labels	24	4	85.7143%
Parcellation	28	id	26	2	92.8571%
Parcellation	28	name	11	17	39.2857%
Parcellation	28	space	10	18	35.7143%
Parcellation	28	labels	8	20	28.5714%
Parcellation	28	modality	8	20	28.5714%
Parcellation	28	slug	8	20	28.5714%
Parcellation	28	source	8	20	28.5714%
Parcellation	28	map_files	6	22	21.4286%
Parcellation	28	space_resolution_ mm	6	22	21.4286%
Parcellation	28	space_variant	5	23	17.8571%
Parcellation	28	scale	4	24	14.2857%
Parcellation	28	variant	3	25	10.7143%
Parcellation	28	resolution_mm	2	26	7.1429%
Parcellation	28	sha256	2	26	7.1429%
Parcellation	28	source_path	2	26	7.1429%
Battery	27	collection_alias	27	0	100.0000%
Battery	27	collection_date_ introduced	27	0	100.0000%
Battery	27	collection_ description	27	0	100.0000%
Battery	27	collection_ publisher	27	0	100.0000%
Battery	27	creation_time	27	0	100.0000%
Battery	27	event_stamp	27	0	100.0000%
Battery	27	flag_for_curator	27	0	100.0000%
Battery	27	id	27	0	100.0000%
Battery	27	id_user	27	0	100.0000%
Battery	27	labels	27	0	100.0000%
Battery	27	last_updated	27	0	100.0000%
Battery	27	name	27	0	100.0000%
Battery	27	relationship	27	0	100.0000%
Battery	27	source	27	0	100.0000%
Battery	27	website	27	0	100.0000%
Parcel	26	index	26	0	100.0000%
Parcel	26	label_raw	26	0	100.0000%
Parcel	26	name	26	0	100.0000%
Parcel	26	parcellation	26	0	100.0000%
Finding	16	confidence	16	0	100.0000%
Finding	16	created_at	16	0	100.0000%

Node label	Nodes	Property	Present	Missing	Present %
Finding	16	description	16	0	100.0000%
Finding	16	source_tool	16	0	100.0000%
Species	14	id	14	0	100.0000%
Species	14	labels	14	0	100.0000%
Species	14	name	14	0	100.0000%
ReviewRuleGroup	13	id	13	0	100.0000%
ReviewRuleGroup	13	key	13	0	100.0000%
ReviewRuleGroup	13	label	13	0	100.0000%
ReviewRuleGroup	13	labels	13	0	100.0000%
ReviewRuleGroup	13	section	13	0	100.0000%
IngestionRun	12	config_hash	12	0	100.0000%
IngestionRun	12	db_name	12	0	100.0000%
IngestionRun	12	db_uri	12	0	100.0000%
IngestionRun	12	duration_sec	12	0	100.0000%
IngestionRun	12	errors	12	0	100.0000%
IngestionRun	12	finished_at	12	0	100.0000%
IngestionRun	12	git_sha	12	0	100.0000%
IngestionRun	12	id	12	0	100.0000%
IngestionRun	12	labels	12	0	100.0000%
IngestionRun	12	run_id	12	0	100.0000%
IngestionRun	12	sources	12	0	100.0000%
IngestionRun	12	started_at	12	0	100.0000%
IngestionRun	12	total_entities	12	0	100.0000%
IngestionRun	12	total_relationships	12	0	100.0000%
IngestionRun	12	neurovault_ summary	9	3	75.0000%
ConceptClass	10	creation_time	10	0	100.0000%
ConceptClass	10	description	10	0	100.0000%
ConceptClass	10	display_order	10	0	100.0000%
ConceptClass	10	id	10	0	100.0000%
ConceptClass	10	labels	10	0	100.0000%
ConceptClass	10	last_updated	10	0	100.0000%
ConceptClass	10	name	10	0	100.0000%
ConceptClass	10	relationship	10	0	100.0000%
ConceptClass	10	source	10	0	100.0000%
Process	10	creation_time	10	0	100.0000%
Process	10	description	10	0	100.0000%
Process	10	display_order	10	0	100.0000%
Process	10	id	10	0	100.0000%
Process	10	labels	10	0	100.0000%
Process	10	last_updated	10	0	100.0000%
Process	10	name	10	0	100.0000%
Process	10	relationship	10	0	100.0000%
Process	10	source	10	0	100.0000%
TemplateSpace	10	id	10	0	100.0000%
TemplateSpace	10	name	10	0	100.0000%
TemplateSpace	10	labels	8	2	80.0000%
TemplateSpace	10	source	8	2	80.0000%
TemplateSpace	10	modality	6	4	60.0000%
TemplateSpace	10	variant	5	5	50.0000%
TemplateSpace	10	resolution_mmm	4	6	40.0000%
TemplateSpace	10	atlas	1	9	10.0000%
TemplateSpace	10	atlas_slug	1	9	10.0000%
TemplateSpace	10	space	1	9	10.0000%
TemplateSpace	10	species	1	9	10.0000%
Region	8	id	8	0	100.0000%

Node label	Nodes	Property	Present	Missing	Present %
Region	8	labels	8	0	100.0000%
Region	8	name	8	0	100.0000%
Region	8	abbreviation	6	2	75.0000%
ReviewReasonTag	8	description	8	0	100.0000%
ReviewReasonTag	8	id	8	0	100.0000%
ReviewReasonTag	8	key	8	0	100.0000%
ReviewReasonTag	8	labels	8	0	100.0000%
GLMVariant	7	contrast	7	0	100.0000%
GLMVariant	7	decision_points	7	0	100.0000%
GLMVariant	7	fitlins_params	7	0	100.0000%
GLMVariant	7	id	7	0	100.0000%
GLMVariant	7	labels	7	0	100.0000%
GLMVariant	7	model_id	7	0	100.0000%
GLMVariant	7	model_x	7	0	100.0000%
GLMVariant	7	spec_path	7	0	100.0000%
GLMVariant	7	spec_sha256	7	0	100.0000%
GLMVariant	7	status	7	0	100.0000%
GLMVariant	7	literature_evidence	6	1	85.7143%
GLMVariant	7	rationale	6	1	85.7143%
GLMVariant	7	references	6	1	85.7143%
GLMVariant	7	selection_reason	6	1	85.7143%
GLMVariant	7	variant_id	6	1	85.7143%
ReviewPolicyDecision	6	decision	6	0	100.0000%
ReviewPolicyDecision	6	id	6	0	100.0000%
ReviewPolicyDecision	6	key	6	0	100.0000%
ReviewPolicyDecision	6	labels	6	0	100.0000%
ReviewPolicyDecision	6	question	6	0	100.0000%
ReviewPolicyDecision	6	rationale	6	0	100.0000%
ReviewLifecycleStatus	5	description	5	0	100.0000%
ReviewLifecycleStatus	5	id	5	0	100.0000%
ReviewLifecycleStatus	5	key	5	0	100.0000%
ReviewLifecycleStatus	5	labels	5	0	100.0000%
ReviewSensitivityTemplate	5	choice	5	0	100.0000%
ReviewSensitivityTemplate	5	id	5	0	100.0000%
ReviewSensitivityTemplate	5	key	5	0	100.0000%
ReviewSensitivityTemplate	5	labels	5	0	100.0000%
ReviewSensitivityTemplate	5	minimum_requirement	5	0	100.0000%
ReviewValidityLayer	5	id	5	0	100.0000%
ReviewValidityLayer	5	key	5	0	100.0000%
ReviewValidityLayer	5	label	5	0	100.0000%
ReviewValidityLayer	5	labels	5	0	100.0000%
ReviewValidityLayer	5	scope	5	0	100.0000%
ExecutionFailure	4	created_at	4	0	100.0000%
ExecutionFailure	4	error_category	4	0	100.0000%

Node label	Nodes	Property	Present	Missing	Present %
ExecutionFailure	4	error_message	4	0	100.0000%
ExecutionFailure	4	failure_id	4	0	100.0000%
ExecutionFailure	4	plan_id	4	0	100.0000%
ExecutionFailure	4	run_id	4	0	100.0000%
ExecutionFailure	4	step_id	4	0	100.0000%
ExecutionFailure	4	tool_id	4	0	100.0000%
ExecutionFailure	4	updated_at	4	0	100.0000%
ExecutionFailure	4	dataset_id	3	1	75.0000%
ExecutionFailure	4	task_family	3	1	75.0000%
ExecutionFailure	4	error_taxonomy	1	3	25.0000%
ExecutionFailure	4	is_retryable	1	3	25.0000%
ExecutionFailure	4	recovered	1	3	25.0000%
ExecutionFailure	4	recovery_action	1	3	25.0000%
ExecutionFailure	4	recovery_actions	1	3	25.0000%
ExecutionFailure	4	tool_version_id	1	3	25.0000%
Run	4	id	4	0	100.0000%
Atlas	2	name	2	0	100.0000%
Atlas	2	source	2	0	100.0000%
Atlas	2	atlas	1	1	50.0000%
Atlas	2	atlas_slug	1	1	50.0000%
Atlas	2	id	1	1	50.0000%
Atlas	2	labels	1	1	50.0000%
Atlas	2	modality	1	1	50.0000%
Atlas	2	short_name	1	1	50.0000%
Atlas	2	space	1	1	50.0000%
Atlas	2	species	1	1	50.0000%
ReviewSeverity	2	default_action	2	0	100.0000%
ReviewSeverity	2	description	2	0	100.0000%
ReviewSeverity	2	id	2	0	100.0000%
ReviewSeverity	2	key	2	0	100.0000%
ReviewSeverity	2	labels	2	0	100.0000%
GLMRun	1	analysis_level	1	0	100.0000%
GLMRun	1	bids_root	1	0	100.0000%
GLMRun	1	created_at	1	0	100.0000%
GLMRun	1	dataset_id	1	0	100.0000%
GLMRun	1	derivatives_root	1	0	100.0000%
GLMRun	1	execute	1	0	100.0000%
GLMRun	1	id	1	0	100.0000%
GLMRun	1	k	1	0	100.0000%
GLMRun	1	labels	1	0	100.0000%
GLMRun	1	provenance_path	1	0	100.0000%
GLMRun	1	runtime	1	0	100.0000%
GLMRun	1	seed	1	0	100.0000%
GLMRun	1	task	1	0	100.0000%
Psych101Dataset	1	dataset_id	1	0	100.0000%
Psych101Dataset	1	description	1	0	100.0000%
Psych101Dataset	1	id	1	0	100.0000%
Psych101Dataset	1	labels	1	0	100.0000%
Psych101Dataset	1	license	1	0	100.0000%
Psych101Dataset	1	n_experiments	1	0	100.0000%
Psych101Dataset	1	n_participants	1	0	100.0000%
Psych101Dataset	1	name	1	0	100.0000%
Psych101Dataset	1	provenance	1	0	100.0000%
Psych101Dataset	1	schema_version	1	0	100.0000%
Psych101Dataset	1	source	1	0	100.0000%
Psych101Dataset	1	tags	1	0	100.0000%

Node label	Nodes	Property	Present	Missing	Present %
Psych101Dataset	1	url	1	0	100.0000%
ResultSummary	1	edges_cypher	1	0	100.0000%
ResultSummary	1	edges_path	1	0	100.0000%
ResultSummary	1	id	1	0	100.0000%
ResultSummary	1	labels	1	0	100.0000%
ResultSummary	1	robustness_json	1	0	100.0000%
ResultSummary	1	robustness_md	1	0	100.0000%
ResultSummary	1	summary_path	1	0	100.0000%
ReviewImplementationRuleCatalog	1	catalog_id	1	0	100.0000%
ReviewImplementationRuleCatalog	1	id	1	0	100.0000%
ReviewImplementationRuleCatalog	1	labels	1	0	100.0000%
ReviewImplementationRuleCatalog	1	rule_count	1	0	100.0000%
ReviewImplementationRuleCatalog	1	source_path	1	0	100.0000%
ReviewImplementationRuleCatalog	1	title	1	0	100.0000%
ReviewPositiveModifier	1	description	1	0	100.0000%
ReviewPositiveModifier	1	effect	1	0	100.0000%
ReviewPositiveModifier	1	id	1	0	100.0000%
ReviewPositiveModifier	1	key	1	0	100.0000%
ReviewPositiveModifier	1	labels	1	0	100.0000%
ReviewPositiveModifier	1	metadata_fields	1	0	100.0000%
ReviewRuleRegistry	1	document_version	1	0	100.0000%
ReviewRuleRegistry	1	execution_boundary	1	0	100.0000%
ReviewRuleRegistry	1	id	1	0	100.0000%
ReviewRuleRegistry	1	labels	1	0	100.0000%
ReviewRuleRegistry	1	registry_id	1	0	100.0000%
ReviewRuleRegistry	1	scope	1	0	100.0000%
ReviewRuleRegistry	1	source_document_type	1	0	100.0000%
ReviewRuleRegistry	1	title	1	0	100.0000%
ReviewRuleRegistry	1	version	1	0	100.0000%
ToolEvidence	1	created_at	1	0	100.0000%
ToolEvidence	1	dataset_family	1	0	100.0000%
ToolEvidence	1	dataset_id	1	0	100.0000%
ToolEvidence	1	fail_count	1	0	100.0000%
ToolEvidence	1	failure_categories	1	0	100.0000%
ToolEvidence	1	latency_ms_samples	1	0	100.0000%
ToolEvidence	1	success_count	1	0	100.0000%
ToolEvidence	1	task_family	1	0	100.0000%
ToolEvidence	1	tool_id	1	0	100.0000%
ToolEvidence	1	tool_version	1	0	100.0000%
ToolEvidence	1	updated_at	1	0	100.0000%

B23. Relationship property coverage by relationship type and property

Relationship type	Edges	Property	Present	Missing	Present %
BELONGS_TO	1,077,573	source	1,077,356	217	99.9799%
HAS_COORDINATE	447,499	source	447,499	0	100.0000%
HAS_TERM	358,050	rank	358,050	0	100.0000%
HAS_TERM	358,050	section	358,050	0	100.0000%
HAS_TERM	358,050	source	358,050	0	100.0000%
HAS_TERM	358,050	weight	358,050	0	100.0000%
IN_REGION	121,261	atlas	121,261	0	100.0000%
IN_REGION	121,261	edge_source	121,261	0	100.0000%
IN_REGION	121,261	etl_version	121,261	0	100.0000%
IN_REGION	121,261	measure	121,261	0	100.0000%
IN_REGION	121,261	n_vox	121,261	0	100.0000%
IN_REGION	121,261	pct_active	121,261	0	100.0000%
IN_REGION	121,261	weight	121,261	0	100.0000%
IN_REGION	121,261	z_thr	121,261	0	100.0000%
ABOUT	75,922	confidence	75,922	0	100.0000%
ABOUT	75,922	created_at	75,922	0	100.0000%
ABOUT	75,922	source	75,922	0	100.0000%
ABOUT	75,922	match_terms	75,888	34	99.9552%
ABOUT	75,922	path_modes	1,025	74,897	1.3501%
ABOUT	75,922	path_support	1,025	74,897	1.3501%
ABOUT	75,922	updated_at	1,025	74,897	1.3501%
ABOUT	75,922	confidence_tier	34	75,888	0.0448%
ABOUT	75,922	method	34	75,888	0.0448%
IN_ONVOC	63,160	confidence	63,160	0	100.0000%
IN_ONVOC	63,160	source	63,160	0	100.0000%
IN_ONVOC	63,160	evidence_json	34,445	28,715	54.5361%
IN_ONVOC	63,160	method	34,445	28,715	54.5361%
IN_ONVOC	63,160	created_at	28,715	34,445	45.4639%
IN_ONVOC	63,160	match_terms	28,715	34,445	45.4639%
IN_DOMAIN	52,117	confidence	52,117	0	100.0000%
IN_DOMAIN	52,117	method	52,117	0	100.0000%
IN_DOMAIN	52,117	source	52,117	0	100.0000%
IN_SPACE	35,199	source	35,189	10	99.9716%
IN_SPACE	35,199	atlas	1	35,198	0.0028%
COMPUTED_WITH	30,880	computed_at	30,880	0	100.0000%
COMPUTED_WITH	30,880	confidence	30,880	0	100.0000%
COMPUTED_WITH	30,880	confidence_components	30,880	0	100.0000%
COMPUTED_WITH	30,880	confidence_version	30,880	0	100.0000%
COMPUTED_WITH	30,880	evidence_type	30,880	0	100.0000%
COMPUTED_WITH	30,880	evidence_type_diversity	30,880	0	100.0000%
COMPUTED_WITH	30,880	prov_base_conf	30,880	0	100.0000%
COMPUTED_WITH	30,880	prov_source	30,880	0	100.0000%
COMPUTED_WITH	30,880	source	30,880	0	100.0000%
COMPUTED_WITH	30,880	source_diversity	30,880	0	100.0000%
COMPUTED_WITH	30,880	support_count_raw	30,880	0	100.0000%

Relationship type	Edges	Property	Present	Missing	Present %
COMPUTED_WITH	30,880	support_count_unique	30,880	0	100.0000%
GENERATED_FROM	30,880	computed_at	30,880	0	100.0000%
GENERATED_FROM	30,880	confidence	30,880	0	100.0000%
GENERATED_FROM	30,880	confidence_components	30,880	0	100.0000%
GENERATED_FROM	30,880	confidence_version	30,880	0	100.0000%
GENERATED_FROM	30,880	evidence_type	30,880	0	100.0000%
GENERATED_FROM	30,880	evidence_type_diversity	30,880	0	100.0000%
GENERATED_FROM	30,880	prov_base_conf	30,880	0	100.0000%
GENERATED_FROM	30,880	prov_source	30,880	0	100.0000%
GENERATED_FROM	30,880	source	30,880	0	100.0000%
GENERATED_FROM	30,880	source_diversity	30,880	0	100.0000%
GENERATED_FROM	30,880	support_count_raw	30,880	0	100.0000%
GENERATED_FROM	30,880	support_count_unique	30,880	0	100.0000%
DERIVED_FROM	30,163	computed_at	30,163	0	100.0000%
DERIVED_FROM	30,163	confidence	30,163	0	100.0000%
DERIVED_FROM	30,163	confidence_components	30,163	0	100.0000%
DERIVED_FROM	30,163	confidence_version	30,163	0	100.0000%
DERIVED_FROM	30,163	evidence_type	30,163	0	100.0000%
DERIVED_FROM	30,163	evidence_type_diversity	30,163	0	100.0000%
DERIVED_FROM	30,163	prov_base_conf	30,163	0	100.0000%
DERIVED_FROM	30,163	source_diversity	30,163	0	100.0000%
DERIVED_FROM	30,163	support_count_raw	30,163	0	100.0000%
DERIVED_FROM	30,163	support_count_unique	30,163	0	100.0000%
DERIVED_FROM	30,163	prov_source	30,120	43	99.8574%
DERIVED_FROM	30,163	source	30,120	43	99.8574%
DERIVED_FROM	30,163	method	43	30,120	0.1426%
DERIVED_FROM	30,163	prov_method	43	30,120	0.1426%
DERIVED_FROM	30,163	provenance	43	30,120	0.1426%
MAPS_TO	17,667	confidence	17,667	0	100.0000%
MAPS_TO	17,667	computed_at	17,652	15	99.9151%
MAPS_TO	17,667	evidence_type	17,652	15	99.9151%
MAPS_TO	17,667	evidence_type_diversity	17,652	15	99.9151%
MAPS_TO	17,667	prov_base_conf	17,652	15	99.9151%
MAPS_TO	17,667	source_diversity	17,652	15	99.9151%
MAPS_TO	17,667	support_count_raw	17,652	15	99.9151%
MAPS_TO	17,667	support_count_unique	17,652	15	99.9151%
MAPS_TO	17,667	confidence_components	12,020	5,647	68.0365%
MAPS_TO	17,667	confidence_version	12,020	5,647	68.0365%
MAPS_TO	17,667	prov_source	9,368	8,299	53.0254%
MAPS_TO	17,667	source	9,180	8,487	51.9613%
MAPS_TO	17,667	canonical_id	9,158	8,509	51.8368%
MAPS_TO	17,667	match_method	9,158	8,509	51.8368%
MAPS_TO	17,667	prov_method	8,509	9,158	48.1632%

Relationship type	Edges	Property	Present	Missing	Present %
MAPS.TO	17,667	method	8,306	9,361	47.0142%
MAPS.TO	17,667	created_at	8,284	9,383	46.8897%
MAPS.TO	17,667	created_by	8,284	9,383	46.8897%
MAPS.TO	17,667	source_load	8,284	9,383	46.8897%
MAPS.TO	17,667	match_config_hash	7,917	9,750	44.8124%
MAPS.TO	17,667	match_profile	7,917	9,750	44.8124%
MAPS.TO	17,667	match_version	7,917	9,750	44.8124%
MAPS.TO	17,667	strategy	7,606	10,061	43.0520%
MAPS.TO	17,667	timestamp	7,606	10,061	43.0520%
MAPS.TO	17,667	match_provenance	4,219	13,448	23.8807%
MAPS.TO	17,667	link_type	678	16,989	3.8377%
MAPS.TO	17,667	normalized_name	203	17,464	1.1490%
MAPS.TO	17,667	canonical_label	15	17,652	0.0849%
MAPS.TO	17,667	experiment_id	15	17,652	0.0849%
MAPS.TO	17,667	parameters.json	7	17,660	0.0396%
HAS.RESOURCE	14,728	source	14,728	0	100.0000%
HAS.TEXT_EMBEDDING	12,958	kind	12,958	0	100.0000%
HAS.TEXT_EMBEDDING	12,958	model	12,958	0	100.0000%
HAS.TEXT_EMBEDDING	12,958	text_section	12,958	0	100.0000%
MEASURES	10,875	computed_at	10,875	0	100.0000%
MEASURES	10,875	confidence	10,875	0	100.0000%
MEASURES	10,875	confidence_components	10,875	0	100.0000%
MEASURES	10,875	confidence_version	10,875	0	100.0000%
MEASURES	10,875	evidence.type	10,875	0	100.0000%
MEASURES	10,875	evidence.type_diversity	10,875	0	100.0000%
MEASURES	10,875	prov_base.conf	10,875	0	100.0000%
MEASURES	10,875	prov_source	10,875	0	100.0000%
MEASURES	10,875	source	10,875	0	100.0000%
MEASURES	10,875	source_diversity	10,875	0	100.0000%
MEASURES	10,875	support_count_raw	10,875	0	100.0000%
MEASURES	10,875	support_count_unique	10,875	0	100.0000%
MEASURES	10,875	method	5,961	4,914	54.8138%
MEASURES	10,875	prov_method	5,961	4,914	54.8138%
MEASURES	10,875	relationship	1,422	9,453	13.0759%
MEASURES	10,875	contrasts	1,218	9,657	11.2000%
MEASURES	10,875	timestamp	8	10,867	0.0736%
IMPLEMENTS_FAMILY	4,294	computed_at	2,127	2,167	49.5342%
IMPLEMENTS_FAMILY	4,294	confidence	2,127	2,167	49.5342%
IMPLEMENTS_FAMILY	4,294	confidence_components	2,127	2,167	49.5342%
IMPLEMENTS_FAMILY	4,294	confidence_version	2,127	2,167	49.5342%
IMPLEMENTS_FAMILY	4,294	evidence.type	2,127	2,167	49.5342%
IMPLEMENTS_FAMILY	4,294	evidence.type_diversity	2,127	2,167	49.5342%
IMPLEMENTS_FAMILY	4,294	prov_base.conf	2,127	2,167	49.5342%
IMPLEMENTS_FAMILY	4,294	source_diversity	2,127	2,167	49.5342%
IMPLEMENTS_FAMILY	4,294	support_count_raw	2,127	2,167	49.5342%
IMPLEMENTS_FAMILY	4,294	support_count_unique	2,127	2,167	49.5342%

Relationship type	Edges	Property	Present	Missing	Present %
HAS.VERSION	4,142	(no properties observed)	0	4,142	0.0000%
CITES	3,397	source	3,396	1	99.9706%
CITES	3,397	timestamp	1	3,396	0.0294%
SUPPORTS_MODALITY	2,655	(no properties observed)	0	2,655	0.0000%
SUGGESTS_MEASURES	2,514	computed_at	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	confidence	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	confidence_components	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	confidence_version	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	evidence_type	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	evidence_type_diversity	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	method	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	prov_base_conf	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	prov_method	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	prov_source	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	source	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	source_diversity	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	support_count_raw	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	support_count_unique	2,514	0	100.0000%
SUGGESTS_MEASURES	2,514	canonical_id	612	1,902	24.3437%
SUGGESTS_MEASURES	2,514	evidence_json	612	1,902	24.3437%
SUGGESTS_MEASURES	2,514	match_label	612	1,902	24.3437%
HAS.CONTRAST	2,464	source	1,978	486	80.2760%
HAS.CONTRAST	2,464	task	486	1,978	19.7240%
HAS.REGION	2,123	source	2,123	0	100.0000%
HAS.REGION	2,123	atlas	1,327	796	62.5059%
PARTICIPATES_IN	1,982	source	1,982	0	100.0000%
HAS.MODALITY	1,839	(no properties observed)	0	1,839	0.0000%
HOSTED.AT	1,694	(no properties observed)	0	1,694	0.0000%
HAS.TASK	1,679	computed_at	1,679	0	100.0000%
HAS.TASK	1,679	confidence	1,679	0	100.0000%
HAS.TASK	1,679	confidence_components	1,679	0	100.0000%
HAS.TASK	1,679	confidence_version	1,679	0	100.0000%
HAS.TASK	1,679	evidence_type	1,679	0	100.0000%
HAS.TASK	1,679	evidence_type_diversity	1,679	0	100.0000%
HAS.TASK	1,679	prov_base_conf	1,679	0	100.0000%
HAS.TASK	1,679	source_diversity	1,679	0	100.0000%
HAS.TASK	1,679	support_count_raw	1,679	0	100.0000%
HAS.TASK	1,679	support_count_unique	1,679	0	100.0000%
HAS.TASK	1,679	prov_source	1,641	38	97.7367%
HAS.TASK	1,679	mapping_method	1,569	110	93.4485%

Relationship type	Edges	Property	Present	Missing	Present %
HAS_TASK	1,679	mapping_version	1,569	110	93.4485%
HAS_TASK	1,679	match_score	1,569	110	93.4485%
HAS_TASK	1,679	normalized_task	1,569	110	93.4485%
HAS_TASK	1,679	prov_method	1,569	110	93.4485%
HAS_TASK	1,679	raw_task	1,569	110	93.4485%
HAS_TASK	1,679	needs_measures	1,241	438	73.9130%
HAS_TASK	1,679	source	72	1,607	4.2883%
HAS_TASK	1,679	task_name	38	1,641	2.2633%
PART_OF	1,425	atlas	1,326	99	93.0526%
PART_OF	1,425	hierarchy_type	1,326	99	93.0526%
PART_OF	1,425	source	1,326	99	93.0526%
HASCITATION	1,293	source	1,293	0	100.0000%
ASSERTS	1,237	source	1,237	0	100.0000%
ASSERTS	1,237	contrasts	1,033	204	83.5085%
HAS_PHENOTYPE	1,218	aggregation_level	1,218	0	100.0000%
HAS_PHENOTYPE	1,218	source	1,218	0	100.0000%
MEASUREDBY	1,218	contrast_id	1,218	0	100.0000%
MEASUREDBY	1,218	name	1,218	0	100.0000%
MEASUREDBY	1,218	source	1,218	0	100.0000%
INVOLVES_SPECIES	990	(no properties observed)	0	990	0.0000%
CLASSIFIED_UNDER	863	source	863	0	100.0000%
CLASSIFIED_UNDER	863	relation	765	98	88.6443%
CLASSIFIED_UNDER	863	scheme	765	98	88.6443%
CLASSIFIED_UNDER	863	confidence	98	765	11.3557%
CLASSIFIED_UNDER	863	ontology_match_method	61	802	7.0684%
CLASSIFIED_UNDER	863	ontology_match_score	61	802	7.0684%
CLASSIFIED_UNDER	863	subfamily_id	35	828	4.0556%
CLASSIFIED_UNDER	863	subfamily_label	35	828	4.0556%
HASCONDITION	807	source	807	0	100.0000%
USES_CONDITION	750	order	750	0	100.0000%
USES_CONDITION	750	source	750	0	100.0000%
USES_CONDITION	750	weight	750	0	100.0000%
HASINDICATOR	584	source	584	0	100.0000%
CLASSIFIEDUNDER	523	source	523	0	100.0000%
KINDOF	429	definition_text	429	0	100.0000%
KINDOF	429	source	429	0	100.0000%
KINDOF	429	alias	46	383	10.7226%
INCLUDES	402	source	402	0	100.0000%
PRODUCES_RESOURCE	362	(no properties observed)	0	362	0.0000%
CONSUMES_RESOURCE	338	(no properties observed)	0	338	0.0000%
HAS_CONDITION	310	order	310	0	100.0000%
HAS_CONDITION	310	source	310	0	100.0000%
AFFILIATED_WITH	292	source	292	0	100.0000%
PARTOF	263	definition_text	263	0	100.0000%

Relationship type	Edges	Property	Present	Missing	Present %
PARTOF	263	source	263	0	100.0000%
PARTOF	263	alias	39	224	14.8289%
AUTHORED_BY	232	source	232	0	100.0000%
HAS_GLM_PRIOR	224	scope	224	0	100.0000%
HAS_GLM_PRIOR	224	dataset_id	78	146	34.8214%
RELATED_TO	190	computed_at	190	0	100.0000%
RELATED_TO	190	confidence	190	0	100.0000%
RELATED_TO	190	confidence_components	190	0	100.0000%
RELATED_TO	190	confidence_version	190	0	100.0000%
RELATED_TO	190	evidence_type	190	0	100.0000%
RELATED_TO	190	evidence_type_diversity	190	0	100.0000%
RELATED_TO	190	prov_base_conf	190	0	100.0000%
RELATED_TO	190	prov_source	190	0	100.0000%
RELATED_TO	190	source	190	0	100.0000%
RELATED_TO	190	source_diversity	190	0	100.0000%
RELATED_TO	190	support_count_raw	190	0	100.0000%
RELATED_TO	190	support_count_unique	190	0	100.0000%
RELATED_TO	190	contrasts	185	5	97.3684%
RELATED_TO	190	timestamp	5	185	2.6316%
BELONGS_TO_FAMILY	176	source	176	0	100.0000%
BELONGS_TO_FAMILY	176	subfamily_id	176	0	100.0000%
BELONGS_TO_FAMILY	176	subfamily_label	176	0	100.0000%
BELONGS_TO_FAMILY	176	match_method	126	50	71.5909%
BELONGS_TO_FAMILY	176	match_score	126	50	71.5909%
BELONGS_TO_FAMILY	176	paradigm_name	126	50	71.5909%
BELONGS_TO_FAMILY	176	confidence	50	126	28.4091%
REQUIRES_FIELD	163	(no properties observed)	0	163	0.0000%
USES_TASK	152	confidence	152	0	100.0000%
USES_TASK	152	source	152	0	100.0000%
USES_TASK	152	computed_at	1	151	0.6579%
USES_TASK	152	confidence_components	1	151	0.6579%
USES_TASK	152	confidence_version	1	151	0.6579%
USES_TASK	152	evidence_type	1	151	0.6579%
USES_TASK	152	evidence_type_diversity	1	151	0.6579%
USES_TASK	152	prov_base_conf	1	151	0.6579%
USES_TASK	152	prov_source	1	151	0.6579%
USES_TASK	152	source_diversity	1	151	0.6579%
USES_TASK	152	support_count_raw	1	151	0.6579%
USES_TASK	152	support_count_unique	1	151	0.6579%
USES_TASK	152	timestamp	1	151	0.6579%
INBATTERY	134	source	134	0	100.0000%
HAS_VALIDITY_LAYER	102	(no properties observed)	0	102	0.0000%
HAS_REASON_TAG	81	(no properties observed)	0	81	0.0000%

Relationship type	Edges	Property	Present	Missing	Present %
CONTAINS_IMPLEMENTATION_RULE	80	implementation_rule_id	80	0	100.0000%
HAS_EXPERIMENT	76	confidence	76	0	100.0000%
HAS_EXPERIMENT	76	source	76	0	100.0000%
DESCRIBES_TASK	72	source	72	0	100.0000%
CONTAINS_RULE	71	(no properties observed)	0	71	0.0000%
HAS_LIFECYCLE_STATUS	71	(no properties observed)	0	71	0.0000%
HAS_SEVERITY	71	(no properties observed)	0	71	0.0000%
IN_RULE_GROUP	71	(no properties observed)	0	71	0.0000%
CONTAINS_CALIBRATION_CASE	60	(no properties observed)	0	60	0.0000%
CALIBRATES_RULE	55	(no properties observed)	0	55	0.0000%
CONTRAST_OF	53	confidence	53	0	100.0000%
CONTRAST_OF	53	method	53	0	100.0000%
CONTRAST_OF	53	source	53	0	100.0000%
CITED_BY	49	source	49	0	100.0000%
MAPPED_TO_IMPLEMENTATION	28	implementation_rule_id	28	0	100.0000%
HAS_PARCEL	26	(no properties observed)	0	26	0.0000%
IN_PARCELLATION	17	(no properties observed)	0	17	0.0000%
VALIDATED_ON	13	(no properties observed)	0	13	0.0000%
HAD_FAILURE	7	(no properties observed)	0	7	0.0000%
HAS_VARIANT	7	(no properties observed)	0	7	0.0000%
ACTIVATES	6	source	6	0	100.0000%
ACTIVATES	6	timestamp	6	0	100.0000%
ACTIVATES	6	confidence	5	1	83.3333%
ACTIVATES	6	source_id	2	4	33.3333%
HAS_POLICY_DECISION	6	(no properties observed)	0	6	0.0000%
TRIGGERS_SENSITIVITY	5	(no properties observed)	0	5	0.0000%
FAILED_ON	3	error_category	3	0	100.0000%
FAILED_ON	3	fail_count	3	0	100.0000%
FAILED_ON	3	last_run_id	3	0	100.0000%
FAILED_ON	3	last_seen	3	0	100.0000%
FAILED_ON	3	task_family	3	0	100.0000%
DOCUMENTED_IN	2	(no properties observed)	0	2	0.0000%
HAS_GLM_RUN	2	(no properties observed)	0	2	0.0000%
HAS_PARCELLATION	2	(no properties observed)	0	2	0.0000%
CALIBRATES_MODIFIER	1	(no properties observed)	0	1	0.0000%
CONTAINS_MODIFIER	1	(no properties observed)	0	1	0.0000%

Relationship type	Edges	Property	Present	Missing	Present %
HAS_EVIDENCE	1	(no properties observed)	0	1	0.0000%
HAS_SUMMARY	1	(no properties observed)	0	1	0.0000%
LOCATED_IN	1	source	1	0	100.0000%
MEASURED_BY	1	source	1	0	100.0000%
MENTIONS_CONCEPT	1	source	1	0	100.0000%
SIMILAR_TO	1	confidence	1	0	100.0000%
SIMILAR_TO	1	source	1	0	100.0000%
SIMILAR_TO	1	timestamp	1	0	100.0000%

B24. Node Type Inventory and Edge Type Inventory

Field	Required content
Node type / label	Exact Neo4j label or canonical exported type.
Definition	One-sentence semantic definition.
Typical instances	2-3 concrete examples from the graph.
Source(s)	Upstream source or loader family.
Primary identifier	Internal ID, external CURIE, UUID, or composite key.
Key properties	Human-readable label, source IDs, dates, coordinates, ontology IDs, etc.
Required vs optional properties	Separate required contract from enrichment fields.
Missingness	Count and percentage missing for each important property.
Multi-label behavior	Whether the label appears alone or in label sets such as Dataset OpenNeuro.
Current count	Latest refreshed count plus snapshot date.
Evidence path	Query, CSV, test, or loader path used to verify the row.

Field	Required content
Edge type	Exact Neo4j relationship type.
Semantics	What the relationship asserts.
Directionality	Source type, target type, and whether inverse traversal is meaningful.
Weighting	Whether edge has weight, strength, score, or count fields.
Evidence	Whether edge carries source paper, source dataset, extraction method, or confidence.
Construction method	Rule, structured parse, statistical association, spatial overlap, ontology mapping, LLM extraction, manual curation, or runtime provenance.
Required properties	Minimal property contract.
Optional properties	Enrichment fields and provenance fields.
Count	Latest refreshed edge count.
Dominant schema triples	Major source-label / target-label pairs for the edge.
Caveats	Direction ambiguity, duplicate variants, weak semantics, or legacy naming.
Evidence path	Query, CSV, test, or loader path used to verify the row.

B25. Property Schema

Object type	Property	Required?	Data type	Example	Missing count	Missing percent	Source of truth
Publication	pmid	optional in live graph	string	PMID value	16,871 / 49,744	33.92%	Live property profile, 2026-05-03
Publication	doi	optional in live graph	string	DOI value	1,245 / 49,744	2.50%	Live property profile, 2026-05-03
Publication	title	optional in live graph	string	paper title	2,702 / 49,744	5.43%	Live property profile, 2026-05-03
Coordinate	x, y, z	required in live graph	number	MNI coordinate	0 / 447,499 for each field	0.00%	Live property profile, 2026-05-03
Coordinate	space	required in live graph	string	template space	0 / 447,499	0.00%	Live property profile, 2026-05-03
edge HAS-COORDINATE	source	required in live graph	string	source dataset string	0 / 447,499	0.00%	Live relationship-property profile, 2026-05-03
edge HAS-COORDINATE	source_paper	not present in live profile	string	source paper pointer	447,499 / 447,499	100.00%	Live relationship-property profile, 2026-05-03

B26. Identifier Schema

Entity class	Primary key	Secondary IDs	Merge key	Collision handling	Evidence path
Publication	id present on 49,580 / 49,744 nodes	DOI present on 48,499; PMID present on 32,873; PMCID not measured as a live property in this pass	DOI/PMID/title/source profile	Missing id: 164 nodes; missing DOI: 1,245; missing PMID: 16,871	Live property profile, 2026-05-03
Coordinate	id present on 447,499 / 447,499 nodes	MNI x, y, z, space each present on 447,499 / 447,499	Coordinate ID plus exact spatial fields	No missing id, x, y, z, or space in live profile	Live property profile, 2026-05-03
Task	id and name present on 34,926 / 34,926 nodes	aliases present on 34,870; source present on 34,914; text/behavior embeddings each present on 44	Task ID/name plus aliases and source	Missing aliases: 56; missing source: 12; embedded Task coverage: 44 / 34,926	Live property profile, 2026-05-03

B27. Ontology Cross-Reference

Ontology	Covered entity types	Link property or edge	Coverage numerator	Coverage denominator	Coverage percent	Caveats
UBERON	anatomy / regions	no live count measured	0 measured	2,140 BrainRegion nodes as denominator candidate	not measured	No UBERON source/property coverage query was found in this pass
Cognitive Atlas	tasks / concepts	source-like node counts and task/concept labels	4,214 cognitive_atlas nodes; 1,772 cognitive_atlas_niclip; 645 cognitive_atlas_cao	34,926 Task; 2,336 Concept	source-count only, not ontology-coverage percent	Live source and label counts, 2026-05-03

Ontology	Covered entity types	Link property or edge	Coverage numerator	Coverage denominator	Coverage percent	Caveats
MeSH	papers / diseases / terms	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass
NIFSTD	neuro concepts / anatomy	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass
ChEBI	drugs / chemicals	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass
HGNC	genes	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass
DOID	disorders / diseases	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass	not measured in current numeric pass
ONVOC	task / concept normalization	IN_ONVOC edges and ONVOC labels	63,160 IN_ONVOC edges; 752 OnvocClass; 752 Ontology-Concept	relevant Task/Concept/StatsMap/Contrast/Dataset surfaces	edge count, not full source coverage denominator	Live relationship and label counts, 2026-05-03

B28. Source Inventory

Source	Entity/edge types contributed	Snap-shot date	Version	URL	License	Required citation	Loader path	Data artifact path	Status
Neurosynth	464,946 nodes with source=neurosynth; 447,498 edges with source=neurosynth_v7; 358,050 edges with source=neurosynth	snap-shot date not stored in live source fields	source string only	up-stream URL not measured in current numeric pass	license not measured in current numeric pass	citation not measured in current numeric pass	loader path not measured in current numeric pass	data artifact path not measured in current numeric pass	live graph source counts verified
NeuroVault	37,467 nodes with source=neurovault; 1,076,839 edges with source=neurovault	snap-shot date not stored in live source fields	source string only	up-stream URL not measured in current numeric pass	license not measured in current numeric pass	citation not measured in current numeric pass	loader path not measured in current numeric pass	data artifact path not measured in current numeric pass	live graph source counts verified
Cognitive Atlas	4,214 nodes with source=cognitive_atlas; 14,591 edges with source=cognitive_atlas; plus 1,772 cognitive_atlas_niclip nodes and 645 cognitive_atlas_cao nodes	snap-shot date not stored in live source fields	source string only	up-stream URL not measured in current numeric pass	license not measured in current numeric pass	citation not measured in current numeric pass	loader path not measured in current numeric pass	data artifact path not measured in current numeric pass	live graph source counts verified
BrainMap	0 live nodes/edges under brainmap	not present in live source scan	configured spine source	up-stream URL not measured in current numeric pass	license likely requires explicit check before redistribution	citation not measured in current numeric pass	gwas not relevant; BrainMap loader status not measured in this pass	data artifact path not measured in current numeric pass	configured but absent from live scan

Source	Entity/ edge types contributed	Snap- shot date	Version	URL	License	Required citation	Loader path	Data artifact path	Status
BIDS	0 live nodes/ edges under bids; BIDS- derived GLM fields appear under openneuro_ glmfitlins	not present as literal source	configured spine source	up- stream URL not mea- sured in cur- rent nu- meric pass	license inher- ited from datasets; not mea- sured	citation not mea- sured in current numeric pass	BIDS/OpenNeuro parser path not fully mapped in this pass	artifact path not measured in current numeric pass	represented in- directly through OpenNeuro/ GLMFitLins lanes
Virtual Brain	0 live nodes/ edges under virtual_brain	not present in live source scan	configured spine source	up- stream URL not mea- sured in cur- rent nu- meric pass	license not mea- sured in current numeric pass	citation not mea- sured in current numeric pass	loader path not measured in current numeric pass	artifact path not measured in current numeric pass	configured but absent from live scan
Wikidata	0 live nodes/ edges under wikidata	not present in live source scan	configured spine source	up- stream URL not mea- sured in cur- rent nu- meric pass	license not mea- sured in current numeric pass	citation not mea- sured in current numeric pass	src/brain_ researcher/services/ BR-KG/etl/loaders/ wikidata_loader.py; wikidata_json_ loader.py exist	artifact path not measured in current numeric pass	configured but absent from live scan
NIDM results	0 live nodes/ edges under nidm_results	on-de mand config- ured source	configured on- demand source	up- stream URL not mea- sured in cur- rent nu- meric pass	license not mea- sured in current numeric pass	citation not mea- sured in current numeric pass	src/brain_ researcher/services/ BR-KG/etl/loaders/ nidm_results_ loader.py exists	artifact path not measured in current numeric pass	adapter/source configured but no persisted live source value
Neuro- Query	0 live nodes/ edges under neuroquery	on-de mand config- ured source	configured on- demand source	up- stream URL not mea- sured in cur- rent nu- meric pass	license not mea- sured in current numeric pass	citation not mea- sured in current numeric pass	on-demand adapter config- ured in configs/ BR-KG/ingestion_ modes.yaml	sample path con- figured as data/ BR-KG/raw/ evidence/neuroquery_ sample.json	adapter/source configured but no persisted live source value
NiMARE	0 live nodes/ edges under nimare	on-de mand config- ured source	configured on- demand source	up- stream URL not mea- sured in cur- rent nu- meric pass	license not mea- sured in current numeric pass	citation not mea- sured in current numeric pass	on-demand adapter config- ured in configs/ BR-KG/ingestion_ modes.yaml	sample path config- ured as data/BR-KG/ raw/evidence/nimare_ sample.json	adapter/source configured but no persisted live source value
Neu- roScout	0 live nodes/ edges under neuroscout	on-de mand config- ured source	configured on- demand source	up- stream URL not mea- sured in cur- rent nu- meric pass	license not mea- sured in current numeric pass	citation not mea- sured in current numeric pass	on-demand adapter config- ured in configs/ BR-KG/ingestion_ modes.yaml	sample path con- figured as data/ BR-KG/raw/ evidence/neuroscout_ features.json	adapter/source configured but no persisted live source value
Allen HBA	no literal allen_hba; Allen Brain Atlas has 1,329 nodes / 1,328 edges and Allen CCFv3 has 1,326 edges	source alias mis- match	configured spine source represented by source aliases	up- stream URL not mea- sured in cur- rent nu- meric pass	license not mea- sured in current numeric pass	citation not mea- sured in current numeric pass	src/brain_ researcher/services/ BR-KG/etl/loaders/ allen_hba_adapter.py not in loader list; allen_hba_adapter/ source config present	data/BR-KG/ raw/allen_hba/ manifest.json exists	present via aliases; normalize naming

Source	Entity/ edge types contributed	Snap- shot date	Version	URL	License	Required citation	Loader path	Data artifact path	Status
Unattributed / missing source	<missing> has 6,889 nodes and 41,393 edges	no source string	provenance gap, not an upstream source	not applicable	not releasable without audit	not applicable	audit by label/type required	audit artifact not yet created	release blocker until attributed/excluded/documented
Internal/generated/test lanes	includes config.text.backfill, capabilities, merged.yaml, taxonomy_rule, taxonomy_surface_rules, task_families, task_family_enrichment, disease_path.backfill, seed, bulk_loader, GraphQL API, Manual, Test*	mixed source-like strings	internal/generated/manual/test, not external upstream	not applicable	not upstream license; derive from underlying sources and local policy	not applicable	see release-ready provenance gap register	see release-ready provenance gap register	must be classified before public release
Gabriel / KGGen	32 Gabriel manifests, 12 Gabriel review queues, 75 KGGen summaries, 81 KGGen JSONL files; 0 live source-marked nodes/edges	raw/run artifacts but no live source marker	LLM/candidate/comparison lanes	not applicable	release depends on candidate/accepted status and upstream paper licenses	citation/provenance not measured in current numeric pass	docs/BR-KG/gabriel.full.pipeline.md; data/BR-KG/raw/gabriel; data/BR-KG/raw/kggen	repo artifacts present	confirm not ingested vs candidate-only vs unmarked ingestion

B29. Provenance Granularity

Edge type	Edge-level provenance?	Node-level provenance only?	Source row trace?	Paper/ dataset trace?	Confidence trace?	Missingness	Evidence path
HAS.COORDINATE	yes: source present on 447,499 / 447,499 edges	no	source property only	indirect via Publication -> Coordinate edge; no separate source-paper property	no confidence; no weight	0 missing source; 447,499 missing source-paper	Live relationship-property profile, 2026-05-03
IN_REGION	yes: edge_source, atlas, etl_version, measure, n_vox, pct_active, weight, and z_thr each present on 121,261 / 121,261 edges	no	source/method properties only	no separate paper/dataset property measured	weight present on 121,261 / 121,261; no confidence property measured	0 missing for the listed properties	Live relationship-property profile, 2026-05-03
ABOUT	yes: source, confidence, created_at present on 75,922 / 75,922 edges	no	source property plus match terms	source entity -> concept; no source-paper property measured	confidence present on 75,922 / 75,922; confidence_tier present on 34 / 75,922	missing match_terms: 34; missing confidence_tier: 75,888	Live relationship-property profile, 2026-05-03

Appendix C. Dataset/resource and readiness card

Purpose

Appendix C should record the dataset-level and asset-level resources used or ruled out in an episode. It answers whether a conceptually suitable dataset was actually reachable, authorized, complete enough, and compatible with the selected workflow. It documents dataset identity, access class, BIDS and derivative roots, phenotype and covariate availability, target variables, feature assets, missing resources, backend reachability, readiness verdicts, blockers, and non-executable candidates.

C2. Dataset identity and access class

Field	Example / value
Dataset ID	HCP-YA
Aliases	Human Connectome Project Young Adult
Episode ID	ep_hcp_predict_001
Dataset role	primary / candidate / rejected / validation
Modality	resting-state fMRI
Data type	FC matrix / time series / statistical map / phenotype
Access class	public / controlled / local / collaborator-provided
Local availability	yes / no / partial
Remote source	OpenNeuro / HCP / local lab / collaborator
Governance note	DUA / license / restricted / internal only

C3. Dataset-level readiness

Readiness field	Status	Notes
dataset record exists	pass / warn / block	Found in registry, BR-KG, catalogue, or local resource card.
access authorized	pass / warn / block	DUA, license, local access, or collaborator permission.
local path reachable	pass / block	Path exists or is unavailable.
BIDS root available	pass / warn / n/a	Raw BIDS available or not required.
derivative root available	pass / block	Required derivatives exist or are missing.
phenotype manifest available	pass / block	Target and covariate fields checked.
backend reachable	pass / warn / block	Local, Neurodesk, container, or Slurm compatibility.

C4. Asset-level ledger

Asset class	Required?	Found?	Path / manifest	Status
feature matrix	yes	yes/no	redacted path or hash	pass/block
target variable	yes	yes/no	phenotype manifest	pass/block
covariates	yes/optional	yes/partial/no	age, sex, motion, site	pass/warn/block
fold manifest	yes for prediction	yes/no	fold_manifest.tsv	pass/block
mask / atlas	yes if used	yes/no	Schaefer-100 x 7 or other atlas	pass/block
connectivity matrix	yes if FC/FNC	yes/no	FC or FNC matrix	pass/block

Asset class	Required?	Found?	Path / manifest	Status
QC report	required/optional	yes/partial/no	MRIQC or motion table	pass/warn
derivative outputs	yes if postprocessing	yes/no	derivative root	pass/block

C5. BIDS/OpenNeuro/derivative discovery record

Field	Meaning
BIDS root	Raw dataset root if used.
BIDS validation status	pass / warn / fail / not run.
Derivative root	fMRIPrep, GLM, connectivity, NeuroMark, or other derivative root.
Derivative type	preprocessed images, time courses, FC/FNC, maps, GLM outputs.
OpenNeuro parsing	tasks, contrasts, condition weights, and contrast-to-concept mappings.
Fallback query used?	Whether file listing failed and entity queries were used instead.
Selective download required?	Whether remote resources had to be downloaded.
Local/remote mismatch	Any mismatch between catalogue metadata and reachable files.

Resolver output snippets

Pass or warn example:

```
resource_resolution:
episode_id: ep_hcp_predict_001
dataset_id: HCP-YA
dataset_readiness: pass
asset_readiness: warn
access_class: controlled_available
required_assets:
- feature_matrix
- target_variable
- covariate_table
- fold_manifest
resolved_assets:
feature_matrix: pass
target_variable: pass
covariate_table: pass
fold_manifest: pass
missing_assets: []
backend_reachability: pass
final_resource_verdict: pass
```

Blocker example:

```
resource_resolution:
dataset_id: candidate_dataset_X
dataset_readiness: pass
asset_readiness: block
missing_assets:
- target_variable
- fold_manifest
final_resource_verdict: block
consequence: "candidate not committed for predictive modeling"
```

C8. NeuroMark resource card

```
neuromark_resource_card:  
dataset_family: FBIRN  
resource_type: NeuroMark_2.2_ICN_time_courses  
episode_type: post_preprocessing_FNC  
raw_preprocessing_required: false  
derivative_assets:  
icn_time_courses: pass  
fnc_matrix: pass  
phenotype_assets:  
diagnostic_label: pass  
covariates: partial  
covariates_bound:  
- age  
- sex  
- site  
final_resource_verdict: pass_with_warnings
```

C9. HCP-YA resource card

```
hcp_ya_resource_card:  
dataset_id: HCP-YA  
sample_size: 326  
feature_space: Schaefer-100_x_7  
component_set: five_Liu_components  
statistic_catalogue: pyspi  
fold_manifest: fixed  
required_assets:  
- feature_matrix  
- behavioral_target  
- covariates  
- fold_manifest  
missingness_rule: subject_intersection_required  
backend_reachability: pass  
non_executable_candidates_recorded: true
```

C10. Resource-readiness verdict legend

Verdict	Meaning
pass	Required dataset and assets are available for the selected workflow.
pass_with_warnings	Executable, but caveats or sensitivity checks are required.
block	Required resource missing; plan cannot be committed.
deferred	Planned for later or external validation, not used in current claim.
non_executable	Considered but not runnable under current resources.
unavailable	Access, path, or governance prevents use.

Appendix D. Tool registry, candidate ranking, and specification ledger card

Appendix D is the tool-registry ledger: it proves how Brain Researcher moved from a scientific intent to candidate tool families, ranked candidates, rejected alternatives, selected canonical tool/workflow IDs, and generated backend-specific recipes.

D1. Purpose and scope

This card records how Brain Researcher selected, ranked, rejected, and resolved candidate tools and workflows during an episode. It links the episode intent and resource ledger to the typed tool registry, family cards, canonical tool IDs, parameter schemas, compatibility checks, backend recipes, and rejected-candidate ledger.

D2. Registry snapshot and inventory level

Field	Recommended entry	Why it matters
Registry snapshot ID	registry_YYYYMMDD or immutable hash	Pins the tool catalog used by the episode.
MCP surface	87 operations / 3 resources if using the checked implementation	Separates public MCP operations from the broader neuroimaging catalog.
Exposed + workflow specs	177 in the checked quick self-test	Records planning and recipe-generation surface.
Total registry specs	2,089 in the checked registry	Documents the broader searchable catalog without printing all specs.
Executable-wrapper inventory	~196 Python-native tools + ~2,100 command-line wrappers	Records executable scale separately from MCP endpoints.
Allowlist mode	interactive-safe / diagnostic / benchmark	Explains which tools could be retrieved.
Embedding index state	loaded / unavailable / not used	Documents whether semantic family retrieval influenced ranking.

D3. Intent and resource-category binding

Field	Example	Source
Episode ID	ep_hcp_predict_001	Appendix A
User intent	prediction	Planner / episode record
Modality	resting-state fMRI / FC	Appendix B/C
Required resources	feature matrix; target vector; co-variate table; fold manifest	Appendix C
Expected outputs	predictions; model artifact; score-card; null-test output	Appendix D/F
Incompatible families	raw preprocessing; whole-brain GLM; map-level meta-analysis	Intent-output mismatch or missing assets

```
family_card_query:
episode_id: ep_hcp_predict_001
intent: prediction
modality: resting_state_fMRI
```

required_resources:
 - feature_matrix
 - target_vector
 - covariate_table
 - fold_manifest
 expanded_families:
 - predictive_modeling
 - functional_connectivity
 rejected_families:
 - family: dynamic_fc
 reason: missing_time_series_asset
 - family: whole_brain_glm
 reason: output_type_mismatch

D4. Family cards queried

Family card	Why queried	Required inputs	Status
predictive_modeling	Matches prediction intent	features; targets; folds	expanded
functional_connectivity	Matches FC feature space	time series or FC matrix	considered
dynamic_fc	Potential alternative feature family	time-series derivative	rejected if missing asset
whole_brain_glm	Common fMRI family but wrong output type	BOLD + events	rejected for prediction episode

D5. Ranked tool/workflow candidates

Rank	Candidate	Compatibility summary	Decision
1	predictive_modeling_path_b	Input pass; output pass; Python/Slurm backend; constraints warn only	selected
2	nested_cv_elastic_net	Input pass; output pass; backend pass; alternative model family	backup or sensitivity
3	dynamic_fc_state_model	Requires time-series derivative not available in this episode	rejected
4	whole_brain_glm	Wrong output type for the prediction question	rejected

D6. Rejected-candidate ledger

Rejected candidate	Reason	Evidence pointer
dynamic_fc_state_model	Missing required time-series derivative	Appendix C asset ledger
whole_brain_glm	Output type does not match prediction question	Intent/output binding

Rejected candidate	Reason	Evidence pointer
map_correlation_workflow	Requires statistical maps rather than FC features	Resource mismatch
tool_X	Backend unavailable or not allowlisted	Backend / policy check
tool_Y	Tool-contract clause violated	Appendix E constraint card

D7. Canonical tool IDs, backend mapping, and selected specification

Canonical ID	Backend target	Runtime mapping	Expected outputs
predictive_modeling_path_b	Python / Slurm	local script or rendered batch script	predictions.tsv; scorecard.json; model artifact
pyspi_catalogue	Python	pyspi statistic runner	statistic matrix; feature table
neuromark_fnc_workflow	Python / container	post-preprocessing FNC workflow	FNC matrix; adjusted model output; report
fsl_randomise	Neurodesk	FSL command template in curated runtime	corrected maps; permutation output

```

selected_specification:
canonical_workflow_id: predictive_modeling_path_b
registry_snapshot: registry_YYYYMMDD
backend_targets:
- python
- slurm
required_inputs:
- feature_matrix
- target_vector
- covariate_table
- fold_manifest
expected_outputs:
- predictions_tsv
- scorecard_json
- model_artifact
- null_test_report

```

D8. Parameter schema and tool-contract clauses

Parameter / clause	Required?	Bound value source	Review relevance
feature_matrix	yes	Appendix C asset ledger	Input compatibility
target_variable	yes	Phenotype manifest	Outcome definition
covariates	yes or conditional	Covariate table	Confound handling
fold_manifest	yes for prediction	Fixed fold manifest	Leakage and reproducibility
no feature selection outside CV	hard clause	Pipeline DAG / tool contract	BLOCK if violated
grouped or family-block null	conditional hard clause	Episode contract	Inference validity

```

tool_contract:
canonical_id: predictive_modeling_path_b

```

required_inputs:

- feature_matrix
- target_vector
- fold_manifest

required_checks:

- no_feature_selection_outside_cv
- grouped_or_family_block_cv
- post_selection_correction_if_configurations_replayed

output_contract:

- predictions
- scorecard
- configuration_manifest

D9. Compatibility checks

Check	Possible result	Consequence
Input resource compatibility	pass / block	Candidate retained or rejected.
Output artifact compatibility	pass / mismatch	Candidate retained or rejected.
Backend availability	pass / warn / block	Recipe generated, fallback selected, or candidate rejected.
Parameter schema completeness	pass / revise	Plan can commit or must return to planning.
Tool-contract clause	pass / warn / block	Passed to Appendix E if active.

D10. Embedding-guided retrieval metadata

Field	Recommended entry
Family-card embedding input	title + description + command tags + resource labels
Embedding lane	tool-description lane
Model identity and dimension	record exact model and vector dimension
Ranking criteria	semantic similarity plus typed compatibility, backend, policy, and constraint fit
Allowlist mode	interactive-safe / diagnostic / benchmark
Cache state	loaded snapshot / invalidated / not used

D11. MCP Public Surface

Brain Researcher MCP exposes a FastMCP("brain-researcher") server with:

- 87 decorated operational tools in server.py.
- 3 resources: tool://{tool_id}, dataset://{dataset_ref}, and workflow://{workflow_id}.
- Tool discovery and execution-recipe helpers over the broader BR tool registry.
- Read-only KG, dataset, literature, grounding, run, artifact, and memory helpers.
- Manual/admin execution paths that are intentionally gated.
- Transport modes: stdio, SSE, and streamable HTTP.

- A Starlette HTTP wrapper that adds /healthz, /resolve, bearer/JWT auth, allowed host/origin checks, and MCP session-bootstrap middleware.

Capability family	Tools
artifact_inspection	artifact_list, artifact_read_text, artifact_get_metadata, artifact_read_bytes
autoresearch_review	run_autoresearch_scientific_review
dataset_resolution	dataset_get_resources
execution_recipe	get_execution_recipe
google_research	google_file_search, google_deep_research_start, google_deep_research_get, google_deep_research
grounding	grounding_resolve, grounding_gate_evidence_basis
kg_explore	kg_search_nodes, kg_get_node, kg_neighbors, kg_search_datasets, kg_related_datasets, kg_behavior_to_fmri_retrieval, kg_list_dataset_onvoc_links
kg_hypothesis	kg_verify_hypothesis, verify_hypothesis_with_kg, kg_sample_ood_hypothesis, kg_hypothesis_workflow, kg_verify_sampled_hypotheses, kg_sample_and_verify_hypotheses, kg_hypothesis_candidate_cards, hypothesis_hot_load_research, hypothesis_run_start, hypothesis_run_get, kg_hypothesis_candidate_cards_start, kg_hypothesis_candidate_cards_get
kg_probe	kg_probe, kg_find_structural_leverage, kg_detect_contradiction_motifs, kg_find_contradiction_frontiers, kg_mine_assumption_cracks, kg_find_analogy_transfers, kg_detect_topology_shifts
kg_reasoning	kg_multihop_qa
literature_search	deepxiv
memory	memory_write, memory_search, memory_get
pipeline_execution	pipeline_plan_validate, pipeline_plan_review, run_code_review, run_scientific_review, pipeline_execute
plan_handoff	get_latest_plan
planning	plan_preflight, plan_create
report_generation	latex_report_render
research_logging	log_research_event, write_session_snapshot, research_session_digest, research_log_summary
research_synthesis	generate_research_trajectory_and_insights, generate_bug_digest, generate_repo_repair_context
run_observability	run_get, run_bundle_get, run_scorecard, run_compare, run_list, run_logs, run_find_latest_reviewable, run_request_summary, run_cancel, run_metrics
scientific_report	scientific_report_generate
scientific_review	request_scientific_review, request_external_scientific_review_directive, submit_external_scientific_review_verdict, refuted_landscape_summary, companion_diagnostic_suggester
server_ops	server_info, loop_profile_get, system_self_test
sherlock	sherlock_guide, sherlock_slurm
tool_discovery	tool_search, workflow_search, tool_search_structured, tool_resolve, tool_get
tool_execution_admin	tool_execute

D11. Neuroimaging Tooling Behind MCP

Layer	What it contains	MCP endpoints	Decision value
Registry-scale tool catalog	The broader BR neuroimaging and workflow registry. The latest quick self-test loaded 177 exposed-plus-workflow specs and 2,089 total registry specs.	<code>'tool_search'</code> , <code>'workflow_search'</code> , <code>'tool_search_structured'</code> , <code>'tool_resolve'</code> , <code>'tool_get'</code>	Lets the LLM/client narrow a broad scientific request into a small set of concrete tools or workflows.
Dataset and BIDS resources	Dataset references, local paths, BIDS-style resources, derivatives, and URLs needed before execution.	<code>'dataset_get_resources'</code> , <code>'dataset://{dataset_ref}'</code>	Prevents the system from planning an analysis around missing or unresolved inputs.
Workflow definitions	Multi-step analysis recipes and exposed workflow specifications that combine lower-level tools.	<code>'workflow_search'</code> , <code>'workflow://{workflow_id}'</code>	Moves from single-method selection to end-to-end analysis plans.
Neuroimaging runtime wrappers	Tool specs that can be converted into Python commands, Neurodesk calls, container invocations, or Slurm scripts.	<code>'get_execution_recipe'</code> with <code>'target_runtime'</code>	Separates scientific tool choice from the machine where the job will run.
Quality and review layer	Run bundles, scorecards, comparisons, code/scientific review, and artifact inspection.	<code>'run_bundle_get'</code> , <code>'run_scorecard'</code> , <code>'run_compare'</code> , <code>'artifact_*</code> ', <code>'run_scientific_review'</code>	Lets the LLM decide whether outputs are complete, reproducible, and scientifically usable.

D12. Execution Backends and Where Work actually runs

Backend	Hosted where	Typical use	Who triggers it	Persistence / evidence
Local Python	Local repo checkout, developer machine, or lightweight container.	Small scripts, wrappers, quick checks, validation jobs, or tools with Python endpoints.	Usually the external coding agent or user shell, using a recipe from MCP.	If run through MCP admin execution, records land under the MCP run root; otherwise MCP observes imported artifacts/logs.
Neurodesk	Containerized neuroimaging runtime with domain software images.	Neuroimaging tools that need a curated imaging software environment rather than the slim MCP container.	External agent/user launches the generated Neurodesk recipe.	Outputs are written to allowed mounted roots and can later be inspected through run/artifact tools.
Generic container	Docker/Singularity-style runtime with controlled mounts.	Reproducible execution when dependencies should not leak into the MCP service image.	External agent/user, or gated admin execution if enabled.	The recipe declares mounts, command, parameters, and expected output locations.
Slurm / HPC	Cluster login/dev node and scheduler-managed compute nodes.	Long-running or high-memory neuroimaging jobs, especially batch or array workloads.	External agent/user submits scripts rendered by <code>'get_execution_recipe'</code> or <code>'sherlock_slurm'</code> .	Job logs, status, and produced artifacts become evidence for run bundles and review.

Backend	Hosted where	Typical use	Who triggers it	Persistence / evidence
MCP admin execution	The MCP service process or a worker process spawned by it.	Explicitly approved manual/admin runs, previews, or policy-tested pipeline execution.	‘tool.execute’ or ‘pipeline.execute’; disabled by default and governed by allowlists, roots, and approval phrases.	Creates persisted run ids, step records, trace logs, metrics, and artifacts in the run root.
Hosted HTTP MCP service	A long-running FastMCP/Starlette service, commonly on port 7000 for streamable HTTP/SSE.	Remote discovery, planning, evidence retrieval, and observability from notebooks/web assistants.	Authenticated clients call ‘/mcp’; browser clients should not hold raw MCP credentials.	Service calls are logged by the research telemetry wrapper when session/run binding is present.

D13. How MCP Decides vs. What the LLM Decides

Step	LLM / client role	MCP role	Evidence returned
1. Interpret request	Turn a natural-language research goal into candidate modalities, datasets, methods, and constraints.	Receives structured or semi-structured query arguments.	Initial query, session context, optional plan hand-off.
2. Discover tools	Ask for candidate tools/workflows and inspect summaries instead of guessing a command.	Searches the registry, applies exposed/default filters, and returns ranked cards/specs.	Tool cards, workflow cards, schema summaries, total counts when requested.
3. Resolve selection	Choose a specific method/software/workflow from returned options.	Resolves method/software/op keys to a concrete registry tool or workflow id.	Resolved tool id, metadata, parameter schema.
4. Preflight inputs	Check whether the proposed analysis has required datasets, modality, and safe roots.	Runs ‘plan.preflight’, ‘dataset.get_resources’, and policy-aware validation.	Blockers, missing inputs, resolved resource paths, execution envelope.
5. Generate recipe	Ask for a backend-specific recipe rather than writing ad-hoc commands from memory.	Builds a stateless recipe for Python, Neurodesk, container, or Slurm.	Recipe text, expected mounts, parameters, and output conventions.
6. Execute work	Normally delegates to an external coding agent, shell, container runtime, or scheduler.	Only executes if the ops/admin gate is enabled, allowlisted, and approved.	Run id, logs, stdout/stderr, Slurm status, produced files.
7. Review outputs	Decide whether the result supports the scientific claim or needs another iteration.	Normalizes bundles, scorecards, comparisons, artifacts, and review results.	Observation bundle, scorecard, metric deltas, artifact metadata, review verdicts.
8. Explain or iterate	Summarize evidence, cite run/artifact support, and choose the next MCP call if needed.	Provides durable logs, memory, grounding checks, and run/session digests.	Final evidence basis or next-step plan.

D14. Downstream Workflow Roles

Downstream use	MCP role	Primary tools
External coding-agent loop	deterministic discovery, recipe, observation, comparison, session logging	loop_profile.get, tool_search, get_execution_recipe, run_bundle.get, run_scorecard, run_compare, write_session_snapshot

Downstream use	MCP role	Primary tools
Neuroimaging planning	read-only facts, blockers, tool candidates, handoff envelope	plan_preflight, plan_create, get_latest_plan
Run review	correctness/completeness/judgment review over BR or external artifacts	run_code_review, run_scientific_review, request_external_scientific_review_directive, submit_external_scientific_review_verdict
NeuroKG reasoning	retrieve nodes, paths, datasets, structural signals, hypotheses	kg_search_nodes, kg_neighbors, kg_multihop_qa, kg_probe, kg_hypothesis_workflow
Hypothesis generation	candidate cards, hot-load research, topology shifts	kg_hypothesis_candidate_cards, hypothesis_hot_load_research, kg_detect_topology_shifts
Literature grounding	file search, deep research, arXiv/PMC search/read	google_file_search, google_deep_research_start, google_deep_research, deepxiv
Evidence-basis gate	validate/downgrade final-answer anchors	grounding_resolve, grounding_gate_evidence_basis
HPC/Sherlock execution support	render Slurm scripts and diagnose jobs/logs	sherlock_guide, sherlock_slurm
Report artifacts	render report templates and inspect produced files	scientific_report_generate, latex_report_render, artifact_*
Neuroimaging execution handoff	Selects a registry tool/workflow and converts it into an executable backend recipe while leaving heavy execution outside MCP by default.	'tool_search', 'tool_resolve', 'tool_get', 'get_execution_recipe', 'sherlock_slurm', 'dataset_get_resources'

D15. MCP Tool Inventory

Tool	Family	Tier	Arguments	Status
server_info	server_ops	ops	none	schema
loop_profile_get	server_ops	ops	profile_id	schema
system_self_test	server_ops	ops	mode, include_kg, include_container, include_script, include_inventory, inventory_limit, kg_query, strict	schema
sherlock_guide	sherlock	advanced	action, topic, intent, sunet, pi_group, partition, qos, time, cpus, mem, gpus, dataset, script_path, job_id, log_path, mount_path	schema
sherlock_slurm	sherlock	advanced	action, cluster_profile, template_kind, job_name, time, partition, qos, account, cpus_per_task, mem, nodes, ntasks_per_node, gpus, gpus_per_node, array, array_concurrency, output, error, mail_user, mail_type, workdir, module_lines, env_lines, command, task_file, launcher, include_export_none, change_request, script_text, script_path, job_id, include_squeue, include_sacct, include_scontrol, log_path, stream, tail, grep, stdout_text, stderr_text, sacct_state	schema
tool_search	tool_discovery	default	query, limit, offset, modalities, kind, phases, exposed_only, include_workflows, include_total	schema
workflow_search	tool_discovery	advanced	query, limit, offset, include_param_schema_summary	schema
tool_search_structured	tool_discovery	advanced	query, primary_intents, softwares, exposed_only, default_only, k_methods, k_softwares, k_candidates, force_fallback	schema
tool_resolve	tool_discovery	advanced	method, software, op_key, prefer_version, exposed_only, default_only, force_fallback	schema
tool_get	tool_discovery	default	tool_id, include_schema	schema
get_execution_recipe	execution_recipe	advanced	tool_id, params, target_runtime, cluster_profile	schema
plan_preflight	planning	default	query, domain, modality, inputs, allowlist_mode, semantic, selection_mode	schema

Tool	Family	Tier	Arguments	Status
plan_create	planning	default	query, domain, modality, inputs, allowlist_mode, query_understanding, include_debug, semantic	schema
get_latest_plan	plan_handoff	default	thread_id	schema
tool_execute	tool_execution_admin	ops	tool_id, params, work_dir, output_dir, preview, allow_remap, allow_fallback	schema
pipeline_plan_validate	pipeline_execution	ops	plan	schema
pipeline_plan_review	pipeline_execution	ops	plan, workflow_id, use_kg	schema
run_code_review	pipeline_execution	ops	run_id, workflow_id, force_recompute, use_kg	schema
run_scientific_review	pipeline_execution	ops	run_id, workflow_id, use_judgment_critic, force_recompute	schema
run_autoresearch_scientific_review	autoresearch_review	ops	autoresearch_dir, logs_dir, task_id, use_judgment_critic, force_recompute	schema
request_scientific_review	scientific_review	advanced	goal, run_id, autoresearch_dir, hints, workflow_id, logs_dir, task_id, use_judgment_critic, force_recompute, session_id, source_client, client_session_id	compat/manual
scientific_report_generate	scientific_report	ops	run_id, autoresearch_dir, workflow_id, logs_dir, task_id, title, authors, analysis_sections, subtitle, institution, date, use_judgment_critic, force_recompute, compile_pdf, halt_on_review_block, revision_instructions, revision_source_artifacts, local_workspace, local_workspace_manifest	schema
request_external_scientific_review_directive	scientific_review	advanced	goal, hints, session_id, source_client, client_session_id	schema
submit_external_scientific_review_verdict	scientific_review	advanced	directive_id, verdict, session_id, reviewer, notes, source_client, client_session_id	schema
refuted_landscape_summary	scientific_review	advanced	findings, session_id, run_ids, top_k	schema
companion_diagnostic_suggester	scientific_review	advanced	metric_name, observed_value, context, top_k	schema
pipeline_execute	pipeline_execution	ops	plan, dry_run, approval_phrase	schema
log_research_event	research_logging	advanced	kind, content, session_id, run_id, source, context, tags, source_client, client_session_id	schema
write_session_snapshot	research_logging	advanced	goal, done, open, next_command, session_id, run_id, source, tags, source_client, client_session_id, context	schema
research_session_digest	research_logging	advanced	run_id, session_id	schema
research_log_summary	research_logging	advanced	top_k, since_days	schema
memory_write	memory	advanced	card_type, card_data	schema
memory_search	memory	advanced	query, card_type, filters, limit	schema
memory_get	memory	advanced	card_id	schema
run_get	run_observability	default	run_id	schema
run_bundle_get	run_observability	advanced	run_id	schema
run_scorecard	run_observability	advanced	run_id, profile_id	schema

Tool	Family	Tier	Arguments	Status
run_compare	run_observability	ad-vanced	baseline_run_id, candidate_run_id, profile_id, metric_keys	schema
generate_research_trajectory_and_insights	research_synthesis	ad-vanced	run_id, candidate_id, agent_log_paths, persist	schema
generate_bug_digest	research_synthesis	ad-vanced	run_id, candidate_id, bug_query, agent_log_paths, persist	schema
generate_repo_repair_context	research_synthesis	ad-vanced	top_n, persist	schema
run_list	run_observability	default	limit, cursor	schema
run_logs	run_observability	ad-vanced	run_id	schema
run_find_latest_reviewable	run_observability	ad-vanced	limit, max_candidates, include_non_succeeded	schema
run_request_summary	run_observability	default	top_k, since_days	schema
run_cancel	run_observability	ad-vanced	run_id, reason	schema
run_metrics	run_observability	ad-vanced	run_id	schema
latex_report_render	report_generation	ad-vanced	title, authors, sections, subtitle, institution, date, compile_pdf, sections_are_latex, template_preset, front_matter_latex, back_matter_latex, bibliography_bibtex, bibliography_style, include_execution_pack	schema
artifact_list	artifact_inspection	ad-vanced	run_id	schema
artifact_read_text	artifact_inspection	ad-vanced	run_id, relpath, max_bytes	schema
artifact_get_metadata	artifact_inspection	ad-vanced	run_id, relpath, include_sha256	schema
artifact_read_bytes	artifact_inspection	ad-vanced	run_id, relpath, max_bytes, offset, start, end	schema
kg_search_nodes	kg_explore	default	query, node_types, limit, allow_degraded	schema
kg_get_node	kg_explore	default	kg_id, allow_degraded	schema
kg_neighbors	kg_explore	default	kg_id, relation_types, direction, limit, allow_degraded	schema
kg_search_datasets	kg_explore	ad-vanced	text, task_ids, modality, min_subjects, species, limit, allow_degraded	schema
dataset_get_resources	dataset_resolution	default	dataset_ref, dataset_version	schema
kg_related_datasets	kg_explore	ad-vanced	kg_id, limit, allow_degraded	schema
kg_behavior_to_fmri_retrieval	kg_explore	ad-vanced	seed_id, label, name, limit, max_maps, max_paths, max_regions_per_map, max_behavior_neighbors, min_behavior_similarity, allow_degraded	schema
kg_list_dataset_onvoc_links	kg_explore	ad-vanced	onvoc_id, page, page_size, allow_degraded	schema
kg_multihop_qa	kg_reasoning	ad-vanced	question, max_hops, mode, max_results, allowed_edge_types, return_subgraph, allow_degraded, semantic	schema
kg_verify_hypothesis	kg_hypothesis	default	hypothesis, entity_hints, allowed_node_types, max_evidence, max_paths, strictness, min_evidence_score, include_subgraph, include_path_details, confidence_scoring_version, candidate_lane_mode, semantic	schema
verify_hypothesis_with_kg	kg_hypothesis	compat	same as kg_verify_hypothesis	compat/manual

Tool	Family	Tier	Arguments	Status
kg_probe	kg_probe	ad- vanced	probe.type, query, entity_hints, seed_kg_ids, start_kg_ids, kg_id, max_results, top_k, max_hops, allowed_edge_types, hypothesis, claim, semantic	schema
kg_find_structural_leverage	kg_probe	com- pat	start_kg_ids, max_hops, top_k, allowed_edge_types, seed_kg_ids, kg_id, semantic	com- pat/ manual
kg_detect_contradiction_motifs	kg_probe	com- pat	hypothesis, entity_hints, max_results, claim, semantic	com- pat/ manual
kg_find_contradiction_frontiers	kg_probe	com- pat	hypothesis, entity_hints, max_results, claim, allowed_edge_types, semantic	com- pat/ manual
kg_mine_assumption_cracks	kg_probe	com- pat	hypothesis, entity_hints, max_results, claim, semantic	com- pat/ manual
kg_find_analogy_transfers	kg_probe	com- pat	hypothesis, entity_hints, max_results, claim, allowed_edge_types, semantic	com- pat/ manual
kg_sample_ood_hypothesis	kg_hypothesis	com- pat	seed_kg_ids, n_samples, max_hops, strategy, start_kg_ids, kg_id, semantic	com- pat/ manual
kg_hypothesis_workflow	kg_hypothesis	ad- vanced	operation, query, sampled_hypotheses, seed_kg_ids, start_kg_ids, kg_id, n_samples, verify_top_k, max_hops, strategy, strictness, allowed_node_types, max_evidence, max_paths, min_evidence_score, include_subgraph, include_path_details, confidence_scoring_version, candidate_lane_mode, use_external_literature, external_literature_top_k, external_literature_recency_days, external_literature_exclude_domains, semantic	schema
kg_verify_sampled_hypotheses	kg_hypothesis	com- pat	sampled hypotheses plus verification options	com- pat/ manual
kg_sample_and_verify_hypotheses	kg_hypothesis	com- pat	query, seed ids, sampling options, verification options	com- pat/ manual
kg_hypothesis_candidate_cards	kg_hypothesis	ad- vanced	query, seed_kg_ids, relation_types, top_n, top_k, taste_mode, controller_mode, candidate_lane_mode, frontier_mode, with_deep_research, deep_research_interaction_id, recency_days, exclude_domains, semantic	schema
hypothesis_hot_load_research	kg_hypothesis	ad- vanced	query, max_cards, depth, candidate_lane_mode, frontier_mode, seed_kg_ids, exclude_domains	schema
hypothesis_run_start	kg_hypothesis	ad- vanced	query, max_cards, depth, candidate_lane_mode, frontier_mode, seed_kg_ids, exclude_domains	schema
hypothesis_run_get	kg_hypothesis	com- pat	run_id	com- pat/ manual
kg_hypothesis_candidate_cards_start	kg_hypothesis	ad- vanced	candidate-card arguments plus background launch	schema
kg_hypothesis_candidate_cards_get	kg_hypothesis	com- pat	run_id	com- pat/ manual
kg_detect_topology_shifts	kg_probe	ad- vanced	mode, baseline_ref, current_ref, scope, approval_phrase	schema
google_file_search	google_research	ad- vanced	operation, store_name, display_name, file_path, file_name, query, metadata_filter, max_tokens_per_chunk, max_overlap_tokens, top_k, page_size, page_token	schema
grounding_resolve	grounding	ad- vanced	ref, document_resolver, kg_resolver, session_resolver	com- pat/ manual

Tool	Family	Tier	Arguments	Status
grounding_gate_evidence_basis	grounding	advanced	evidence_basis, anchors, document_resolver, kg_resolver, session_resolver, alignment_mode, partial_action, min_claim_chars	compat/ manual
google_deep_research_start	google_research	advanced	input, agent, file_search_store_names, store_names, previous_interaction_id, query, scientific_mode	schema
google_deep_research_get	google_research	compat	interaction_id, include_raw	compat/ manual
google_deep_research	google_research	advanced	query, model, max_output_tokens, temperature, recency_days, exclude_domains, scientific_mode, include_raw	schema
deepxiv	literature_search	default	action, query, paper_id, section_name, limit, search_mode, categories, date_from, date_to, authors, min_citation, days	schema

Appendix E. Constraint compiler, pre-execution validity checks, and commitment-gate card

Appendix E is the pre-execution validity gate: it proves which hard and soft constraints were active, how they were bound to the selected plan, and who allowed, revised, or blocked execution.

E1. Purpose and scope

This card records the active validity boundary between planning and execution. It documents the hard and soft constraints compiled for the selected workflow, their provenance, bound episode fields, pass/warn/block outcomes, required sensitivity analyses, caveat requirements, and commitment-gate decision.

E2. Constraint sources

Constraint source	What it contributes	Example
Tool contract	Required inputs, parameter restrictions, output requirements	Fold manifest required for prediction.
BR-KG method-condition record	Known failure modes or assumption dependencies	Feature selection outside CV violates prediction validity.
Community validator	BIDS, metadata, coordinate-space, naming rules	BIDS validation failure or warning.
Resource resolver	Missing assets, access blockers, backend compatibility	Missing target variable blocks a prediction workflow.
Review-rule registry	Known BLOCK/WARN families	GSR requires sensitivity analysis or caveat.
Episode contract	Benchmark or autonomous constraints	Fixed budget, fixed folds, validation ladder.

E3. Active constraint set

Constraint ID	Type	Source	Bound field / result
fold_manifest_required	hard	tool contract	fold_manifest.tsv found -> pass
no_feature_selection_outside_cv	hard	method-condition record	pipeline DAG satisfies clause -> pass
family_block_permutation_required	hard	episode contract	null test specified -> pass
gsr_sensitivity_required	soft	review-rule registry	unresolved choice -> warn
external_validation_missing	soft	validation ladder	external gate deferred -> caveat

```
constraint_compiler_output:
episode_id: ep_hcp_predict_001
selected_workflow: predictive_modeling_path_b
compiled_from:
- tool_contracts
- method_condition_records
- resource_resolver
- review_rule_registry
```

active_constraints:
hard: 4
soft: 2
final_pre_execution_verdict: approve_with_warnings

E4. Hard constraints

Hard check	Why it matters	Consequence
Required resource exists	Workflow cannot run without required asset.	Block if missing.
Fold manifest exists	Prevents CV leakage and irreproducible partitions.	Block or revise if missing.
Standardization inside CV	Prevents test-set leakage.	Block if violated.
Valid null / exchangeability	Required for permutation inference.	Block or revise if invalid.
Design-output compatibility	Selected workflow must match intended claim.	Reject candidate if mismatch.

E5. Soft constraints and caveat requirements

Soft constraint	Trigger	Required action / claim consequence
GSR choice	Preprocessing includes or excludes GSR	Run with/without GSR or require caveat.
Dynamic-FC window	Sliding-window choice	Window sensitivity or surrogate-null check.
Graph thresholding	Density threshold chosen	Multi-threshold curve or caveat.
Missing external validation	No independent dataset	Internal-validation claim only.
Small sample	Unstable estimate risk	Confidence intervals/permutation; no biomarker claim.

soft_constraints:
- id: external_validation_missing
source: validation_ladder
status: unresolved
verdict: warn
required_caveat: "Internal validation only; no external generalization claim."

E6. Verdict legend and commitment gate

Verdict	Meaning	Next action
pass	Constraint satisfied.	Proceed.
warn	Issue does not block execution but requires caveat or sensitivity.	Proceed conditionally.
revise	Plan is fixable but not ready.	Return to planning.
block	Hard validity or resource failure.	Stop or redesign.
deferred	Not required for current execution but limits claim strength.	Record caveat.

Gate field	Recommended entry
Gate decision	approve / approve with warnings / revise / block
Gate authority	human researcher / autonomous critic / evaluation harness
Approved workflow	canonical workflow ID from Appendix D
Approved resources	links to Appendix C resource IDs
Open warnings	list of unresolved soft constraints
Required caveats	claim restrictions passed to Appendix G/H
Benchmark pass-through?	yes/no; pass-through does not remove constraints
Execution allowed?	yes/no

```

commitment_gate:
episode_id: ep_hcp_predict_001
authority: evaluation_harness
decision: approve_with_warnings
approved_workflow: predictive_modeling_path_b
open_warnings:
- external_validation_missing
required_caveats:
- "Do not describe the result as externally validated."
execution_allowed: true

```

E7. Gate authority by episode mode

Mode	Gate authority	What Appendix E records
Interactive	human researcher	Approval, modification, or rejection of the proposed plan.
Collaborator case	researcher plus system preview	Collaborator-approved plan and caveats.
Bounded autonomous	critic/supervisor	Approve, revise, block under declared budget and policy.

E8. Example constraint/commitment excerpts

```

constraint_commitment_card:
episode_id: ep_hcp_predict_001
workflow: HCP_YA_predictive_modeling
mode: bounded_replay_or_benchmark

```

```

hard_constraints:
- id: fixed_fold_manifest_required
status: pass
verdict: pass
- id: family_block_permutation_required
status: pass
verdict: pass
- id: no_post_selection_unadjusted_claim
status: pass
verdict: pass

```

soft_constraints:
- id: external_validation_deferred
status: unresolved
verdict: warn
required_caveat: "HCP-Aging validation remains deferred."

commitment_gate:
decision: approve_with_warnings
authority: evaluation_harness
execution_allowed: true
memory_policy: qualified_claim_only
constraint_commitment_card:
episode_id: ep_neuromark_fnc_001
workflow: post_preprocessing_fnc_analysis

hard_constraints:
- id: derivative_timecourses_required
status: pass
- id: diagnostic_label_required
status: pass
- id: raw_preprocessing_not_claimed
status: pass

soft_constraints:
- id: site_covariate_required_or_caveated
status: partial
verdict: warn
required_action: include_site_covariate_or_report_limitation

commitment_gate:
decision: approve_with_warnings
authority: human_researcher
execution_allowed: true

Appendix F. Execution trace, artifact manifest, and provenance card

Appendix F is the execution evidence ledger: it proves what actually ran, where it ran, what artifacts were expected and produced, and whether failures, retries, checksums, and environment versions were recorded.

F1. Purpose and scope

This card records the authoritative execution provenance for an episode. It stores the run bundle, event trace, trajectory document, observation record, analysis bundle, artifact manifest, logs, checksums, backend and software versions, preflight results, failures, retries, recovery events, and final execution status. Scientific interpretation and claim eligibility are handled in Appendix G.

F2. Run-bundle identity

Field	Recommended entry
Episode ID	ep_hcp_predict_001
Run ID	run_hcp_pathb_001
Run mode	benchmark / collaborator / bounded autonomous
Selected workflow	canonical workflow ID from Appendix D
Backend	local Python / Slurm / Neurodesk / container
Run root	redacted path or artifact root
Start/end time	timestamps
Final execution status	completed / failed / recovered / partial

F3. Run-bundle components

Component	What it records
Event trace	Ordered execution events, timestamps, session IDs, and phase markers.
Trajectory document	Plan decisions, tool calls, parameter bindings, and <u>constraint verdicts</u> .
Observation record	Intermediate results, QC outputs, diagnostics, and <u>observed runtime state</u> .
Analysis bundle	Final artifacts, metadata, checksums, and <u>provenance chains</u> .
Run card	Policy hashes, model references, registry versions, and <u>evaluation configuration</u> .

```
run_bundle:
episode_id: ep_hcp_predict_001
run_id: run_hcp_pathb_001
components:
event_trace: trace.jsonl
trajectory: trajectory.yaml
observation_record: observation.json
analysis_bundle: analysis_bundle.json
run_card: run_card.yaml
authoritative: true
```

F4. Execution backend and recipe record

Field	Example
Backend target	Python / Neurodesk / container / Slurm
Recipe source	get_execution_recipe
Command or script path	redacted path or script hash
Runtime environment	local / container / cluster
Container or module ID	image digest / Neurodesk module / Slurm module
Submission method	shell / Slurm job / container runtime
Job ID	if applicable

```
execution_backend:  
target_runtime: slurm  
recipe_source: get_execution_recipe  
script: run_predictive_modeling.sbatch  
job_id: 123456  
container_or_module: python_env_YYYYMMDD  
output_root: redacted
```

F5. Preflight execution record

Preflight item	Result	Evidence
Filesystem access	pass	path/output-root check
Backend availability	pass	module/container check
Required assets	pass	Appendix C asset IDs
Parameter schema	pass	Appendix D tool schema
Hard constraints	pass	Appendix E active constraints
Soft warnings	warn	caveat carried forward to G/H

```
preflight_record:  
operational:  
filesystem_access: pass  
backend_available: pass  
output_root_writable: pass  
scientific:  
fold_manifest: pass  
subject_intersection: pass  
covariate_table: pass  
final_preflight_status: pass_with_warnings
```

F6. Expected vs produced artifact manifest

Expected artifact	Produced?	File/object	Used by
Predictions table	yes	predictions.tsv + check-sum	scorecard / review
Scorecard	yes	scorecard.json + check-sum	Appendix G

Expected artifact	Produced?	File/object	Used by
Model artifact	yes or n/a	model file or manifest	reproducibility
Configuration manifest	yes	config.yaml	replay
Null-test result	yes/no	permutation.json	statistical review
QC summary	complete/partial	qc_summary.json	measurement review

F7. Logs, traces, environment versions, and recovery

Record type	Recommended entries
Log manifest	stdout.log; stderr.log; trace.jsonl; tool_trace.jsonl; scheduler log; research_events.jsonl
Environment versions	Python version; package lock hash; container digest; Neurodesk module; Slurm module; tool versions; random seeds
Failure/retry ledger	failed step; cause; recovery action; outcome; whether original failure remains preserved
Scorecard summary	expected artifacts; produced artifacts; artifact-completeness ratio; required statistic present; logs complete
Final execution status	completed / completed with warnings / failed / recovered / partial / terminated

```

recovery_events:
- event_id: rec_001
failed_step: artifact_validation
failure_reason: missing_scorecard
recovery_action: rerun_scorecard_generation
outcome: recovered

```

F8. Example run-bundle excerpts

```

run_bundle_card:
episode_id: ep_hcp_predict_001
run_id: run_hcp_pathb_001
workflow: HCP_YA_predictive_modeling
backend: python_or_slurm
final_execution_status: completed

```

```

expected_artifacts:
- predictions_tsv
- scorecard_json
- configuration_manifest
- family_block_permutation_result
- post_selection_correction_report

```

```

produced_artifacts:
predictions_tsv: pass
scorecard_json: pass
configuration_manifest: pass
family_block_permutation_result: pass

```

```
post_selection_correction_report: pass

provenance:
registry_snapshot: registry_YYYYMMDD
brkg_snapshot: brkg_YYYYMMDD
software_versions: software_versions.yaml
checksum_manifest: sha256_manifest.txt

recovery_events: []
ready_for_review: true

run_bundle_card:
episode_id: ep_neuromark_fnc_001
run_id: run_neuromark_fnc_001
workflow: post_preprocessing_fnc_analysis
backend: python_or_container
final_execution_status: completed_with_warnings

expected_artifacts:
- fnc_matrix
- covariate_adjusted_model_output
- group_difference_table
- qc_summary
- report_figure

produced_artifacts:
fnc_matrix: pass
covariate_adjusted_model_output: pass
group_difference_table: pass
qc_summary: partial
report_figure: pass

warnings:
- qc_summary_partial
ready_for_review: true
```

Appendix G. Full scientific-review rule registry

G9. Full scientific review rule registry

This appendix gives the reader-facing view of the scientific review rule registry used to define, calibrate, and prioritize Brain Researcher’s verification layer. The full machine-readable registry is part of the release package. The tables below preserve the audit surface without printing the raw registry file in the supplement. A rule’s presence here does not mean that it is enforced in every episode; the lifecycle table in G4 distinguishes implemented rules, deterministic candidates, schema-dependent candidates, text-interpretation candidates, and calibration-only cases.

G9.1 Registry organization

Registry field	Values	Use in review
Severity	BLOCK; WARN; allow or positive modifier	Sets the default review action. BLOCK rules require revision before acceptance; WARN rules require reviewer attention, sensitivity analysis, reporting repair, or claim narrowing.
Lifecycle status	Implemented; deterministic candidate; schema-dependent candidate; NLP/LLM candidate; calibration-only	Separates rules currently enforced from rules that require additional metadata, text interpretation, or calibration before enforcement.
Validity layer	Statistical, measurement, construct, generalization, claim	Explains which part of the scientific inference is at risk. A single rule may touch more than one layer.
Reason tags	leakage; circularity; confound; null mismatch; low reliability; claim inflation; prior conflict; controversial choice	Provides a compact vocabulary for review cards, summary statistics, and post-hoc calibration.
Detection surface	Review-context schema, manifests, pipeline logs, provenance records, result metadata, claim text, or reviewer annotations	Defines what evidence a rule can inspect. Missing fields cause most candidate rules to skip rather than block.

G9.2 Rule families

Rule family	Representative checks	Default treatment
-------------	-----------------------	-------------------

Split, cross-validation, and leakage integrity	Nested-CV manifest completeness; subject and selection-manifest coverage; feature selection, standardization, PCA, harmonization, or confound regression fitted outside the training fold; test-set model selection; temporal splits that ignore run, story, or segment structure.	BLOCK when train-test isolation or generalization claims are invalid.
Design-model mismatch	Repeated-measures designs analyzed as independent samples; factorial designs reported as isolated simple effects without the interaction; longitudinal analyses with no subject effect; fixed-effects analyses making population claims; runs or time points treated as independent subjects.	BLOCK for invalid p-values or population claims; WARN when the error is repairable through model specification or claim narrowing.
Multiple comparisons and inference framework	Uncorrected whole-brain inference; spatial-domain mismatch between data and correction method; unrestricted permutation in hierarchical data; brain-map correlation without a spatial null; arbitrary cluster extent; ROI multiple comparisons without correction.	BLOCK for deterministic null-framework errors; WARN for soft reporting or sensitivity issues.
Circularity	ROI or feature definition from the same data and model used for effect testing; less obvious reuse such as deriving a mean activation feature and then testing the same effect.	BLOCK unless an independent localizer, atlas, or other non-circular provenance is supplied.
Confounds and measurement quality	Group motion imbalance in resting-state or FC analyses; uncontrolled demographic confounds; missing motion distribution reporting; missing MRIQC or visual QA; EPI distortion or dropout not addressed; VBM or cortical-thickness analyses without ICV/TIV; multi-site diagnosis-by-site confounding.	BLOCK for high-confidence confounds that invalidate inference; WARN for reporting gaps or correctable quality-control omissions.
Multivariate, RSA, and predictive modeling	RSA model comparisons without correction; missing hierarchical uncertainty; within-run trial-pattern correlations without appropriate regularization; voxelwise centering or scaling that induces similarity differences; post-hoc layer selection; above-chance claims without permutation or confidence intervals; fold-wise standard errors treated as independent.	Usually WARN, upgraded to BLOCK when leakage or null-framework errors are deterministic.

Reliability and sample size	Strong biomarker or BWAS claims at small sample size; “replicable” or “predictive” language without external validation; individual-difference claims without test-retest reliability or ICC.	WARN with required claim narrowing or validation language.
Claim inflation	Reverse inference from region to process; correlation described as prediction without out-of-sample evidence; model fit treated as mechanism without ablation or alternatives; fixed-stimulus effects generalized to stimulus populations; cognitive interpretation without behavioral covariates.	WARN even when the numerical analysis is formally correct, because the claim exceeds the evidence.
Controversial methodological choices	Global signal regression without on/off sensitivity; dynamic FC or TVFC without null or window sensitivity; graph thresholding without density sweeps; task FC or PPI without adequate task-evoked removal; canonical HRF in aging or clinical samples without sensitivity; single harmonization method with no comparison.	WARN plus a minimum sensitivity template; BLOCK only if the same workflow also triggers a deterministic hard error.
Prior conflict and novelty triggers	Extreme effects without robustness checks; single-ROI outliers; activation patterns dominated by unexpected regions relative to the task and literature.	WARN and audit. Prior conflict alone is not a veto.
Reporting and reproducibility	Missing COBIDAS fields; BIDS Validator failure; missing random seed for stochastic methods; missing software versions; BIDS Stats Models JSON as a positive reporting modifier.	WARN for missing reproducibility fields; allow or positive modifier for complete structured reporting.

G9.3 Policy decisions encoded in the registry

Policy question	Decision	Reason
Can a generic statistic value, such as a t, F, z, or p value, serve as an effect-size prior?	No	These values are not comparable across sample sizes, models, or outcome scales. The registry uses standardized effects, predictive scores, or modality-stratified summaries where possible.
Can synthetic or demonstration data enter automated priors?	No	Synthetic data can support unit tests, but including them in priors would contaminate novelty and prior-conflict checks.

Can prior conflict alone block a claim?	No	A result can be novel or surprising without being invalid. Prior conflict is an audit trigger unless paired with a hard methodological error.
Can novelty override a hard method error?	No	Novel findings do not rescue invalid p-values, leakage, circularity, or broken generalization claims.
Can claim-inflation rules warn when the statistics are formally correct?	Yes	Reverse inference, prediction language without out-of-sample evidence, and mechanism language without ablation are inferential problems rather than arithmetic errors.
Should controversial preprocessing choices be blocked automatically?	Usually no	Global signal regression, dynamic FC, graph thresholding, HRF choices, and harmonization choices usually trigger sensitivity analysis rather than automatic rejection.

G9.4 Implementation priority queue

Tier	Rules or rule groups	Implementation requirement
Tier 1: deterministic	Standardization outside CV; harmonization outside CV; confound regression outside CV; test-set model selection; temporal split autocorrelation; double dipping; uncorrected whole-brain inference; spatial-domain mismatch; brain-map correlation without spatial null.	Uses existing review-context fields, manifests, pipeline logs, and result metadata. These rules follow the same deterministic pattern as the nested-CV checks.
Tier 2: schema extensions	Pseudoreplication; motion group imbalance; motion reporting gaps; COBIDAS mandatory fields.	Requires additional structured fields, such as observation counts, subject counts, group labels, QC summaries, and reporting checklists.
Tier 3: text and critic layer	Correlation described as prediction; model fit described as mechanism; extreme-effect audit; controversial-choice sensitivity requirements.	Requires claim extraction, semantic comparison, or LLM-assisted review of narrative language.

G9.5 Calibration case library

The registry includes 60 calibration cases (C01–C60) for annotator training and regression testing. They are summarized by rule family here; the machine-readable release contains the case-level labels, severity, novelty flag, and rule identifiers.

Cases	Scenarios covered	Typical label
-------	-------------------	---------------

C01–C06	Repeated-measures, mixed-design, longitudinal, and spatial-domain errors, plus valid paired-test and motor-task controls.	BLOCK for invalid design or spatial correction; allow for valid controls.
C07–C12	Extreme or unexpected effects, ordinary morphometry effects, motion-confounded small-sample FC, and expected task activation.	WARN for prior conflict or confounding; allow for expected, well-scoped effects.
C13–C24	Whole-brain correction, cluster-threshold, localization, double-dipping, ROI multiplicity, analytic flexibility, pseudoreplication, fixed-effects population claims, exchangeability, and spatial-null examples.	BLOCK for hard inference errors; WARN for soft multiplicity or reporting issues.
C25–C36	Motion reporting, motion imbalance, global signal regression, dynamic FC, graph thresholding, task FC/PPI, EPI distortion, MRIQC, small sample, missing external validation, and multiband QC.	WARN except for high-confidence motion confounds that block inference.
C37–C41	Reverse inference, stimulus generalization, missing behavioral covariates, ICV/TIV omission, and multi-site site confounding.	WARN with claim narrowing or added covariates/sensitivity.
C42–C55	Harmonization, feature selection, standardization, test-set selection, split grouping, permutation, fold-wise error, prediction language, temporal splits, post-hoc layer selection, RSA multiplicity and hierarchy, and missing ICC.	BLOCK for leakage and test-set use; WARN for missing reliability, uncertainty, or permutation support.
C56–C60	Extreme effects, single-pipeline significance, COBIDAS fields, BIDS validation, and BIDS Stats Models reporting.	WARN for incomplete robustness or reporting; allow/positive modifier for structured reporting.

G9.6 Review-context fields and sensitivity templates

Field group	Examples	Rules enabled
Statistical model specification	Model type; subject effect; interaction terms; HRF model; confound regressors; stimulus effect.	Design-model mismatch, pseudoreplication, HRF sensitivity, confound checks, fixed-stimulus generalization.
Multiple-comparison metadata	Correction method, correction level, null framework, primary threshold, correction domain, spatial-null method.	Whole-brain correction, cluster-threshold, localization overclaim, spatial-domain mismatch, brain-map correlation.
Measurement and QC	MRIQC and visual-QA flags; group FD comparison; EPI-to-T1w mask overlap; site labels.	Motion, distortion, QC, and multi-site confounding rules.
Pipeline DAG and fit scope	Step name and fit scope for standardization, PCA, ComBat, feature selection, and confound regression.	Leakage rules that require train-fold fitting.
Provenance and reproducibility	ROI provenance; random seed; software versions; BIDS Validator status; BIDS Stats Models JSON.	Circularity, random-seed, software-version, BIDS-validation, and structured-reporting rules.

Trigger	Minimum sensitivity requirement
Global signal regression	Report the main findings with and without GSR; report global-signal correlation with motion or physiology; discuss alternative denoising when anticorrelation is central.
Dynamic FC or TVFC	Report results across at least three window lengths; include a surrogate null; report motion or physiology under stricter denoising; report state-feature test-retest when available.
Graph thresholding	Report key graph metrics across at least three density thresholds; include a weighted-network comparison; report overall FC level differences between groups.
HRF model	Compare canonical, canonical plus derivatives, and FIR models for key results; discuss neurovascular-coupling assumptions in aging or clinical samples.
Harmonization	Compare inside-CV and full-data fitting where relevant; compare at least two harmonization methods; report residual site predictability after harmonization.

Appendix H. Memory card, claim-relation events, and BR-KG promotion status

Appendix H is the memory-writeback ledger: it records which reviewed claims become memory cards, which do not, how new claims relate to prior memory, and whether a BR-KG promotion path exists. The memory workflow is summarized in Supplementary Fig. S10.

H1. Purpose and scope

This card records reviewed claims and their relation to prior memory. It stores accepted, qualified, rejected, blocked, and speculative outcomes separately so readers can see what enters accepted memory, what remains only in the run bundle, and what may later become a candidate BR-KG update.

H2. Memory object inventory

Memory object	What it stores	Appendix H role
Claim card	Reviewed claim text, polarity, scope, conditions, evidence level, caveats, provenance pointer	Main memory object.
Episodic run card	Question, plan, tool sequence, successful/failed tactics, blockers, resume hints	Supports future loop closure.
Review-card pointer	Review level, verdict, risk tags, fixes, eligibility	Links back to Appendix G.
Claim-relation event	support, contradiction, refinement, conditioning, supersession	Connects new claim to prior memory.

H3. Claim-card schema and writeback decision

Claim-card field	Meaning
Claim text	Final calibrated claim, not the raw candidate wording.
Polarity	support / contradiction / null / refinement.
Scope	dataset, population, modality, task, feature space.
Condition vector	Analytic conditions under which the claim holds.
Evidence level	L1 exploratory, L2 robustness, L3 internal validation, etc.
Review verdict	accept or accept with qualifications for writeback.
Caveats	Limits required by Appendix G.
Provenance pointer	run bundle, review card, artifact IDs.
Memory namespace	benchmark / collaborator / autonomous / dev.
BR-KG promotion status	none / candidate / validated / promoted.

Review outcome	Memory action	Reason
accept	create accepted claim card	claim passed review
accept with qualifications	create qualified claim card with caveats	claim usable only under conditions
revise	no accepted claim card yet	more evidence or repair needed

Review outcome	Memory action	Reason
block	no accepted claim card	hard validity failure
reject	no accepted claim card	unsupported or contradicted
speculative / exploratory without qualification	no accepted claim card	insufficient claim status

claim_card:
 claim_id: claim_hcp_pathb_001
 claim: "The frozen HCP-YA predictor remained above chance under grouped cross-validation and declared null controls."
 polarity: support
 scope:
 dataset: HCP-YA
 sample: N326_subject_intersection
 feature_space: Schaefer100x7
 condition_vector:
 fold_manifest: fixed
 null_control: family_block_permutation
 post_selection_correction: applied
 external_validation: deferred
 evidence_level: L3_internal_validation
 review_verdict: accepted_with_qualifications
 caveats:
 - "Internal validation only."
 - "No external generalization claim."
 - "No mechanistic or clinical-biomarker claim."
 memory_namespace: benchmark_isolated
 writeback_eligibility: qualified_claim_only
 brkg_promotion_status: candidate_not_promoted

H4. Condition vector, relation events, and memory partitioning

Condition field	Example
dataset	HCP-YA
sample	N = 326 subject intersection
modality / feature space	resting-state fMRI FC; Schaefer-100 x 7
fold policy	fixed fold manifest
null policy	family-block permutation
validation level	L3 internal only
caveat	external validation deferred

New claim	Prior claim	Relation	Conditioning variable
claim_042	claim_017	conditioning	preprocessing pipeline
claim_043	claim_021	refinement	sample definition
claim_044	claim_009	contradiction	null model
claim_045	claim_032	support	same feature family

memory_partition:
namespace: benchmark_isolated
shared_memory_allowed: false
prior_claim_retrieval: disabled_for_target_answers
cross_namespace_access: none
leakage_status: pass

H5. BR-KG promotion status

Promotion status	Meaning
not eligible	Rejected, blocked, or speculative.
memory only	Useful for future retrieval but not a graph candidate.
candidate graph update	May be proposed to BR-KG.
schema validated	Graph schema check passed.
provenance checked	Source/run provenance checked.
curation approved	Human or curation gate passed.
promoted	Accepted BR-KG record created.

brkg_promotion:
status: candidate_graph_update
proposed_nodes:
- Finding
proposed_edges:
- SUPPORTED_BY_RUN_BUNDLE
- CONDITIONED_ON
schema_validation: pending
source_provenance_check: pending
curation_approval: not_requested

H6. Accepted / qualified / rejected outcome examples

Candidate claim	Review verdict	Memory outcome
Predictor is above chance under grouped CV.	accept with qualifications	qualified claim card
This is externally validated.	reject / claim inflation	no accepted claim card
This is a clinical biomarker.	reject / claim inflation	no accepted claim card
Unsupported lead is confirmed.	rejected/falsified	retained only in run bundle
External validation remains deferred.	caveat	stored as condition/caveat, not success claim

Appendix J. NeuroimageKnowledge item manifest

Table 99: Full NeuroimageKnowledge groundable benchmark item set. The table reports only the open-ended agent-facing scenarios used by the benchmark harness.

Task ID	Domain	Diff.	Task type	Agent-facing scenario
NIK-NK-E-002	Neuroscientific Knowledge	easy	claim critique	risk profile for reverse inference as an inferential/logical step in cognitive neuroscience
NIK-NK-E-003	Neuroscientific Knowledge	easy	claim critique	evidence strategy to reduce reverse inference risk
NIK-NK-M-001	Neuroscientific Knowledge	medium	conceptual explanation	To distinguish prediction error from salience/surprise in model-based fMRI
NIK-NK-M-002	Neuroscientific Knowledge	medium	claim critique	You have strong RT differences between conditions. Deciding whether to include RT as covariate requires
NIK-NK-H-001	Neuroscientific Knowledge	hard	method design	Functional connectivity (correlation between regional BOLD signals) primarily
NIK-NK-H-002	Neuroscientific Knowledge	hard	conceptual explanation	Major confounds in resting-state functional connectivity include
NIK-NK-H-003	Neuroscientific Knowledge	hard	conceptual explanation	To claim a region 'implements' a computation (e.g. uncertainty), you need
NIK-ME-H-001	Methods	hard	method design	Gradient-echo BOLD is criticized for 'vein dominance' because
NIK-ME-H-002	Methods	hard	method design	To reduce macrovascular (large vessel) contribution to BOLD using established approaches
NIK-ME-H-003	Methods	hard	method design	Session effects in longitudinal fMRI can arise from
NIK-ME-H-004	Methods	hard	preprocessing review	Prospective motion correction offers advantages over post-hoc realignment because it
NIK-PP-M-001	Preprocessing	medium	reproducibility review	When reporting results with GSR in resting-state you should
NIK-PP-M-002	Preprocessing	medium	statistical review	Filtering data before regression but not filtering regressors
NIK-PP-H-001	Preprocessing	hard	preprocessing review	When comparing groups with different motion levels, the PRIMARY tradeoff between motion-matching vs covarying
NIK-PP-H-002	Preprocessing	hard	reproducibility review	Best practice for group differences with unequal motion
NIK-PP-H-003	Preprocessing	hard	preprocessing review	Surface-based analysis primarily improves
NIK-PP-H-004	Preprocessing	hard	troubleshooting memo	When using surface-based analysis with clinical groups
NIK-PP-H-005	Preprocessing	hard	preprocessing review	In diffusion MRI preprocessing, topup and eddy have distinct roles
NIK-PP-H-006	Preprocessing	hard	troubleshooting memo	Wrong phase-encoding direction in topup parameters
NIK-ST-E-003	Stats	easy	reproducibility review	When reporting fMRI results, you should include
NIK-ST-M-001	Stats	medium	statistical review	You preregistered a confirmatory family (whole-brain + 10 ROIs). You also ran 15 brain-behavior correlations labeled as exploratory. The MOST defensible multiplicity correction strategy
NIK-ST-M-002	Stats	medium	troubleshooting memo	decision point: scenario represents double dipping
NIK-ST-M-003	Stats	medium	statistical review	Valid strategies to avoid double dipping in ROI analysis
NIK-ST-M-004	Stats	medium	troubleshooting memo	Your model-based regressor correlates 0.7 with condition labels and 0.5 with RT. Signs of problematic collinearity include

Continued on next page

Task ID	Domain	Diff.	Task type	Agent-facing scenario
NIK-ST-M-005	Stats	medium	troubleshooting memo	Task t-statistics dropped after adding 24 motion parameters + CompCor. The PRIMARY diagnostic to determine if you removed artifact vs over-controlled signal
NIK-ST-M-006	Stats	medium	statistical review	To make cluster-based inference transparent and reproducible, you should report
NIK-ST-M-007	Stats	medium	statistical review	Filtering fMRI data before nuisance regression, but NOT filtering the nuisance regressors,
NIK-ST-M-008	Stats	medium	statistical review	Permutation tests are MOST valuable in fMRI when
NIK-ST-M-009	Stats	medium	statistical review	You want to control false positives while maintaining reasonable power for exploratory whole-brain analysis. Between FWE (family-wise error) and FDR (false discovery rate), the MOST appropriate choice
NIK-ST-M-010	Stats	medium	statistical review	A result survives FDR $q \leq 0.05$ but not FWE $p \leq 0.05$. This
NIK-ST-M-011	Stats	medium	statistical review	Type S (sign) error
NIK-ST-M-012	Stats	medium	statistical review	Type M (magnitude) error is MOST likely when
NIK-ST-H-001	Stats	hard	troubleshooting memo	You normalized features using full dataset mean/variance before cross-validation. This is problematic because
NIK-ST-H-002	Stats	hard	statistical review	To properly avoid leakage in cross-validated classification
NIK-ST-H-003	Stats	hard	troubleshooting memo	You report 85% accuracy on the same dataset used for feature selection and hyperparameter tuning. The PRIMARY statistical issue
NIK-ST-H-004	Stats	hard	statistical review	If an external held-out dataset is available, the gold-standard evaluation setup for high-dimensional brain-behavior prediction
NIK-ST-H-005	Stats	hard	statistical review	To make a causal mediation claim (condition $\rightarrow$ brain signal $\rightarrow$ behavior) in fMRI, you must assume
NIK-ST-H-006	Stats	hard	statistical review	To claim 'no meaningful group difference in BOLD response', you must
NIK-ST-H-007	Stats	hard	statistical review	In an underpowered fMRI study (power = 20%) that finds a 'significant' effect, you should be concerned about
NIK-ST-H-008	Stats	hard	statistical review	A Bayes Factor $BF_{10} = 0.1$ (or $BF_{01} = 10$)
NIK-ST-H-009	Stats	hard	statistical review	ROPE (Region of Practical Equivalence) in Bayesian analysis is used to
NIK-TR-H-001	Troubleshooting	hard	troubleshooting memo	One subject drives the group effect; removing them flips the conclusion. The scientifically correct approach
NIK-TR-H-002	Troubleshooting	hard	troubleshooting memo	Reviewers suspect motion-driven distance bias in your connectivity group differences. To test this, you should
NIK-TR-H-003	Troubleshooting	hard	troubleshooting memo	decision point: pattern would MOST reassure you that group connectivity differences are not motion artifacts
NIK-TR-H-004	Troubleshooting	hard	troubleshooting memo	After preprocessing, some subjects have NaN or extreme outliers in ROI timeseries, but the pipeline didn't crash. Silent causes include
NIK-TR-H-005	Troubleshooting	hard	troubleshooting memo	To locate the source of NaN values in ROI timeseries, the MOST SYSTEMATIC diagnostic approach
NIK-TR-H-006	Troubleshooting	hard	troubleshooting memo	To diagnose scanner drift separate from participant motion in longitudinal data
NIK-IN-E-002	Interpretation	easy	claim critique	Regarding scientific interpretation, a significant contrast 'A $\neq$ B' does NOT demonstrate
NIK-IN-E-004	Interpretation	easy	reproducibility review	When reporting 'deactivation', the MOST scientifically accurate phrasing
NIK-IN-M-001	Interpretation	medium	claim critique	The same region activates in Task A (working memory) and Task B (cognitive control). The MOST scientifically defensible interpretation

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Task ID	Domain	Diff.	Task type	Agent-facing scenario
NIK-IN-M-002	Interpretation	medium	claim critique	To strengthen claims beyond simple overlap, you need
NIK-IN-M-005	Interpretation	medium	claim critique	Clinical group shows higher BOLD activation than controls. Before claiming neural differences, consider
NIK-IN-M-006	Interpretation	medium	claim critique	To interpret hyperactivation as 'compensation' rather than artifact or inefficiency, you MUST show
NIK-IN-M-008	Interpretation	medium	statistical review	You found a strong brain-behavior correlation after exploring many ROIs and behavioral metrics. To keep it in the paper defensibly, you should
NIK-IN-H-001	Interpretation	hard	claim critique	BOLD group differences could reflect
NIK-IN-H-002	Interpretation	hard	claim critique	To strengthen the interpretation of BOLD group differences as neural (not vascular), you could
NIK-IN-H-003	Interpretation	hard	claim critique	To claim a region 'implements' a computation (e.g. uncertainty estimation), beyond simple correlation with a model regressor, you need
NIK-IN-H-004	Interpretation	hard	claim critique	Your main effect appears only with one preprocessing choice (e.g. only with GSR, not without). The MOST appropriate interpretation and reporting strategy
NIK-IN-H-005	Interpretation	hard	claim critique	Your behavioral effect replicates across labs but the fMRI effect does not. The MOST scientifically defensible interpretation
NIK-IN-H-006	Interpretation	hard	claim critique	Causal language about brain-behavior relationships (e.g. 'Region X generates behavior Y') is MOST justified when
NIK-BP-E-004	Best Practices	easy	conceptual explanation	Your team says 'let's try different pipelines and pick the one with strongest activation in expected regions.' This is problematic because
NIK-BP-E-006	Best Practices	easy	reproducibility review	When reporting fMRI results, you should include behavioral performance
NIK-BP-M-001	Best Practices	medium	reproducibility review	When preregistering an fMRI analysis, the PRIMARY distinction between confirmatory and exploratory
NIK-BP-M-002	Best Practices	medium	reproducibility review	A preregistration for fMRI should include
NIK-BP-M-003	Best Practices	medium	statistical review	You have 10 ROIs and 5 behavioral measures. To avoid 'garden of forking paths' while remaining useful for theory, the BEST strategy
NIK-BP-M-004	Best Practices	medium	reproducibility review	The PRIMARY purpose of sensitivity analyses (e.g. with/without GSR, different motion thresholds) is to
NIK-BP-M-005	Best Practices	medium	reproducibility review	To prevent 'it ran on my machine' reproducibility failures when sharing code
NIK-BP-M-006	Best Practices	medium	troubleshooting memo	To ensure QC-based exclusions are defensible and not biased, you should
NIK-BP-M-007	Best Practices	medium	reproducibility review	When comparing groups with different motion levels, best practice reporting includes
NIK-BP-M-008	Best Practices	medium	reproducibility review	Your conclusion changes substantially with different normalization settings. You should
NIK-BP-H-001	Best Practices	hard	reproducibility review	You tried five preprocessing variants. The MOST defensible reporting approach to avoid p-hacking while enabling exploration
NIK-BP-H-002	Best Practices	hard	reproducibility review	When reporting results across multiple pipelines, include
NIK-BP-H-003	Best Practices	hard	method design	To CONTROL FOR leakage/memorization in a neuroimaging benchmark using published literature (preventative measures AND detection diagnostics)
NIK-BP-H-004	Best Practices	hard	method design	Before claiming clinical relevance from a neuroimaging prediction model, you must
NIK-BP-H-005	Best Practices	hard	reproducibility review	The PRIMARY purpose of multiverse analysis (testing all reasonable analytic combinations) is to

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Task ID	Domain	Diff.	Task type	Agent-facing scenario
NIK-BP-H-006	Best Practices	hard	conceptual explanation	When facing unavoidable researcher degrees of freedom (multiple defensible preprocessing choices), the approach that BEST balances rigor and discovery

Appendix J. Representative paired NeuroimageKnowledge answers

Example 1: content tie where no-BR recalls strong canonical sources. Task and model. `codex_gpt55`, NIK-BP-E-004.

Question. Your team says, “let’s try different pipelines and pick the one with strongest activation in expected regions.” Why is this problematic?

without-BR answer excerpt. Trying many preprocessing or analysis pipelines and then choosing the one with the strongest activation in expected regions is a form of analytic circularity or researcher degrees of freedom. The same prior expectation is being used both to select the analysis and to support the result, so the final activation is no longer an independent test of the hypothesis.

with-BR proper answer excerpt. This is problematic because it turns pipeline choice into a results-contingent decision. If the team tries many preprocessing, modeling, or thresholding variants and then reports the one with the strongest activation in expected regions, the nominal statistical evidence no longer reflects a single planned test.

Evidence and scoring. The without-BR answer cited real canonical sources from memory, including DOI:10.1038/nm.2303 / PMID:19396166, DOI:10.1177/0956797611417632, and DOI:10.1038/s41586-020-2314-9. The support judge marked 3 without-BR evidence rows as yes. The with-BR answer used retrieved-document anchors; the support judge marked 4 rows as yes and 1 as partial. Both pairwise judges marked content as a tie and audit-aware judging as without-BR.

Takeaway. no-BR is not treated as wrong merely because it did not call BR. If a memory-based DOI or PMID resolves and supports the claim, it receives credit.

Example 2: content tie where BR improves evidence closure. Task and model. `codex_gpt55`, NIK-NK-E-003.

Question. Which evidence strategy most reduces reverse-inference risk?

without-BR answer excerpt. To reduce reverse inference risk, avoid claiming that activation in a region proves engagement of a specific mental process. A safer interpretation is that the observed pattern is consistent with the proposed process, but not diagnostic unless supported by converging evidence.

with-BR proper answer excerpt. The manuscript should not infer a specific mental process solely from regional activation. A safer interpretation is that the task condition was associated with activity in regions or networks previously implicated in a broader functional domain, while the specific cognitive process remains a hypothesis unless supported by converging task, behavioral, decoding, meta-analytic, or perturbation evidence.

Evidence and scoring. The without-BR answer used general-principle rows without concrete references, so the support-judge denominator contained no claimed-grounded source rows. The with-BR answer used retrieved-document anchors, and the support judge marked 5 rows as yes. Both pairwise judges marked content as a tie and audit-aware judging as with-BR.

Takeaway. The answers make the same conservative scientific point, but BR turns that point into resolvable evidence rows.

Example 3: BR improves both content and auditability. Task and model. `codex_gpt55`, NIK-BP-H-002.

Question. When reporting results across multiple pipelines, what should be included?

without-BR answer excerpt. For multi-pipeline neuroimaging results, report enough detail for each pipeline that another lab can rerun, compare, and audit the analyses. At minimum, provide a pipeline manifest, preprocessing and modeling choices, software or container versions, parameter settings, inclusion and exclusion rules, quality-control outputs, harmonized outcome definitions, and both per-pipeline and aggregate results.

with-BR proper answer excerpt. For a multi-pipeline neuroimaging study, report every pipeline as an auditable analysis path: software and versions, preprocessing steps, subject-level model, group model, nuisance regressors, smoothing, HRF basis, thresholding, multiple-comparison correction, excluded data, and runtime environment. Publish the analysis code, configuration files, provenance metadata, QC outputs, primary statistical maps, and where feasible, all alternative-pipeline results.

Evidence and scoring. The without-BR answer used general-principle rows without concrete references. The with-BR answer used retrieved-document anchors; the support judge marked 5 rows as yes and 2 as partial. Both Codex and Claude pairwise judges selected with-BR for content-only and audit-aware criteria.

Takeaway. This is a case where BR improves the checklist itself and also attaches that checklist to source-grounded evidence rows.

Example 4: no-BR wins on content and audit-aware judgment. Task and model. `opencode_gemini31_pro`, NIK-BP-H-003.

Question. How should a neuroimaging benchmark based on published literature control for leakage or memorization, including both preventative measures and detection diagnostics?

without-BR answer excerpt. To appropriately control for leakage and memorization in a neuroimaging benchmark based on published literature, adopt a hybrid approach using temporal holdouts for prevention and n-gram overlap or membership-inference diagnostics for detection. Relying solely on temporal cutoffs is insufficient due to preprint ingestion and undocumented rolling updates in foundation models.

with-BR proper answer excerpt. To effectively control for leakage and memorization in neuroimaging benchmarks, implement nested cross-validation with strict subject-level or site-level splitting to prevent training on repeated scans from the same individuals. Ensure all preprocessing and feature selection occurs exclusively within training folds to avoid double-dipping, and validate against an independent out-of-distribution set.

Evidence and scoring. The without-BR answer used general-principle rows. The with-BR answer had one retrieved-document yes row plus uncertain and general-principle rows. Both pairwise judges selected without-BR for content-only and audit-aware criteria.

Takeaway. This counterexample shows why the evaluation does not claim that BR always improves answer quality. The no-BR answer is better matched to the literature-benchmark leakage question, while the with-BR answer drifts toward predictive-model leakage.

Example 5: weak-model case where BR reduces off-target citations. Task and model. `opencode_glm51`, NIK-NK-H-003.

Question. What evidence is needed to claim that a brain region implements a computation such as uncertainty?

without-BR answer excerpt. To claim a region implements a computation such as uncertainty, you need convergent evidence across multiple criteria: selective activation by conditions that parametrically vary the target computation, multivariate or model-based fMRI evidence that the pattern carries information about the computational variable, and controls for alternative explanations.

with-BR proper answer excerpt. To claim a brain region implements a computation such as uncertainty, you need substantially more than showing the region activates more under uncertain conditions. The core requirement is evidence that the region selectively and necessarily carries out that computational transformation, something activation alone cannot establish.

Evidence and scoring. The without-BR answer used specific citations and general-principle rows, but the support judge marked 3 claimed-grounded rows as no_unrelated. The with-BR answer used retrieved-document anchors; the support judge marked 6 rows as yes and 1 as partial. Codex selected with-BR for content and audit; Claude marked content as a tie and audit-aware judging as with-BR.

Takeaway. The no-BR answer content is plausible, but the evidence basis is noisy. BR keeps the conservative scientific point while replacing off-target evidence with resolvable support.

Appendix J. Unified benchmark-bundle inventory

J2d. Unified benchmark bundle artifact inventory

The Appendix J benchmark tables and examples are grounded in the unified bundle at `benchmarks/UNIFIED_BENCHMARK_BUNDLE`. The appendix includes public manifests, source-safe task descriptions, metric denominators, and selected examples. Raw episode payloads, private answer-key mappings, judge logs, and rerun scripts remain in the bundle and are referenced as provenance rather than copied into the manuscript source.

Table 100: Unified benchmark bundle artifacts used to support Appendix J. File paths are relative to `benchmarks/UNIFIED_BENCHMARK_BUNDLE`.

Bundle component	Canonical files	Appendix role and guardrail
Tool-calling traces and metrics	<code>tool_routing/traces/INDEX.csv</code> ; <code>tool_routing/eval/results/.</code>	Trace index records 840 episode paths, matching 60 tasks times 7 models times 2 modes. Action-budget metrics are reported as trajectory-level quantities, not execution success claims.
Tool-calling canonical provenance	<code>tool_routing/CANONICAL_PROVENANCE.md</code> ; <code>tool_routing/canonical_versions/.</code>	Documents the v0 to v7 lineage: parser repair, clean-finish reruns, multi-model consensus, Claude expert review, and the v7 dual-reviewer intersection used for the main result.
NeuroimageKnowledge task manifests	<code>neuroimage_knowledge/tasks/source_v3_canonical.csv</code> ; <code>neuroimage_knowledge/tasks/task_index.csv</code> ; <code>neuroimage_knowledge/tasks/agent_prompts.jsonl</code> ; <code>neuroimage_knowledge/tasks/schema.json</code> .	Source for the 76-item groundable manifest in Table 99. The private answer-key mapping remains in the benchmark bundle and is not exposed in Appendix J.
NeuroimageKnowledge traces and metrics	<code>neuroimage_knowledge/traces/INDEX.csv</code> ; <code>neuroimage_knowledge/eval/methodology.md</code> ; <code>neuroimage_knowledge/eval/results/comparison_paper_grade.json</code> ; <code>neuroimage_knowledge/eval/results/.</code>	Trace index records 1,064 episode paths, matching 76 tasks times 7 models times 2 modes. The 50-question analysis files define the evidence-citation aggregate used for the Results.
NeuroimageKnowledge ground-truth audit	<code>neuroimage_knowledge/gt_cross_review/README.md</code> ; <code>nik_review_packet.json</code> ; <code>gemini_nik_review.json</code> ; <code>codex_nik_review.json</code> ; <code>v2_to_v3_edits.json</code> .	Records the independent ground-truth cross-review and two wording-only v2 to v3 clarifications. It supports task validity, not model performance.
Out-of-scope supplementary NIK content	<code>../UNIFIED_BENCHMARK_BUNDLE_supplementary/nik_definitional_61/.</code>	Preserves 61 definitional items where BR was not expected to help and the empirical effect was mixed. These are excluded from the main positive-results bundle and main Appendix J metrics.

Table 101: Bundle denominators used by Appendix J.

Surface	Task set	Episode index rows	Paper-facing subset
Tool calling	60 canonical v7 routing tasks	840	Full 60-task surface; 420 model-item trajectories per mode.

Surface	Task set	Episode index rows	Paper-facing subset
NeuroimageKnowledge	76 groundable open-ended tasks	1,064	Full 76-item auditability surface; 50-item BR-separation cut for the main paper-grade grounding result.

Appendix J. Representative benchmark-bundle cases

J2e. Representative NeuroimageKnowledge benchmark cases

The cases below are selected from the NeuroimageKnowledge task-level audit files to show how Appendix J connects aggregate evidence-citation metrics to concrete benchmark items. They are illustrative examples and do not add new denominators beyond Tables 99 and 101.

Table 102: Representative NeuroimageKnowledge task-level cases from the 50-question analysis set.

Task ID	Scenario focus	Observed task-level pattern	Interpretation boundary
NIK-NK-H-003	Evidence required to claim that a region implements a computation.	Largest-separation case; verified groundedness increased from 0.025 to 0.529; all 7 models had positive verified-groundedness deltas.	A high-separation case where BR materially improves source-backed auditability for an inferentially difficult claim.
NIK-ME-H-003	Session effects in longitudinal fMRI.	Rank 2; verified groundedness increased from 0.000 to 0.525, while mean correctness score decreased by 0.054; all 7 models had positive verified-groundedness deltas.	Demonstrates why Appendix J separates auditability metrics from raw correctness.
NIK-TR-H-001	One subject drives a group effect and removing them flips the conclusion.	Rank 3; verified groundedness increased from 0.000 to 0.297; mean correctness score increased by 0.381; with-BR citation-spam rate was 0.000.	A case where BR improved both the scientific answer and the evidence basis.
NIK-BP-E-004	Trying multiple pipelines and selecting the strongest expected-region activation.	Rank 11; verified groundedness increased from 0.073 to 0.484; mean correctness score delta was 0.000.	Content can already be strong without BR, but BR changes the evidence boundary by adding resolvable anchors.
NIK-BP-H-003	Leakage and memorization controls for literature-based neuroimaging benchmarks.	Rank 42; verified groundedness increased from 0.000 to 0.054; only 2 models had positive verified-groundedness deltas.	Included as a weak-separation boundary case; it cautions against claiming that BR always improves the answer or evidence audit.